

Pre-empting Competitive Sales: Private Offers, Early Resolution, and Seller Information

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Abstract

Buyers sometimes make private offers to cancel announced competitive sales. Norwegian, Australian, and New Zealand housing markets exemplify settings allowing pre-emption. We study when it occurs and why sellers reject some attempts. Waiting preserves the seller's chance of an outcome worth more than the buyer's valuation, creating a competition wedge. Early resolution or risk aversion can offset this wedge, while private seller information about late-participation prospects, urgency, or outside options can make acceptance state-dependent. Equilibrium can combine waiting, rejected probing offers, and successful pre-emption. A ban preserves competitive allocation and avoids effort costs of early bargaining but sacrifices gains from early resolution.

Keywords: auctions, competitive bidding, pre-emptive offers, housing, pre-auction bargaining, private information, screening, early resolution.

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1 Introduction

Competitive sales are often announced before the seller is committed to holding them. A buyer can then attempt to end the process with a private, pre-emptive offer. In the Norwegian housing market this practice is known as *kupping* (Finansdepartementet, 2021, p. 66); Australian and New Zealand residential auction rules likewise permit sellers to accept offers before the scheduled auction (NSW Fair Trading, nd; Real Estate Authority, nd). The same strategic sequence may arise in announced sales of art, industrial equipment, or corporate assets when the seller remains free to accept a private offer before bidding begins. Acceptance excludes other prospective buyers and abandons the price discovery and allocation provided by open competition, while giving the buyer early certainty and saving the seller the costs of continuing the sale.

We study a buyer who values securing a purchase before an announced auction but does not know how attractive the auction is to the seller. Making an early offer is costly, and the buyer chooses the price. We show when the buyer prefers an offer that succeeds only when the seller’s auction prospects are weak, rather than waiting or paying enough to secure acceptance in either state. Such an offer can secure early trade without paying the higher price needed to persuade a seller with strong auction prospects. Rejection leaves the buyer free to compete in the auction.

In our one-offer game, the early buyer privately knows its object-specific match value and its benefit from early resolution. The seller privately observes a binary state that determines its net value of continuing to the English auction. The buyer’s private offer is binding. Because the buyer remains a bidder after rejection, its offer also informs the seller about the revenue to expect from the auction. The price therefore both signals the buyer’s counterfactual bid and screens the seller’s continuation state. We characterize pure-strategy perfect Bayesian equilibria (PBE), use locally passive beliefs for price selection, and apply Criterion D1 (Cho and Kreps, 1987; Banks and Sobel, 1987) to assess the relevant signaling equilibria. Figure 1.1 summarizes the sequence.

Our first result identifies the opportunity cost of cancelling competition. Before direct resolution gains and offer costs, an accepted price p gives the buyer $b - p$ and the seller p , for a combined payoff of b . The announced round instead gives them $b + \Gamma_s(b)$ in expectation. The excess

$$\Gamma_s(b) = \mathbb{E}_s[\max\{\max\{v_s, Y_{2N}\} - b, 0\}] \geq 0,$$

is the *competition wedge*: the extra expected payoff that the round provides to the seller and early buyer jointly. Here s is the seller’s continuation state, v_s its no-sale value, and Y_{2N} the second-highest late value. The wedge arises when competing late buyers can push the price above b , or the seller’s no-sale value exceeds it. What the seller knows about

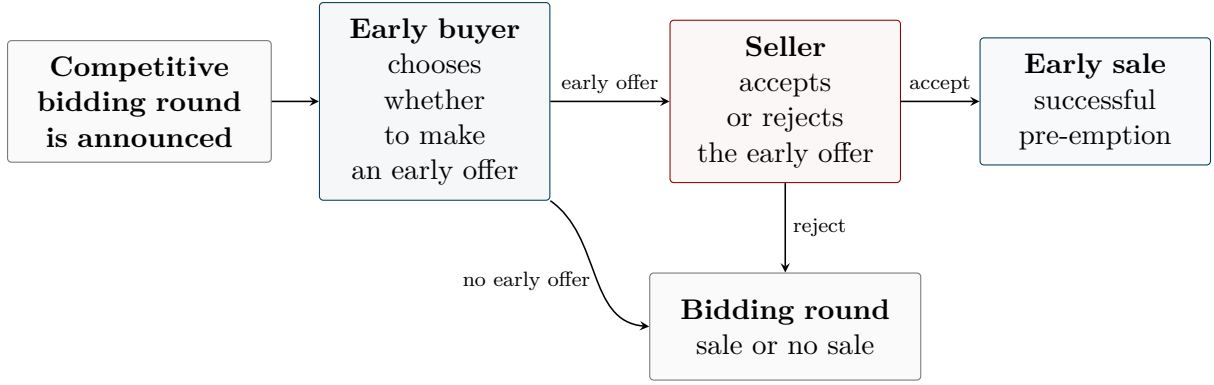


Figure 1.1: Baseline sequence of decisions and outcomes. The bidding round is announced before the early buyer chooses whether to make an offer. Acceptance produces an early sale; rejection and no early offer lead to the bidding round.

its auction prospects, and what the offer reveals about the buyer’s value, affects whether an early sale occurs. For that sale to benefit the two parties jointly, however, the gains from early resolution must cover what they give up by cancelling the auction. In every risk-neutral PBE, accepted trades must collectively finance this wedge and the costs of successful and rejected attempts; with a positive attempt cost and no direct resolution gains, nobody attempts pre-emption.

In the risk-neutral baseline, an early purchase gives the buyer a resolution benefit and saves the seller continuation costs.¹ Housing transactions illustrate both sides. A household that has sold its current home may need to secure another before moving out, making a suitable home available in time more attractive than continuing to search for a better match. An early agreement can help it coordinate the move and avoid committing to temporary accommodation or storage. A seller who has already bought another home may likewise value resolving the sale of its existing property to plan financing and relocation.²

Similar considerations arise outside housing. A contractor buying used equipment may need to arrange transport and assign a crew before work begins. Securing the equipment early can allow the contractor to keep its planned start date rather than book a temporary rental or reschedule the work. An acquirer purchasing a production facility may be coordinating the transaction with a new customer contract or an investment at another site. An early agreement can allow it to commit to delivery schedules and complementary investments; waiting may require an interim supply arrangement or postponing those commitments. An investor may need to secure one company before committing to a second, complementary acquisition. When the second deal has a deadline, resolving the first early can preserve the opportunity to realize synergies between them.

¹Belief errors have also been proposed as an explanation for pre-auction agreement (Ivanova-Stenzel and Kröger, 2008); with heterogeneous beliefs, welfare comparisons depend on how beliefs are evaluated (Brunnermeier et al., 2014).

²Nordea (nd) highlights the potential need for temporary accommodation when selling first and the burden of overlapping housing costs when buying first. See “Bør man kjøpe eller selge bolig først?”, <https://www.nordea.no/privat/din-okonomi/bolig/kjope-selge-bolig.html> (accessed 8 September 2026).

Our second result shows how seller information determines which potential gains are realized. A seller may privately observe weak or strong continuation prospects from participation signals, its no-sale fallback, or its own urgency. Under explicit primitive conditions, sufficiently patient buyers wait, while more urgent buyers make a moderate *probing offer* that the low-continuation seller accepts and the high-continuation seller rejects. The buyer thereby secures early agreement when the seller's continuation is weak without paying enough to pre-empt a strong continuation; after rejection, it retains access to the announced competition. Because match value and urgency jointly determine who attempts, the offer is an informative but imperfect signal of the buyer's counterfactual bid. Waiting and probing alone produce no early offer, rejection, and successful pre-emption with positive probability, while the most urgent buyers may add a higher *knockout offer* accepted in both states.

The ordered binary seller state can reflect stronger late participation, a higher no-sale fallback, or greater seller patience. Each admits probing, but rejection conveys different news: about competing bids, the seller's terminal threshold, or its gains from early agreement. These distinctions also affect whether D1 rules out equilibria without rejection. The main quantitative example combines all three components; the general results also cover each separately.

Our third result compares a ban on pre-emption with discretion to trade early. Holding late participation fixed, a ban preserves the terminal allocation process and avoids the effort costs of attempting pre-emption; discretion to trade early instead delivers the buyer's resolution benefit and saves seller continuation costs and any bidder preparation not yet incurred. A ban is therefore more attractive when competition often reallocates the object to a higher-value late buyer or to the seller's no-sale use, and when rejected attempts consume substantial resources; discretion is more attractive when completed early trades rarely change the eventual allocation and deliver substantial resolution benefits or save preparation resources. Two equilibria that differ only in their common, mean-matched distribution of buyer values show that moving probability into the upper tail can reverse this welfare ranking: a rare high late value can raise the allocative loss from pre-emption without comparably raising the auction price, which is governed by the runner-up.

Committed and voluntary disclosure need not coincide. Verifiable disclosure of the seller's continuation state before the buyer acts eliminates on-path rejected offers in pure-strategy equilibria when attempts are costly. Yet, under the canonical continuation equilibria, the seller weakly prefers opacity ex ante, and any positive certification cost makes silence in both seller states the unique pure reporting equilibrium. The numerical disclosure exercise isolates late participation, the component most naturally represented by verifiable market information.

Risk aversion supplies another source of gains: parties may surrender expected payoff to secure early agreement instead of facing the auction lottery. We also extend the model

to risk-averse buyers. Buyers privately know their degree of risk aversion, which affects whether they make an early offer, while the seller’s private auction prospects determine whether the offer is accepted. A D1-compatible equilibrium retains waiting, rejected attempts, and early sales without direct resolution benefits or seller timing costs.

No early offer, rejected attempts, and successful pre-emption can also coexist with continued bargaining before the auction or with a seller counteroffer after bidding (Appendices H–I). Continued bargaining before the auction can induce early offers even from buyers with no direct benefit from early resolution. After bidding, a seller counteroffer can make the final purchase price depend on whether the buyer previously waited or made an early offer.

Section 2 develops the institutional applications and protocol comparisons with related work. Sections 3–4 establish the auction payoffs, gains from early agreement, and information structure. Sections 5–6 construct and refine equilibria, followed by the example in Section 7. Section 8 studies welfare and disclosure; Section 9 examines insurance and bargaining.

2 Institutional Applications and Related Literature

2.1 Housing-Market Institutions and Evidence

Residential property markets provide a transparent application of buyer-initiated pre-emption because a competitive stage is commonly announced before the seller is committed to holding it. Norwegian residential listings normally announce an asking price, a public viewing, and a subsequent ascending bidding round. Recent Norwegian bid-log evidence describes these rounds as digital ascending auctions in which the broker relays binding bids to the participants in real time (Anundsen et al., 2026). The seller may reject any bid and need not choose the highest one. A buyer seeking to pre-empt the process can approach the seller directly, often after a private viewing and with a short acceptance deadline; acceptance cancels the announced viewing and bidding round (Finansdepartementet, 2021, pp. 58, 66).

Since 1 July 2025, an estate agent may not relay a bid conditioned on keeping the bid or its conditions secret from other bidders and interested parties (Stortinget, 2025; Finanstilsynet, 2025). The reform did not prohibit direct agreement between buyer and seller. It does, however, require the broker to urge the seller to refer bidders to the broker instead of contracting directly, and to consult the broker when receiving a direct bid (Stortinget, 2025, Section 6-8(2)). The legislative proposal notes that prohibiting direct agreements would require a broader intervention in freedom of contract and might redirect attempts toward direct contact (Finansdepartementet, 2025, ch. 7). This direct route corresponds most closely to the baseline’s private early offer, subject to the broker’s

advisory role.

In a sample of 84,203 Oslo sales from 2007 to 2017, [Flått et al. \(2022\)](#) classify 2,018 transactions (2.4 percent) as pre-empted because they closed within five days of listing. They estimate an average price premium of about 4.5 percent and find that pre-emption is more common for smaller, newer, and higher-priced dwellings and when market activity is high. They suggest that the premium may reflect payment by risk-averse buyers to avoid auction uncertainty.³ This conditional price moment provides a useful comparison with the model’s selected price distribution. Under the maintained independent-type specification, the baseline supports its sign: high-value buyers are more likely to attempt, seller continuation information sorts attempts between acceptance and the planned round, and the main calibration yields a higher mean price among completed pre-emptions than among completed bidding-round sales. Risk aversion provides one mechanism for this pattern, as suggested by [Flått et al. \(2022\)](#); heterogeneous benefits from early resolution provide another.

The classification is constructed from completed transactions and therefore does not identify rejected attempts. Eiendomsmeglingsutvalget reports that an analysis of bidding data indicates an average incidence of around 5 percent of home sales, but stresses that pre-emption is not systematically registered ([Finansdepartementet, 2021](#), ch. 6.6). The observed incidence of early sales thus leaves open both why sellers sometimes accept and how often early offers are rejected.

Norwegian practice matches the baseline’s seller discretion over early agreement but not its terminal commitment. Asking prices are nonbinding, and the seller may reject every bid ([Finansdepartementet, 2021](#), ch. 6.6). Professional practice nevertheless disciplines them. Finanstilsynet defines the asking price as the broker’s assessment of market value and the price at which the seller is willing to sell ([Finanstilsynet, 2022](#)); the official review states that a broker may not advertise a price the seller has not approved and that lure pricing violates good brokerage practice ([Finansdepartementet, 2021](#), p. 220). In mapping the baseline to this setting, we absorb the asking price’s public information into the common primitives and represent the seller’s private no-sale alternative by v_s . Appendix I studies one final counteroffer.

Australian and New Zealand residential auctions combine the same possibility of pre-auction agreement with stronger terminal commitment: the reserve is confidential, and once it is met the highest bidder wins under the announced auction rules ([NSW Fair Trading, nd](#); [Real Estate Authority, nd](#)). Across these housing institutions, the common margin is the seller’s discretion to replace announced competition with early bilateral trade; they differ in how tightly the seller is bound once bidding begins. The baseline isolates the common early-agreement margin and uses a committed terminal round as its benchmark.

³Section 9.1 shows how risk aversion changes bilateral gains from early agreement.

2.2 Related Literature

The mechanism we study is a private binding offer initiated and priced by a buyer before an announced competitive sale. Grebe et al. (2016) study the same Sell-It-Now sequence: one designated buyer makes the offer and, after rejection, enters a second-price auction with one rival. Their risk-neutral benchmark has a publicly known seller value of zero, fixed participation, no reserve or attempt cost, and no direct gains from early resolution. Under their tie-breaking rule, every offer is rejected. Yet 37 percent of Sell-It-Now offers were accepted in their laboratory experiment. Their working-paper version refers to Grebe’s dissertation for an analysis with risk-averse buyers and/or sellers (Grebe et al., 2015, fn. 7). We study how private seller continuation prospects interact with voluntary costly offers and heterogeneous resolution benefits to support no offer, rational rejection, and state-dependent acceptance.

The Sell-It-Now label has also been used more broadly: Vragov et al. (2008) use it for temporary buyer-submitted prices during an ongoing online auction, while developing their formal pricing analysis in seller-posted dynamic-buyout form. The Best Offer label denotes a different sequential haggling game between one seller and a series of potential buyers (Huang et al., 2013). In Khezr and Menezes (2018), the seller posts a common asking price and buyers simultaneously either accept it or submit counteroffers. In their baseline, the seller then learns its reservation value and may accept the highest counteroffer or retain the object; an auction occurs only when multiple buyers accept the asking price. Acceptance thus depends on seller information, but rejected counteroffers do not lead to an auction. Here the seller knows its continuation state before responding, and rejection preserves the announced auction in which the buyer remains eligible to bid.

Seller-initiated pre-auction offers and buyout prices are studied by Kirkegaard and Overgaard (2008), Mathews (2004), Ivanova-Stenzel and Kröger (2008), and Khezr (2018). Choné and Linnemer (2025) let the seller offer a preferred buyer an option whose rejection informs a later tailored auction; here the informed buyer chooses the price and screens the seller. Indicative bids can privately convey preliminary willingness to pay before costly due diligence, but they are nonbinding and select entrants rather than terminate the sale (Quint and Hendricks, 2018). Public buyer-initiated pre-emptive bids can deter rival information acquisition or entry, as in Fishman (1988) and Cramton and Schwartz (1991); here rivals do not observe the offer. Private pre-auction messages in Yan (2026) are cheap talk; here the proposed price binds the buyer, and competition ends only if the seller accepts it.

Gains from early trade are already formalized in models with seller-posted prices. Mathews (2004) and Gallien and Gupta (2007) derive buyer equilibria and analyze the seller’s price choice with time-sensitive participants. Budish and Takeyama (2001) and Mathews and Katzman (2006) model insurance gains from buyout options under risk

aversion. In these models, the seller sets the early price and buyers choose whether to exercise the option. Reference-dependent preferences can also make a seller-posted price affect subsequent bids (Shunda, 2009); experimental departures from risk-neutral predictions have been attributed to mistaken beliefs or bounded rationality (Ivanova-Stenzel and Kröger, 2008; Grebe et al., 2016). In bidder-initiated takeover negotiations, Calcagno et al. (2021) combine bilateral bargaining and learning with latent rival entry; rejection continues bargaining rather than triggering an announced auction, and their published baseline has symmetric information and abstracts from time discounting and negotiation costs. Our baseline combines additive private resolution benefits with a buyer-proposed price that signals the buyer’s counterfactual auction bid and screens the seller’s private continuation state; risk aversion is an extension.

Private target information also appears in takeover models, but at different points in the sequence. In Baron (1983), management responds to the terminal bid after public tender-offer competition; in Dimopoulos and Sacchetto (2014), an initial public bid may deter rival entry, and the winner learns the target’s minimum acceptable offer only after the auction. A related screening logic outside auctions appears in Jullien et al. (2017): an uninformed buyer offers an early contract to a seller who privately knows future gains from trade. Rejection affects the buyer’s incentive to invest in bypassing the seller, and early contracting can make that threat credible. Here the alternative is an announced auction with competing buyers. The offer also reveals information about the early buyer, who remains a bidder after rejection, so the seller’s assessment of the offer changes its assessment of the auction. This interaction generates rejected attempts and the competition-versus-resolution welfare tradeoff.

Our environment is related to broader comparisons of auctions and negotiations (Bulow and Klemperer, 1996). In Bulow and Klemperer (2009), sequential buyers can use public pre-emptive bids to influence costly rival entry; our offer is private and late participation is fixed. McAdams and Schwarz (2007) study public buyer offers over repeated rounds when the seller cannot credibly close the sale. Other work on non-commitment, intermediation, and endogenous auction initiation includes McAfee and Vincent (1997), Akbarpour and Li (2020), Skreta (2015), and Gorbenko and Malenko (2024). Lauermann and Wolinsky (2026) study recruitment, entry, and limited commitment in the formation of an informal auction. Their central margin is whether recruitment sustains an auction; ours is whether a privately initiated, binding offer terminates a competitive sale that has already been announced. We hold late participation fixed to isolate buyer selection and the seller’s private continuation value.

Post-auction negotiation can change bidding and extraction (Koska and Stähler, 2021). Appendix I studies how identifying the winner’s earlier offer decision affects terminal counteroffers and the survival of probing.

Housing-market evidence gives concrete interpretations to several model primitives.

Residential transactions can involve extended bilateral bargaining (Merlo and Ortalo-Magné, 2004), making the time saved by early agreement economically relevant. Bidding wars vary with market conditions (Han and Strange, 2014), giving the late-participation component of the seller’s continuation state a housing-market interpretation. Glower et al. (1998) find that sellers with a planned move date, and especially an earlier one, sell more quickly; such circumstances correspond to a higher seller-side cost of waiting in our urgency microfoundation. Finally, auction prices primarily reflect buyer values, while negotiated prices place greater weight on slow-moving seller values (Genesove and Hansen, 2025), supporting the model’s separation between the buyer’s counterfactual bid and the seller’s continuation value.

The maintained environment uses independent private match values. The Norwegian application offers one rationale for treating common quality as public: before marketing, the seller faces strong disclosure duties and is encouraged to obtain a standardized condition report from a building expert. The report and other sales documents give buyers a shared account of technical condition; although voluntary, omitting it increases the seller’s exposure to later claims (Kommunal- og distriktsdepartementet, 2023); see also Finansdepartementet (2021, ch. 13.4). In applying the baseline, we treat inspection as revealing to the early buyer its dwelling-specific match value, while any common quality component is common knowledge. This independent-private-values benchmark lets the offer reveal the buyer’s counterfactual bid while preserving truthful late bidding; private information about defects or resale value would require an interdependent-values model.

3 Model and the Planned Bidding Round

One seller offers an object for sale through an announced bidding round. Before that round, one early buyer can either wait or incur a cost to submit a binding offer. The seller accepts or rejects it. Acceptance transfers the object to the early buyer and ends the sale process. Rejection and waiting both lead to the announced round, in which the early buyer competes with any late buyers who arrive. The baseline has risk-neutral buyers and a risk-neutral seller.

Before making an offer, the early buyer privately knows its value for the object b and its benefit from securing the purchase early. The seller privately knows a continuation state $s \in \{L, H\}$, which describes its prospects from continuing the sale: how many late buyers it expects, the value of retaining the object if it remains unsold, and its own cost of reaching the round. State H makes continuation more attractive to the seller. Each buyer’s value remains private to that buyer; the seller knows the distribution of late participation conditional on s , while the number of late buyers is drawn only if the round is reached.

To evaluate an early offer, the parties must compare it with the bidding round that

follows rejection. For each early value b and seller state s , we first calculate auction payoffs averaged over the remaining uncertainty about late participation and values. These conditional payoffs let us identify the gains available from early agreement. The buyer's decision then averages across seller states, while the seller's decision averages across the buyer values inferred from the offer; Section 4 derives these decisions under private information.

Conditional on s , the number of late buyers N is drawn from a distribution Q_s on the nonnegative integers, with finite mean and $Q_s(N \geq 1) > 0$. All match values are independent draws from the same continuous distribution F on $[\underline{b}, \bar{b}]$, with positive density f . Let $Y_{1N} \geq Y_{2N}$ denote the two highest late values. When fewer than two late buyers arrive, set

$$Y_{1,0} = Y_{2,0} = \underline{b}, \quad Y_{2,1} = \underline{b}.$$

If the round ends without a sale, the seller retains the object. Its no-sale continuation value v_s is the expected payoff from its best subsequent use or disposal of the object, expressed in the same monetary units as buyer values, where

$$\underline{b} < v_L \leq v_H < \bar{b}.$$

For a dwelling, this value can represent continued occupation, rental income, or resale at a later date. For another asset, it can reflect continued production or an alternative use. It can also include the value of retaining the option to choose among future uses as circumstances change. The best continuation accounts for feasible combinations of use and eventual sale, net of costs not already included in the timing cost below. The buyer knows (Q_L, Q_H, v_L, v_H) but not the realized state.

For the terminal auction, the maintained aligned specification assumes that Q_H nontrivially first-order stochastically dominates Q_L and that $v_H > v_L$. Once seller timing costs are introduced below, state H is also weakly less urgent. Section 4.2 develops this composite and the alternatives in which only one component varies. If the round begins, the realized participant set is observed and the early buyer remains a bidder; N counts only the other participants. Its value b may still be too low to induce a sale. Participation and the opportunity to approach first are exogenous.

The terminal benchmark is an English-auction outcome implemented as a second-price mechanism that treats v_s as a bid entered on the seller's behalf and not observed by buyers. The object is allocated to the highest-value buyer if its value reaches v_s , at the larger of v_s and the second-highest buyer value; otherwise the seller retains it and receives v_s . These rules and v_s are fixed before bidding begins: after observing the bids, the seller can neither revise the no-sale threshold nor reject the prescribed outcome and counteroffer.

Lemma 3.1 (Truthful bidding with a private no-sale threshold). *If v_s is observed by the*

seller but not by buyers, truthful bidding is weakly dominant for every buyer.

Proof. Fix v_s and all rival bids. Let h be the highest rival bid, setting $h = \underline{b}$ if no rival exists, and set $t = \max\{v_s, h\}$. The buyer's own bid does not affect t . If $b > t$, winning yields $b - t > 0$, whereas losing yields zero, and the truthful bid b wins. If $b < t$, winning yields $b - t < 0$, whereas losing yields zero, and the truthful bid loses. If $b = t$, the buyer is indifferent. Thus bidding b is optimal for every realization of v_s and the rival bids, and hence is weakly dominant although the buyer does not observe v_s . $\square$

Define

$$V_{2N}(b) = \max\{Y_{2N}, \min\{b, Y_{1N}\}\}.$$

The seller's realized gross terminal payoff—the cash price after a sale and the continuation value after no sale—is

$$P_s(b) = \max\{v_s, V_{2N}(b)\} = \max\{v_s, Y_{2N}, \min\{b, Y_{1N}\}\}. \quad (3.1)$$

Write $\mathbb{E}_s$ for expectation over $N \sim Q_s$ and the late values. For a value threshold t , the probability that both of two late buyers have values below t is $F(t)^2$. With n late buyers it is $F(t)^n$; when their number is uncertain, the probability is $\mathbb{E}_{Q_s}[F(t)^N]$. To express such probabilities compactly, define

$$\phi_s(x) = \mathbb{E}_{Q_s}[x^N] = \sum_{n=0}^{\infty} Q_s(n)x^n, \quad x \in [0, 1],$$

the probability-generating function of N in state s . Its argument x is an individual late buyer's probability of satisfying the event of interest, and $\phi_s(x)$ is the probability that all late buyers satisfy it, averaged over their uncertain number. Here $x^0 = 1$. In particular, $\phi_s(F(v_s)) = \Pr_s(Y_{1N} \leq v_s)$ is the probability that no late buyer values the object above the seller's no-sale continuation. Conditional on (b, s) , the no-sale probability is

$$\Pr(\text{no sale} \mid b, s) = \mathbf{1}\{b < v_s\} \phi_s(F(v_s)). \quad (3.2)$$

If no late buyer arrives, the early buyer is the only bidder and the round ends without a sale exactly when $b < v_s$. Before the early value is observed, the ex ante no-sale probability is $F(v_s)\phi_s(F(v_s)) = \mathbb{E}_{Q_s}[F(v_s)^{N+1}]$. More late participants reduce no-sale risk, whereas a higher v_s raises it. We therefore impose no cross-state ordering on no-sale probabilities.

The early buyer wins only if its value reaches both the seller's threshold and the highest late value. It then receives its value less the price; otherwise its auction surplus is zero. Conditional on (b, s) , the early buyer's expected auction surplus is

$$u_s(b) = \mathbb{E}_s[\max\{b - \max\{v_s, Y_{1N}\}, 0\}]. \quad (3.3)$$

Subtracting this auction surplus from the buyer's value gives its conditional willingness to pay for early ownership, before the additional benefit of early resolution:

$$\begin{aligned} w_s(b) &= b - u_s(b) = \mathbb{E}_s[\min\{b, \max\{v_s, Y_{1N}\}\}] \\ &= \begin{cases} b, & b \leq v_s, \\ v_s + \int_{v_s}^b \{1 - \phi_s(F(t))\} dt, & b > v_s. \end{cases} \end{aligned} \quad (3.4)$$

Conditional on buyer value b and seller state s , the seller's expected gross terminal payoff, averaging over late participation and values, is

$$R_s(b) = \mathbb{E}_s[P_s(b)]. \quad (3.5)$$

Before accounting for direct gains from early resolution, the seller and early buyer jointly obtain $R_s(b) + u_s(b)$ from the bidding round, compared with b from an early sale: the early price is a transfer between them. Since $w_s(b) = b - u_s(b)$, the difference $R_s(b) - w_s(b)$ measures the extra expected payoff the bidding round gives these two parties. We call this difference the *competition wedge*.

Lemma 3.2 (Competition wedge). *For every b and state s ,*

$$R_s(b) - w_s(b) = \mathbb{E}_s[\max\{\max\{v_s, Y_{2N}\} - b, 0\}] \geq 0. \quad (3.6)$$

The inequality is strict exactly when $\Pr_s(\max\{v_s, Y_{2N}\} > b) > 0$.

Proof. Let $Z = \max\{v_s, Y_{2N}\}$ and $M = \min\{b, \max\{v_s, Y_{1N}\}\}$. Then $w_s(b) = \mathbb{E}_s[M]$ and $P_s(b) = \max\{Z, M\}$. If $Z \leq b$, the order $Y_{2N} \leq Y_{1N}$ implies $Z \leq M$. If $Z > b$, then $M = b$ and $P_s(b) = Z$. Thus $P_s(b) - M = \max\{Z - b, 0\}$ pointwise. $\square$

Write the competition wedge as

$$\Gamma_s(b) = \mathbb{E}_s[\max\{\max\{v_s, Y_{2N}\} - b, 0\}]. \quad (3.7)$$

To see why the second-highest late value matters, consider an early buyer with value 100 and a seller's no-sale continuation value below 100. If one late buyer arrives with value 140, it wins and pays 100: the seller and early buyer jointly receive 100. If two late buyers arrive with values 140 and 120, the winner pays 120. In this realization, the seller alone receives 20 more than the entire value that the seller and early buyer can divide through an early sale. The competition wedge averages such gains from preserving the bidding round over possible realizations.

For $b \geq v_s$, the wedge reduces to $\mathbb{E}_s[\max\{Y_{2N} - b, 0\}]$: late buyers must compete with each other to push the price above the early buyer's value. For $b < v_s$, it also

includes the gap between that value and the seller's no-sale continuation. In either case it is nonnegative and weakly decreasing in b . Thus the bidding round and the option of retaining the object create a nonnegative opportunity cost of early agreement for the two contracting parties; absent timing or insurance gains, they do not create bilateral gains. Conditional on zero or one late buyer, the wedge is only $\max\{v_s - b, 0\}$.

State H combines a stochastically larger number of late buyers with a higher no-sale continuation value. These changes produce three orderings:

$$R_H(b) > R_L(b), \quad w_H(b) \geq w_L(b), \quad \Gamma_H(b) \geq \Gamma_L(b).$$

The first says that an H -seller is harder to buy out. The second says that the buyer is also willing to pay weakly more to avoid the more competitive terminal round. The third says that the seller's additional continuation value is at least as large as this increase in buyer willingness, so the bilateral gap is no smaller in H . Formally, these inequalities follow by representing the participation draws so that $N_H \geq N_L$ almost surely; $v_H > v_L$ makes the seller-payoff ordering strict.

3.1 No Sale and Terminal Commitment

The no-sale continuation value v_s is the seller's privately known, offer-stage conditional expected payoff from its best alternative if the planned round produces no sale. It plays the role of a seller reservation value in the terminal allocation rule, but it is neither the advertised asking price nor a public binding reserve. The benchmark nevertheless assumes that the terminal mechanism applies v_s as a fixed threshold once bidding begins.

We maintain terminal commitment to isolate pre-emption-stage screening; Appendix I examines one final counteroffer.

3.2 Payoffs and Bilateral Gains from Early Agreement

Let $d_B \geq 0$ be the buyer's intrinsic benefit from securing an early and certain purchase, measured in monetary units. Before the offer cost, an early agreement at price p gives $b - p + d_B$, while the bidding round gives expected surplus $u_s(b)$. Only a completed early purchase delivers this benefit; withdrawing without buying does not. In the risk-neutral baseline, subtracting d_B from every payoff of a given buyer type leaves all choices unchanged. We use this normalization, writing early payoff as $b - p$ and terminal payoff as $u_s(b) - d_B$. A buyer with $d_B = 0$ has no direct resolution benefit. Whereas [Mathews \(2004\)](#) and [Gallien and Gupta \(2007\)](#) discount transaction surplus multiplicatively, our additive benefit does not scale with match value or surplus: securing an acquisition date can be valuable separately from the match with the object.

Let $d_{S,s} \geq 0$ be the seller's state-dependent cost of reaching the round, with $d_{S,H} \leq d_{S,L}$.

Holding (b, d_B, s) fixed and omitting the offer cost introduced below, the seller obtains $R_s(b) - d_{S,s}$ from continuation and p from early sale. The transfer cancels from the parties' combined gain:

$$\underbrace{b - p - [u_s(b) - d_B]}_{\text{buyer gain}} + \underbrace{p - [R_s(b) - d_{S,s}]}_{\text{seller gain}} = d_B + d_{S,s} - \Gamma_s(b). \quad (3.8)$$

The maintained condition $R_s(b) - d_{S,s} > 0$ keeps every relevant early price nonnegative.

Proposition 3.3 (Gains from early resolution). *An early price makes both parties weakly better off than the terminal bidding round if and only if*

$$d_B + d_{S,s} \geq \Gamma_s(b) = \mathbb{E}_s[\max\{\max\{v_s, Y_{2N}\} - b, 0\}]. \quad (3.9)$$

Strict gains exist when the inequality is strict.

Proof. The buyer weakly prefers early sale when $p \leq w_s(b) + d_B$, and the seller does so when $p \geq R_s(b) - d_{S,s}$. Such a price exists exactly when $w_s(b) + d_B \geq R_s(b) - d_{S,s}$. Lemma 3.2 reduces this inequality to (3.9). $\square$

Corollary 3.4 (State-contingent bilateral gains from early resolution). *Fix a buyer type (b, d_B) and abstract from the offer cost introduced below. Define the state-specific early-agreement threshold*

$$\tau_s(b) := \Gamma_s(b) - d_{S,s}. \quad (3.10)$$

Under the aligned state ordering, $\tau_L(b) \leq \tau_H(b)$. Conditional on seller state s , some price makes both parties weakly prefer early agreement to the terminal bidding round if and only if $d_B \geq \tau_s(b)$. Hence:

- (i) *if $d_B < \tau_L(b)$, no such price exists in either seller state;*
- (ii) *if $\tau_L(b) \leq d_B < \tau_H(b)$, such a price exists conditional on state L , but not conditional on state H ;*
- (iii) *if $d_B \geq \tau_H(b)$, such a price exists conditional on either seller state.*

In state s , a price that makes both parties strictly better off exists if and only if $d_B > \tau_s(b)$.

Proof. The ordering follows from

$$\tau_H(b) - \tau_L(b) = \Gamma_H(b) - \Gamma_L(b) + d_{S,L} - d_{S,H} \geq 0.$$

Applying Proposition 3.3 in each state then gives the three cases. $\square$

The middle region, $\tau_L(b) \leq d_B < \tau_H(b)$, is the key screening margin. For the same buyer type, early agreement can create nonnegative bilateral value in state L , while state H rationally prefers continuation at every price acceptable to the buyer.

4 Private Information and the One-Offer Game

The conditional payoff comparisons in Section 3 identify when early agreement can create bilateral value for a given buyer type and seller state. Whether an offer realizes those gains depends on decisions made with private information: the buyer knows its value and resolution benefit but not the seller state, while the seller knows its state and infers the buyer's value from the offer. The same price therefore determines the terms of early trade and conveys information about the buyer who would return to the bidding round after rejection.

4.1 Buyer Types and Seller Information

Before acting, the early buyer privately observes its match value,

$$b \sim F \quad \text{on } [b, \bar{b}], \quad (4.1)$$

and its benefit from early resolution

$$d_B \sim G \quad \text{on } [0, \bar{d}]. \quad (4.2)$$

Both distributions have continuously differentiable positive densities, and b and d_B are independent. We use *urgency* as shorthand for the buyer's benefit from early resolution. Securing an acquisition date or coordinating a linked transaction can be valuable separately from the match with the particular object. The upper bound $\bar{d}$ limits how much that benefit can compensate for a poor match. The value support extends below both no-sale continuation values, so the model allows no-sale risk and private agreement with buyers who could not clear the terminal threshold.

The seller privately observes

$$s \in \{L, H\}, \quad \Pr(s = H) = \pi \in (0, 1). \quad (4.3)$$

The state is independent of (b, d_B) . It determines the seller's net value of proceeding to the round through the distribution Q_s of late participation, the no-sale continuation value v_s , and the timing cost $d_{S,s}$. Across states, Q_H weakly first-order stochastically dominates Q_L , $v_H \geq v_L$, and $d_{S,H} \leq d_{S,L}$. Every buyer's match value is drawn from the same distribution F . State H is the high-net-continuation state. The early buyer knows π and both state-contingent terminal environments, but not s .

All primitives are common knowledge. The seller privately observes s , the realized late count is drawn subsequently from Q_s , and each match value remains private to its buyer.

Write

$$C_s(b) = R_s(b) - d_{S,s}, \quad \bar{w}(b) = (1 - \pi)w_L(b) + \pi w_H(b), \quad (4.4)$$

where $C_s(b)$ is the seller's net continuation value. The seller's acceptance decision depends on its private information only through this scalar. Since $d_{S,H} \leq d_{S,L}$, the ordering $R_H(b) \geq R_L(b)$ implies $C_H(b) > C_L(b)$ whenever either inequality is strict. At the offer stage, the prior-weighted term $\bar{w}(b)$ is the early buyer's expected willingness to pay across the two continuation states.

Assumption 1 (Ordered continuation values).

- (i) *Regularity.* The lowest seller continuation value is positive, $C_L(b) > 0$. Each w_s is continuous and strictly increasing; each C_s is continuous, nondecreasing, and strictly increasing on a top interval.
- (ii) *State ordering.* For every b ,

$$w_L(b) \leq w_H(b), \quad C_L(b) < C_H(b). \quad (4.5)$$

- (iii) *Selection shapes.* The willingness gap $w_H(b) - w_L(b)$ is weakly increasing, while the low-state wedge $\Gamma_L(b) = C_L(b) + d_{S,L} - w_L(b)$ is weakly decreasing.

Regularity supports the cutoff and fixed-point arguments below. State ordering makes seller H harder to buy out even though the buyer may also be more willing to avoid its terminal environment. The selection shapes order which buyer values are most attracted to a deviation.

4.2 Microfoundations of the Continuation State

Assumption 1 is a reduced form: the seller's private information enters the one-offer game only through the ordered continuation pair (C_L, C_H) and the willingness pair (w_L, w_H) . Three one-dimensional information structures generate it within the same terminal auction. The equilibrium analysis applies to all three; only the primitive moved by the state changes.

Late participation. Let $v_L = v_H = v_0$ and $d_{S,L} = d_{S,H}$, while Q_H first-order stochastically dominates Q_L , with $\Pr_H(N \geq k) > \Pr_L(N \geq k)$ for at least one integer $k \geq 2$. This tail condition makes the seller's continuation strictly higher even when the early buyer's value is below the fallback: a second sufficiently high late value can raise the auction price above v_0 , whereas a shift only from zero to one late buyer cannot. Thus $C_H(b) > C_L(b)$ for every b , as shown in Appendix B. For $b > v_0$,

$$w_H(b) - w_L(b) = \int_{v_0}^b [\phi_L(F(t)) - \phi_H(F(t))] dt$$

is strictly increasing. After rejection, the early buyer returns to the round and pays only the strongest competing claim. With few late buyers, it may therefore win below its rejected offer; weak prospective participation makes acceptance more attractive, while strong participation makes rejection more attractive.

Fallback. Let $Q_L = Q_H = Q$ with generating function ϕ and $d_{S,L} = d_{S,H}$, while $v_H > v_L$. Then

$$C_H(b) - C_L(b) = \int_{v_L}^{v_H} \Pr(\max\{Y_{2N}, \min\{b, Y_{1N}\}\} \leq t) dt$$

is positive and weakly decreasing in b , while for $b > v_L$ the willingness gap $w_H(b) - w_L(b) = \int_{v_L}^{\min\{b, v_H\}} \phi(F(t)) dt$ rises to a plateau at v_H and is constant above it: the buyer's stake in the state stops growing once its value clears the higher fallback.

Urgency. Let Q and v be common across states while the timing cost varies, $d_{S,L} > d_{S,H} \geq 0$: state L is the more urgent seller. Then $C_H(b) - C_L(b) = d_{S,L} - d_{S,H}$ is constant and $w_H \equiv w_L$: the seller's timing cost never enters the buyer's auction payoff, so the state is payoff-irrelevant to the buyer conditional on reaching the round.

Late-participation and fallback information affect the announced round itself and therefore matter to the buyer after rejection. Urgency and seller risk aversion instead change the seller's preferences without changing the buyer's auction payoff. They are not dynamically identical: the modeled urgency cost is saved only by avoiding delay, whereas risk aversion remains relevant whenever the seller still faces a payoff lottery.

The maintained *aligned composite* bundles the microfoundations under the market-strength interpretation: state H has stochastically stronger late participation, a weakly higher no-sale value, and a weakly less urgent seller. The composite auction gives

$$w_H(b) - w_L(b) = \begin{cases} 0, & b \leq v_L, \\ \int_{v_L}^b \phi_L(F(t)) dt, & v_L < b \leq v_H, \\ \int_{v_L}^{v_H} \phi_L(F(t)) dt + \int_{v_H}^b [\phi_L(F(t)) - \phi_H(F(t))] dt, & b > v_H, \end{cases} \quad (4.6)$$

so the difference is weakly increasing, and strictly increasing above v_L when the late-participation shift is nontrivial. Away from the kink at v_s ,

$$R'_s(b) = \begin{cases} 0, & b < v_s, \\ [1 - F(b)]\phi'_s(F(b)) > 0, & b > v_s. \end{cases} \quad (4.7)$$

Under $Q_s(N \geq 1) > 0$ and $d_{S,L} < R_L(\underline{b})$, the three one-dimensional structures and the aligned composite satisfy Assumption 1. Equations (3.4), (3.6), (4.6), and (4.7), together with the state-specific formulas above, establish the required regularity and selection shapes; monotone coupling and the primitive orderings establish the state ordering. The

terminal auction also gives $\Gamma_s(\bar{b}) = 0$, or equivalently $C_s(\bar{b}) = w_s(\bar{b}) - d_{s,s}$, pinning down the top-type continuation values used in the offer thresholds below. Thus the equilibrium analysis can be stated once in terms of (C_s, w_s) and then applied to each source of seller information.

The binary state also bounds the number of relevant positive prices. Under ordered seller payoffs and pure responses, the two nonempty acceptance patterns are acceptance by L alone and acceptance by both states; H cannot accept when L rejects. A costly offer rejected by both states is dominated by no offer, and for either nonempty pattern the buyer prefers the lowest price. Lemma 5.1 combines this price bound with a lowest-type participation condition and reduces the equilibrium analysis to four action supports: waiting alone, waiting with either accepted-offer profile, and waiting with both. More seller types could generate more acceptance patterns and therefore more on-path prices; the binary specification isolates the simplest nontrivial screening problem.

The seller's realized state does not affect what an offer reveals about the buyer's value. Because the buyer does not observe s and s is independent of (b, d_B) , the posterior distribution of b conditional on p is the same for $s = L$ and $s = H$. Pointwise ordering therefore implies

$$\mathbb{E}[C_L(b) \mid p] < \mathbb{E}[C_H(b) \mid p].$$

Seller responses are consequently nested—both reject, only seller L accepts, or both accept—even when the ranges of C_L and C_H overlap. This common-posterior ordering is the screening restriction used below.

4.3 Timing, Beliefs, and Acceptance

Figure 1.1 gives a nontechnical overview. Formally, the timing is as follows.

1. Nature draws b , d_B , and s .
2. The buyer observes (b, d_B) ; the seller observes s .
3. The buyer makes no early offer, denoted $\emptyset$, or submits a binding pre-emption offer at price $p \geq 0$. Any submitted offer costs $\kappa \geq 0$, whether accepted or rejected.
4. The seller observes p , updates about b , and accepts or rejects. In the baseline, rejection activates the terminal bidding process.

Late buyers do not observe the early exchange, and rejection leaves their participation and value distributions unchanged. Thus waiting and rejection activate the same terminal environment.

The one-offer protocol captures a short early window. Continued bargaining, studied in Appendix H, gives rejection an additional negotiation value.

The offer cost κ is the buyer's additional time and effort in preparing and submitting an early offer, valued in monetary units. It can include contacting the seller, reviewing information about the object, checking financing arrangements, and preparing a binding offer before the announced sale. This effort is incurred whether the offer is accepted or rejected. Preparation that would also be required for the bidding round is common to both routes and is normalized away. The distinction matters in comparison with indicative-bidding models, where entry costs buy information needed to participate in the auction (Quint and Hendricks, 2018). Thus κ governs the extensive margin of attempting pre-emption, whereas d_B measures the benefit of completing the purchase before the bidding round.

An offer changes the seller's assessment of the auction as well as the payment available from accepting. If it leads the seller to infer a higher buyer value, the same buyer's return to the auction can make rejection more attractive. Higher inferred b weakly raises the seller's expected terminal payoff, strictly once inferred values reach the relevant upper region.

For each observed price p , let $\mu(\cdot | p)$ be the seller's posterior on the buyer's type space $[b, \bar{b}] \times [0, \bar{d}]$, and let $\mu_B(\cdot | p)$ be its marginal distribution of b . Because s is independent of buyer type and the buyer does not observe s , the seller's realized state does not affect this posterior. Define seller type s 's posterior continuation value by

$$\mathcal{C}_s(p) = \int_{\underline{b}}^{\bar{b}} C_s(b) d\mu_B(b | p). \quad (4.8)$$

The integral is taken with respect to the posterior distribution rather than a posterior density because an off-path belief may assign positive probability to one specific value of b . The offer can reveal both value and urgency, but only the marginal posterior over b enters this continuation value: a rejected early buyer enters the bidding round and bids b , whereas the benefit d_B of an early purchase is no longer available and does not affect bidding. The seller accepts when

$$p \geq \mathcal{C}_s(p), \quad (4.9)$$

with acceptance at equality. We restrict attention to pure seller responses and break indifference in favor of immediate agreement.

Conditional on seller state s , a buyer of type (b, d_B) obtains $u_s(b) - d_B$ by waiting. If it instead submits price p , acceptance yields $b - p - \kappa$, while rejection yields $u_s(b) - d_B - \kappa$. Hence, relative to waiting, acceptance yields $w_s(b) + d_B - p - \kappa$ and rejection yields $-\kappa$.

4.4 Two Benchmarks

Before constructing equilibria, two benchmarks pin down the sources of attempts and rejections. Summing buyer and seller equilibrium gains eliminates the price transfer and

yields a necessary condition for every PBE of the one-offer game.

Proposition 4.1 (Aggregate financing of early offers). *Suppose agents are risk neutral and share a common prior, $\kappa \geq 0$, and rejection leads to the same terminal auction as waiting. Let I be the event that the buyer submits an offer and $A_c \subseteq I$ the event that an offer is accepted. In every PBE, allowing arbitrary price menus and mixed strategies,*

$$\mathbb{E}[\{d_B + d_{S,s} - \Gamma_s(b)\} \mathbf{1}_{A_c}] \geq \kappa \Pr(I). \quad (4.10)$$

Inequality (4.10) says that the timing gains realized by accepted early trades, net of the competition they eliminate, must cover the attempt cost incurred by every buyer who tries. Rejected attempts contribute cost but no timing gain, so they raise the burden on accepted trades.

Proof. The proof uses three implications of PBE. The buyer's equilibrium action must be optimal relative to waiting; the seller's response must be sequentially optimal after every on-path price; and Bayes' rule must make the seller's on-path posterior the conditional distribution generated by the equilibrium strategies. Off-path beliefs do not enter the aggregate calculation.

Normalize both parties' payoffs relative to waiting for the bidding round. Let g_B be the buyer's incremental payoff under its equilibrium action, including the offer cost, and set $g_B = 0$ when the buyer waits. At the buyer's information set, deviating to waiting yields zero. Sequential rationality of the buyer's equilibrium strategy therefore implies

$$\mathbb{E}[g_B \mid \text{buyer type}] \geq 0$$

almost surely, where the conditional expectation averages over the unobserved seller state and any equilibrium mixing. Hence $\mathbb{E}[g_B] \geq 0$.

Now consider a price p that occurs on path. Given seller state s , Bayes' rule identifies the conditional distribution of b used in $\mathcal{C}_s(p)$. Sequential rationality implies that the seller assigns positive probability to acceptance only if

$$\mathbb{E}[p - C_s(b) \mid s, p] \geq 0.$$

Let g_S be the seller's incremental payoff, with $g_S = 0$ after rejection. If $a_s(p)$ is its acceptance probability, then

$$\mathbb{E}[g_S \mid s, p] = a_s(p) \mathbb{E}[p - C_s(b) \mid s, p] \geq 0,$$

where the expression is zero when the seller rejects for sure. Averaging over the equilibrium distribution of on-path (s, p) gives $\mathbb{E}[g_S] \geq 0$. This argument conditions separately on the

seller's private state and therefore does not require independence between the buyer's type and s .

After rejection, the pointwise joint incremental payoff is $g_B + g_S = -\kappa$. After acceptance in state s , it is

$$\begin{aligned} g_B + g_S &= w_s(b) + d_B - p - \kappa + p - R_s(b) + d_{S,s} \\ &= d_B + d_{S,s} - \Gamma_s(b) - \kappa, \end{aligned}$$

where Lemma 3.2 gives the equality. Combining acceptance, rejection, and no offer gives the pointwise identity

$$g_B + g_S = \{d_B + d_{S,s} - \Gamma_s(b)\} \mathbf{1}_{A_c} - \kappa \mathbf{1}_I.$$

The first term is present only after acceptance, while the offer cost is incurred after every attempt. Taking expectations gives

$$\mathbb{E}[g_B + g_S] = \mathbb{E}[\{d_B + d_{S,s} - \Gamma_s(b)\} \mathbf{1}_{A_c}] - \kappa \Pr(I).$$

The preceding buyer and seller PBE conditions imply $\mathbb{E}[g_B + g_S] \geq 0$. Combining this inequality with the accounting identity proves (4.10). $\square$

If $\Pr(A_c) > 0$, Proposition 4.1 implies

$$\mathbb{E}[d_B + d_{S,s} - \Gamma_s(b) \mid A_c] \geq \kappa \frac{\Pr(I)}{\Pr(A_c)}.$$

In particular, if $\kappa > 0$, $d_B = 0$ almost surely, and $d_{S,L} = d_{S,H} = 0$, every PBE has $\Pr(I) = 0$. Pooling may give a patient buyer an information rent, but only when gains elsewhere in the accepted pool finance that rent, the competition wedge, and the costs of unsuccessful attempts.

The second benchmark isolates the role of seller private information.

Lemma 4.2 (Rejections require seller private information). *Suppose the continuation state is public: the buyer observes s before choosing its action. If $\kappa > 0$, then in every pure-strategy PBE of the one-offer game, every offer submitted with positive probability is accepted.*

Proof. Fix a pure-strategy PBE and a state s . Suppose a price p is submitted with positive probability and the seller's strategy rejects it. Every submitting buyer type pays κ and then faces the same bidding round as after waiting, for an incremental payoff of $-\kappa < 0$, whereas waiting yields zero. Sequential rationality rules out submitting p , a contradiction. $\square$

Under pure seller responses, seller private information is therefore necessary for a costly offer to be rejected on path. We next characterize equilibrium offer menus and the seller's acceptance decisions.

5 Equilibrium

5.1 Buyer Choices and Price Selection

Let W denote waiting. With pure seller responses, an on-path offer is rejected in both states, accepted only in L , or accepted in both. We call the latter two categories *probing* and *knockout*. A probe produces an early sale in L and a rejected attempt in H . Initially each submitted price is a distinct action; the lemma below restricts each accepted-offer category to at most one price.

Waiting and one probing price already generate no early offer, rejection, and successful pre-emption. Knockout adds the option of securing early agreement in either state. We first construct probing equilibria and then the full menu; Section 6.1 examines the rejection-free waiting–knockout alternative.

Write f and g for the densities of F and G . Since buyer urgency lies in $[0, \bar{d}]$, we use the usual CDF convention $G(x) = 0$ for $x \leq 0$ and $G(x) = 1$ for $x \geq \bar{d}$. Thus formulas involving G continue to apply when a cutoff lies below or above the feasible urgency range.

We maintain the following lowest-type participation condition:

$$\chi_L := C_L(\underline{b}) - w_L(\underline{b}) + \frac{\kappa}{1 - \pi} > 0, \quad \chi_H := C_H(\underline{b}) - \bar{w}(\underline{b}) + \kappa > 0. \quad (5.1)$$

It says that, even at the lowest seller continuation value compatible with either accepted offer, the least-valued fully patient buyer does not recover the offer cost. In the terminal-auction microfoundations, $w_L(\underline{b}) = w_H(\underline{b}) = \underline{b}$, so the two inequalities read $\Gamma_L(\underline{b}) - d_{S,L} + \kappa/(1 - \pi) > 0$ and $\Gamma_H(\underline{b}) - d_{S,H} + \kappa > 0$. They exclude cases in which extreme seller timing gains make early agreement attractive even to the least-valued buyer at $d = 0$. All calibrations in Sections 5–7 satisfy (5.1). In the aligned example, $\underline{b} = 0$ and $w_L(0) = \bar{w}(0) = 0$, so $\chi_L = C_L(0) + \kappa/(1 - \pi) > 0$ and $\chi_H = C_H(0) + \kappa > 0$. The condition gives positive probability to choosing W , and hence to reaching the announced round without a prior offer. Without it, supports with no waiting action are possible, although a rejected probe can still lead to the round.

Lemma 5.1 (On-path prices and action support). *Suppose $\kappa > 0$. Restrict attention to pure-strategy PBE, with seller indifference resolved in favor of acceptance. All on-path probing prices then coincide almost surely, and all on-path knockout prices coincide almost surely. No offer rejected in both states is submitted with positive probability. Under (5.1), every such PBE has positive waiting mass. Thus, denoting the unique probing and knockout*

price actions by P and K when present, the action support is $\{W\}$, $\{W, P\}$, $\{W, K\}$, or $\{W, P, K\}$.

The proof is in Appendix D.1.

A probing price may leave seller L strictly better off selling early than proceeding to auction. Such a price can be sustained if a slightly lower offer leads the seller to infer a much higher buyer match value, making the auction attractive enough to reject the lower offer. Locally passive beliefs rule out this jump in the seller's inference.

Assumption 2 (Locally passive beliefs). For price selection, let $q > 0$ be an active offer and let $\mu_B(\cdot \mid q)$ be the seller's on-path posterior over the buyer's match value. After unused prices p approaching q from below, the posterior is locally passive:

$$\mu_B(\cdot \mid p) \Rightarrow \mu_B(\cdot \mid q) \quad \text{as } p \uparrow q,$$

where $\Rightarrow$ denotes weak convergence.

Assumption 2 implies that every active price leaves the seller state at the acceptance margin indifferent. To see this, suppose that state instead earned strictly positive expected rent at an active price q . For all sufficiently close undercuts $p < q$, local passivity would keep that state's expected continuation payoff below p , so it would still accept. A probing undercut would retain acceptance by seller L and rejection by seller H ; a knockout undercut would retain acceptance by both states. Any buyer using q could then induce the same seller responses at the lower price p , a profitable deviation. Hence the probing price satisfies $q_L = \mathcal{C}_L(q_L)$, while the knockout price satisfies $q_H = \mathcal{C}_H(q_H)$.

Write q_L and q_H for the probing and knockout prices when present, and P and K for submitting them. Without Assumption 2, fixed primitives may support different prices, buyer pools, posteriors, and seller rents.

We construct the ordered case $0 < q_L < q_H$. Both prices compensate the accepting seller for giving up the bidding round; q_H also secures acceptance in state H .

Probing lets the buyer obtain early agreement when the seller's continuation is weak and proceed to the auction when it is strong. Paying q_H instead secures acceptance in the strong state, but also raises the payment in the weak state, where q_L would have sufficed. The buyer therefore weighs the additional chance of early agreement against the higher price.

Write d for a realization of d_B , and denote making no early offer by action 0. For a buyer with match value b , the expected incremental payoffs relative to waiting are

$$V_0(b, d) = 0, \tag{5.2}$$

$$V_L(b, d) = (1 - \pi) [w_L(b) + d - q_L] - \kappa, \tag{5.3}$$

$$V_H(b, d) = \bar{w}(b) + d - q_H - \kappa. \tag{5.4}$$

If probing is accepted, the buyer realizes the early-resolution benefit and obtains the object at q_L . If it is rejected, the buyer proceeds to the same auction as after waiting, with the offer cost κ as its only additional loss relative to waiting in that state. The offer cost is incurred in either event, so the gains when accepted must cover the cost of making the attempt. A knockout offer delivers the benefit in both continuation states.

For a given match value, a larger benefit from early resolution makes successful preemption more valuable. Probing delivers that benefit only in state L , so its expected payoff rises with d at slope $1 - \pi$; knockout delivers it in both states, giving slope one. Thus higher urgency makes certain early agreement relatively more attractive. The subscripts L and H on the payoff and cutoff notation record which seller responses the offer induces: a probe is accepted only in L , whereas a knockout is accepted in both states. The urgency level at which probing first matches waiting, solving $V_L(b, d) = V_0(b, d)$, is

$$D_L(b; q_L) = q_L - w_L(b) + \frac{\kappa}{1 - \pi}. \quad (5.5)$$

Probing and knockout are equally attractive where $V_L(b, d) = V_H(b, d)$. Since $\bar{w}(b) = (1 - \pi)w_L(b) + \pi w_H(b)$, their indifference cutoff is

$$D_H(b; q_L, q_H) = \frac{q_H - (1 - \pi)q_L}{\pi} - w_H(b). \quad (5.6)$$

When only waiting and knockout are available, their indifference cutoff and the knockout probability conditional on b are

$$\widetilde{D}_H(b; q) = q - \bar{w}(b) + \kappa, \quad m_K(b; q) = 1 - G(\widetilde{D}_H(b; q)). \quad (5.7)$$

The same indifference cutoff will locate the waiting–knockout crossing when probing is also available.

5.2 Waiting Only and Waiting–Probing

The corner type $(\bar{b}, \bar{d})$ determines the transition from waiting only to waiting–probing: once it finds probing profitable, waiting only cannot be sustained; while it still prefers probing to knockout, the higher offer remains inactive.

Call an equilibrium *no-early-offer* when every buyer waits and the planned bidding round therefore occurs with probability one. A buyer prescribed to wait may deviate by submitting either a probing or a knockout offer. Either deviation is least attractive when sellers interpret the off-path offer as coming from $b = \bar{b}$, because this belief maximizes their continuation payoff and hence the price required for acceptance. Under this belief, the urgency levels at which a buyer with value $\bar{b}$ breaks even between waiting and deviating

to probing or knockout, respectively, are

$$\bar{D}_L = C_L(\bar{b}) - w_L(\bar{b}) + \frac{\kappa}{1 - \pi}, \quad (5.8)$$

$$\bar{D}_H = C_H(\bar{b}) - \bar{w}(\bar{b}) + \kappa, \quad (5.9)$$

and let

$$\Delta_D = \bar{D}_H - \bar{D}_L = C_H(\bar{b}) - C_L(\bar{b}) - \pi[w_H(\bar{b}) - w_L(\bar{b})] - \frac{\pi}{1 - \pi}\kappa. \quad (5.10)$$

Under this belief, the minimum acceptable prices are $C_L(\bar{b})$ and $C_H(\bar{b})$. Willingness to attempt is greatest at the top value and urgency, so deterring $(\bar{b}, \bar{d})$ deters every buyer. Moreover,

$$\bar{D}_L + \frac{\Delta_D}{\pi} = \frac{C_H(\bar{b}) - (1 - \pi)C_L(\bar{b})}{\pi} - w_H(\bar{b}) = D_H(\bar{b}; C_L(\bar{b}), C_H(\bar{b})), \quad (5.11)$$

so the upper endpoint used below is the highest-value buyer's probing-knockout cutoff when the two prices equal the sellers' endpoint continuation values. It is therefore the point at which the most urgent buyer can first prefer knockout to probing. For a candidate probing price q , a buyer with value b probes when $d \geq D_L(b; q)$. The mass of probing buyers at that value is therefore proportional to $[1 - G(D_L(b; q))]f(b)$, and seller L 's expected continuation payoff conditional on observing the probe is

$$T_L(q) = \frac{\int_{\underline{b}}^{\bar{b}} C_L(b)[1 - G(D_L(b; q))]f(b) db}{\int_{\underline{b}}^{\bar{b}} [1 - G(D_L(b; q))]f(b) db}, \quad (5.12)$$

whenever the denominator is positive. A fixed point $q = T_L(q)$ makes seller L indifferent between accepting q and proceeding to the bidding round. We call this a *boundary price*: the participation constraint of the seller state that accepts the probe binds for the buyer pool selected into probing. Assumption 2 selects such a fixed point whenever probing is active.

Proposition 5.2 (Waiting-only and waiting-probing equilibria). *Under Assumption 1, a no-early-offer PBE exists if and only if*

$$\bar{d} \leq \min\{\bar{D}_L, \bar{D}_H\}. \quad (5.13)$$

Suppose instead that

$$\Delta_D > 0, \quad \bar{D}_L < \bar{d} \leq \bar{D}_L + \frac{\Delta_D}{\pi}, \quad C_L(\underline{b}) - w_L(\underline{b}) + \frac{\kappa}{1 - \pi} > 0. \quad (5.14)$$

Then a no-early-offer PBE does not exist, while T_L has a fixed point $q_L \in (C_L(\underline{b}), C_L(\bar{b}))$. There is a PBE in which the buyer waits when $d < D_L(b; q_L)$ and offers q_L otherwise, seller

L accepts, and seller H rejects. No knockout deviation is profitable. Consequently, no early offer, a rejected attempt, and successful pre-emption all occur with positive probability.

The lower bound rules out waiting only; the upper bound deters knockout even for the highest-value, most urgent buyer. Within this window, buyers wait or probe, and seller L accepts while H rejects. Locally passive beliefs support the boundary price: a sufficiently small undercut retains the probing posterior and is rejected in both states. Appendix D gives the proof.

Criterion D1 restricts inferences after unused prices according to buyer types' incentives to deviate (Cho and Kreps, 1987; Banks and Sobel, 1987). We use it to test whether probing survives when the seller's beliefs must satisfy this incentive-based restriction.

For an unused price, write the seller's possible response as a state-contingent acceptance plan $a = (a_L, a_H) \in [0, 1]^2$. This outcome- and payoff-equivalent representation keeps equilibrium seller responses pure. D1 deletes a buyer type when every posterior-rationalizable plan that makes it weakly willing to deviate makes another type strictly willing to deviate. Because the buyer does not observe s , the same deviation must induce the same posterior over buyer types in both seller states. Appendix D verifies these incentive comparisons for the continuous two-dimensional buyer type.

For a boundary-price waiting–probing equilibrium with $\kappa > 0$, let

$$b_L^0 = \sup\{b : D_L(b; q_L) \geq 0\}.$$

Thus b_L^0 is the highest match value for which a patient buyer does not strictly gain from probing.

Corollary 5.3 (D1 survival). *The on-path prices and strategies of the waiting-only PBE in Proposition 5.2 admit a D1-compatible belief completion in the contingent-plan representation. For $\kappa > 0$, the on-path profile of a boundary-price waiting–probing PBE in that proposition admits such a completion if and only if $q_L \leq C_L(b_L^0)$. The following parameter condition is sufficient:*

$$\bar{D}_L \geq 0. \tag{5.15}$$

This D1-compatible completion need not be locally passive. Appendix D also constructs nonboundary and nearby positive-rent probing equilibria: attributing an undercut to a sufficiently high-value buyer can induce rejection. Local passivity excludes this discontinuous inference and selects zero marginal rent, while allowing multiple boundary-price fixed points.

For scale, take values $F = U[0, 1]$, deterministic late participation $N = n_s$, and the aligned state $(n_L, v_L, d_{S,L}) = (2, 0.38, 0.02)$, $(n_H, v_H, d_{S,H}) = (3, 0.42, 0)$, with $\pi = 1/2$ and $\kappa = 0.02$. The exact waiting-only boundary is $\bar{D}_L = 0.02$. The highest-value buyer's probing–knockout cutoff is $\bar{D}_L + \Delta_D/\pi = 0.092821907$, giving the sufficient $\{W, P\}$

interval $0.02 < \bar{d} \leq 0.092821907$. At $\bar{d} = 0.045$, let urgency have the truncated exponential distribution $G(d) = (1 - e^{-10d})/(1 - e^{-10\bar{d}})$ on $[0, \bar{d}]$. The probing price is $q_L = 0.659155058$ and attempts occur with probability 0.073608826. No early offer, a rejected attempt, and successful pre-emption then have probabilities 0.926391174, 0.036804413, and 0.036804413.

5.3 Waiting, Probing, and Knockout

With both prices active, buyers sort into waiting, probing, and knockout. These selected pools determine the continuation values that each price must cover, so (q_L, q_H) solves a two-dimensional fixed point. Unlike the one-price case, both action cutoffs move with the endogenous price pair.

At a fixed match value b , a positive share of buyers can probe only if the wait–probe cutoff $D_L(b)$ lies below the wait–knockout cutoff $\widetilde{D}_H(b)$. Their distance is

$$Z(b; q_L, q_H) = \widetilde{D}_H(b; q_H) - D_L(b; q_L) = q_H - q_L - \pi[w_H(b) - w_L(b)] - \frac{\pi}{1 - \pi}\kappa. \quad (5.16)$$

and direct substitution gives

$$D_H - \widetilde{D}_H = \frac{1 - \pi}{\pi}Z, \quad D_H - D_L = \frac{1}{\pi}Z, \quad (5.17)$$

where the arguments are suppressed. Thus one condition, $Z(b; q_L, q_H) > 0$, orders all three indifference points:

$$D_L(b) < \widetilde{D}_H(b) < D_H(b).$$

The buyer therefore waits below $D_L(b)$, probes between $D_L(b)$ and $D_H(b)$, and makes a knockout offer above $D_H(b)$. The intermediate cutoff $\widetilde{D}_H(b)$ only marks where knockout first beats waiting; probing remains better until $D_H(b)$. Both action cutoffs decrease with b , so a higher-value buyer reaches probing and knockout at a lower urgency. As the population includes more urgent buyers, the highest-value buyers are therefore the first to add knockout to the action support.

Figure 5.1 illustrates these comparisons for a fixed match value. The thick segments trace the upper envelope of the three payoff lines.

Because $G(x)$ is the probability that urgency is at most x , the action probabilities conditional on match value b are

$$m_W(b; q) = G(D_L(b; q_L)), \quad (5.18)$$

$$m_P(b; q) = G(D_H(b; q)) - G(D_L(b; q_L)), \quad (5.19)$$

$$m_K(b; q) = 1 - G(D_H(b; q)). \quad (5.20)$$

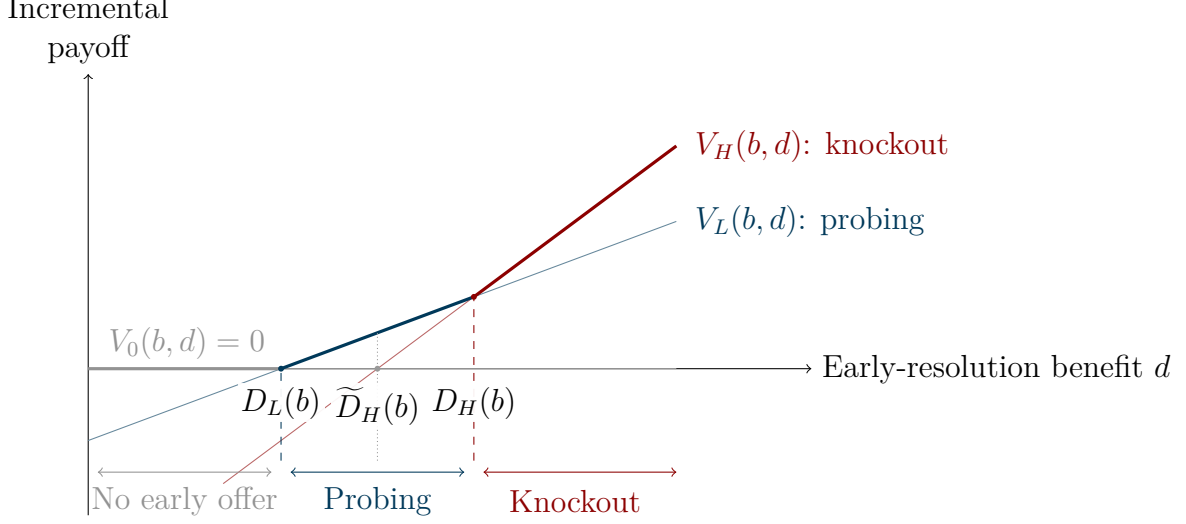


Figure 5.1: Incremental buyer payoffs relative to waiting as urgency d varies, holding b and prices (q_L, q_H) fixed. Thick segments trace the optimal action when $Z(b; q_L, q_H) > 0$. The intermediate crossing $\tilde{D}_H(b)$ is not an action switch: probing still dominates both alternatives there.

Averaging over match values gives the unconditional action probabilities

$$M_a(q) = \int_{\underline{b}}^{\bar{b}} m_a(b; q) f(b) db, \quad a \in \{W, P, K\}. \quad (5.21)$$

Whenever action $a \in \{P, K\}$ has positive probability, Bayes' rule gives the seller's posterior density of the early buyer's match value after observing that offer:

$$h_a(b; q) = \frac{m_a(b; q) f(b)}{M_a(q)}, \quad a \in \{P, K\}. \quad (5.22)$$

Given prices $q = (q_L, q_H)$, the induced probing pool determines seller L 's expected continuation value, while the induced knockout pool determines seller H 's. These two posterior continuation values form the price map $T(q) = (T_P(q), T_K(q))$, whose coordinates are

$$\begin{aligned} T_P(q) &= \int_{\underline{b}}^{\bar{b}} C_L(b) h_P(b; q) db, \\ T_K(q) &= \int_{\underline{b}}^{\bar{b}} C_H(b) h_K(b; q) db. \end{aligned} \quad (5.23)$$

A *boundary-price* pair solves $q = T(q)$: each price equals its marginal seller state's expected continuation value for the selected buyer pool, as required by Assumption 2. Seller L earns a rent at the knockout price because its continuation value is lower. Reducing that price enough to remove the rent would make seller H reject, turning knockout into a probe.

Because $T_P(q)$ and $T_K(q)$ are posterior means of $C_L(b)$ and $C_H(b)$, respectively, any

boundary-price fixed point must lie in the rectangle

$$\mathcal{Q} = [C_L(\underline{b}), C_L(\bar{b})] \times [C_H(\underline{b}), C_H(\bar{b})]. \quad (5.24)$$

To obtain an equilibrium, a candidate price pair must reproduce the buyer pools that justify its acceptance. The proposition uses a price rectangle where all three actions occur at every value and each coordinate of the price map points inward at the corresponding boundaries. Appendix B illustrates this geometry and proves existence.

Proposition 5.4 (Three-action boundary-price equilibrium). *Suppose Assumption 1 holds. Suppose there is a compact price rectangle*

$$\mathcal{Q}_0 = [\ell_L, u_L] \times [\ell_H, u_H] \subseteq \mathcal{Q}, \quad \ell_L < u_L, \quad \ell_H < u_H,$$

such that, for every $q \in \mathcal{Q}_0$ and $b \in [\underline{b}, \bar{b}]$,

$$0 < D_L(b; q_L) < D_H(b; q) < \bar{d} \quad \text{and} \quad Z(b; q) > 0. \quad (5.25)$$

Suppose also that each mapped price lies above its lower candidate boundary and below its upper candidate boundary:

$$\begin{aligned} T_P(\ell_L, q_H) &> \ell_L, & T_P(u_L, q_H) &< u_L & \text{for every } q_H \in [\ell_H, u_H], \\ T_K(q_L, \ell_H) &> \ell_H, & T_K(q_L, u_H) &< u_H & \text{for every } q_L \in [\ell_L, u_L]. \end{aligned} \quad (5.26)$$

Then T has a fixed point $(q_L, q_H) \in \text{int } \mathcal{Q}_0$. At any such fixed point, the following strategies and beliefs form a perfect Bayesian equilibrium:

(i) *a buyer with type (b, d) chooses*

$$p(b, d) = \begin{cases} \emptyset, & d < D_L(b), \\ q_L, & D_L(b) \leq d < D_H(b), \\ q_H, & d \geq D_H(b); \end{cases} \quad (5.27)$$

(ii) *on-path posterior densities are given by (5.22); after sufficiently small undercuts of q_L and q_H , the seller retains the corresponding on-path posterior, and after every remaining unused price assigns $b = \bar{b}$;*

(iii) *seller L accepts both q_L and q_H , seller H rejects q_L and accepts q_H ; at every other price p , seller type s accepts if and only if $p \geq C_s(p)$.*

Boundary buyer types follow the displayed weak inequalities, and sellers accept at equality.

The proof is in Appendix B.

Combining the unconditional action probabilities with the seller state gives the observable outcome probabilities:

$$\Pr(\text{no early offer}) = M_W, \quad (5.28)$$

$$\Pr(\text{rejected attempt}) = \pi M_P, \quad (5.29)$$

$$\Pr(\text{successful pre-emption}) = (1 - \pi)M_P + M_K. \quad (5.30)$$

A probe is rejected only in state H , whereas a knockout offer succeeds in either state. Figure 5.2 displays this decomposition.

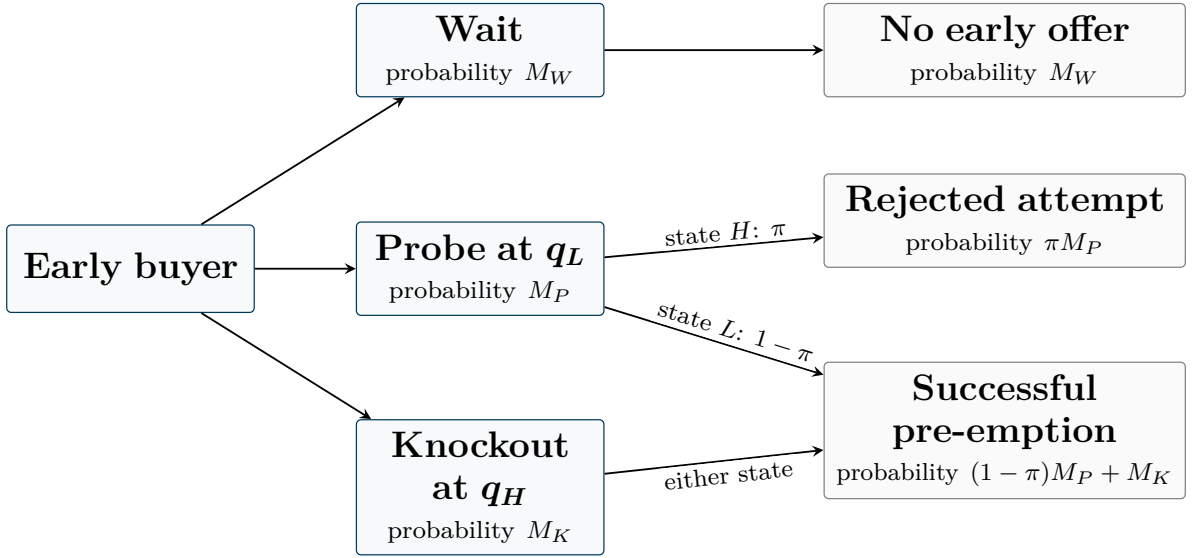


Figure 5.2: Buyer actions and observable outcomes under support $\{W, P, K\}$. The middle boxes report unconditional action probabilities. A probe is accepted in state L and rejected in state H , while a knockout offer is accepted in both states. When no buyer chooses knockout ($M_K = 0$), the diagram reduces to the waiting-probing support $\{W, P\}$.

Condition (5.25) requires waiting, probing, and knockout intervals at every match value, uniformly over the candidate rectangle. Even the lowest-value buyers sometimes choose knockout, and even the highest-value buyers sometimes wait. This is stronger than positive aggregate action masses. Appendix B gives the endpoint inequalities used to check this sufficient condition.

Let $\zeta = F^{-1}$ denote the common match-value quantile and T_ζ the associated price map. Call a quantile admissible if it has the maintained support and satisfies the continuity and positive-density assumptions. Holding the support endpoints, participation distributions, no-sale continuation values, and seller timing costs fixed, the full-menu fixed point in Proposition 5.4 persists under sufficiently small sup-norm perturbations of ζ within this class.

Corollary 5.5 (Local distributional robustness and the surrounding PBE family). *Let q^0 be a fixed point under an admissible quantile ζ^0 at which the pointwise inequalities in*

(5.25) are strict. Define $\Psi(\zeta, q) = T_\zeta(q) - q$, and suppose the price Jacobian $D_q\Psi(\zeta^0, q^0)$ is nonsingular. Then, for every admissible quantile ζ sufficiently close to ζ^0 in the sup norm and every sufficiently small rent vector $r = (r_L, r_H) \in \mathbb{R}_+^2$, there is a unique nearby price pair $q(\zeta, r)$ satisfying

$$q(\zeta, r) - T_\zeta(q(\zeta, r)) = r.$$

It implements a full-menu PBE satisfying (5.25) for every buyer value. In particular, $r = 0$ gives the locally unique nearby boundary-price equilibrium selected by Assumption 2.

Seller L 's marginal probing rent is $r_L = q_L - T_{\zeta,P}(q)$, and seller H 's marginal knockout rent is $r_H = q_H - T_{\zeta,K}(q)$. The selected boundary-price equilibrium sets both to zero. The positive-rent members show that the same action support and seller acceptance pattern persist locally in the larger PBE correspondence, but they are excluded by Assumption 2. Moving the rent vector also moves buyer cutoffs, selected buyer pools, and outcome probabilities, so the surrounding family is not a pure transfer of surplus. Appendix C proves the local result. The aligned example in Section 7 separately supplies an outward-rounded interval certificate for full-menu existence at the displayed primitives.

The same refinement test preserves the two-price equilibrium when all three buyer actions remain active for every match value.

Corollary 5.6 (Full-menu D1 survival). *When $\kappa > 0$, the on-path price and strategy profile of any PBE in Proposition 5.4 that satisfies (5.25) for every buyer value admits a D1-compatible belief completion in the contingent-plan representation.*

Global interiority provides a D1-surviving marginal buyer at every match value, allowing D1 beliefs to preserve the locally passive posterior marginal over b . Both restrictions can therefore hold in the full-menu assessment. The nearby positive-rent PBEs of Corollary 5.5 also admit D1 completions; local passivity selects the zero-marginal-rent member. Appendix D gives the proof.

6 Seller Information and the Role of Rejection

Probing survives D1 under the conditions above. Whether rejection is necessary whenever buyers attempt pre-emption depends on what the seller privately knows. We first examine rejection-free equilibria and then compare the information conveyed by a rejected probe.

6.1 Equilibria without Rejection

The remaining support, $\{W, K\}$, combines waiting with an offer accepted in both seller states. Appendix D.3 gives its boundary-price construction: the knockout price must reproduce seller H 's continuation value in the selected pool, and both distant

and infinitesimal probing deviations must be deterred. These checks are additional to the existence of a price fixed point. The following result excludes this support after D1 refinement under the aligned continuation state, including nonboundary-price constructions.

Proposition 6.1 (No refined waiting–knockout equilibrium under the aligned state). *Suppose early and late match values have a common continuous CDF with positive density on $[\underline{b}, \bar{b}]$. Maintain the aligned composite terminal specification: Q_H nontrivially first-order stochastically dominates Q_L and $v_L < v_H$. Suppose $\pi \in (0, 1)$, $\kappa > 0$, urgency is independent of b and has full interval support $[0, \bar{d}]$, the seller state is independent of buyer type, and Assumption 1 holds. Every pure PBE with action support $\{W, K\}$ fails criterion D1, in its strict-gain form, in the corresponding contingent-plan representation.*

The support $\{W, K\}$ can exist in a PBE before refinement, so D1 drives the exclusion. The same conclusion holds under late-participation variation alone. Under fallback variation, however, a top willingness plateau can support a refined $\{W, K\}$ equilibrium: above v_H , the higher no-sale continuation no longer changes buyer willingness. Appendix E proves the proposition and gives this counterexample. Under urgency variation the willingness gap is flat everywhere, and an exact survival condition determines when $\{W, K\}$ survives. For a knockout price q_H , let $b_K^0 = \min\{\bar{b}, w^{-1}(q_H + \kappa)\}$ denote the highest feasible wait–knockout marginal value.

Proposition 6.2 (D1 survival conditions for waiting–knockout under urgency variation). *Maintain the urgency-only specification with $\kappa > 0$ and $d_{S,L} > d_{S,H} \geq 0$, and fix a pure PBE with support $\{W, K\}$ and price q_H . The equilibrium has a D1-compatible completion in the contingent-plan representation if and only if*

$$d_{S,L} - d_{S,H} \leq C_H(b_K^0) - q_H + \frac{\pi\kappa}{1 - \pi} \quad \text{and} \quad q_H \leq C_H(b_K^0).$$

When $b_K^0 < \bar{b}$, the first condition reads $d_{S,L} \leq \Gamma(b_K^0) + \kappa/(1 - \pi)$, and it implies the second whenever $\Delta_D \geq 0$.

Appendix F proves both directions and reports an interval-certified numerical instance. At the primitives of Example 6.4, the boundary price $q_H = .637358$ satisfies both survival conditions, with margins .003044 and .013044. D1 therefore permits a refined $\{W, K\}$ equilibrium alongside the $\{W, P, K\}$ equilibrium at these urgency primitives. At these primitives, the survival boundary is $d_{S,L}^* = .023044$; on $(.023044, .060642]$, raw $\{W, K\}$ PBEs exist but fail D1. Conditional on positive attempt mass, refinement therefore makes rejections necessary under aligned or late-participation information, whereas fallback and urgency information can preserve attempts without rejection.

6.2 Selection and the Information in Rejection

In supports containing probing, the first attempt cutoff is strictly decreasing in b . At differentiability points,

$$\frac{\partial D_L(b; q_L)}{\partial b} = -w'_L(b) < 0. \quad (6.1)$$

Consequently, in $\{W, P\}$ and $\{W, P, K\}$, attempt probability is weakly increasing in b and strictly increasing wherever the cutoff is interior; at differentiability points with an interior cutoff,

$$\frac{\partial \Pr(p > 0 \mid b)}{\partial b} = g(D_L(b))w'_L(b) > 0. \quad (6.2)$$

The same monotonicity holds in $\{W, K\}$ because $\partial \widetilde{D}_H(b; q_H)/\partial b = -\bar{w}'(b) < 0$. Under independence of match values and urgency, observing an attempt therefore shifts the seller's posterior toward higher b : a buyer with a better match can justify the fixed offer cost and equilibrium price at a lower level of urgency.

Within $\{W, P, K\}$, the switch from probing to knockout is also reached at lower urgency by higher-value buyers. At differentiability points,

$$\frac{\partial D_H(b; q)}{\partial b} = -w'_H(b) < 0. \quad (6.3)$$

The effect of b on probing itself is generally ambiguous, because a higher value both enters the probing region at its lower boundary and exits it toward knockout at its upper boundary. At differentiability points with interior cutoffs,

$$\frac{\partial \Pr(p = q_L \mid b)}{\partial b} = g(D_L(b))w'_L(b) - g(D_H(b))w'_H(b). \quad (6.4)$$

Thus probing becomes more likely with b exactly when the inflow at $D_L(b)$ exceeds the outflow at $D_H(b)$. If urgency is uniform on $[0, \bar{d}]$ and both cutoffs are interior, then

$$\frac{\partial \Pr(p > 0 \mid b)}{\partial b} = \frac{w'_L(b)}{\bar{d}} > 0, \quad \frac{\partial \Pr(p = q_L \mid b)}{\partial b} = \frac{w'_L(b) - w'_H(b)}{\bar{d}} \leq 0, \quad (6.5)$$

where the inequality follows from $w'_H(b) \geq w'_L(b)$. For any maintained positive urgency density, attempt incidence is weakly increasing in b and strictly increasing wherever the cutoff is interior; the shape of G determines the strength of this selection.

The buyer's terminal surplus is $u_s(b) = \mathbb{E}_s[\max\{b - \max\{v_s, Y_{1N}\}, 0\}]$. Under late-participation variation the state enters through Y_{1N} , so rejection of a probe is bad news about the round the rejected buyer now enters. Under fallback variation the state matters only when the fallback prevents the buyer from clearing the terminal threshold or sets the price under weak late competition; above v_H , the willingness gap is constant in b . Under urgency variation the state never enters u_s : rejection reveals a patient seller but leaves

the buyer's auction payoff unchanged.⁴

Selection also changes the interval of prices that separates the seller states. Conditional on probing, this interval has width $\int [C_H(b) - C_L(b)] h_P(b) db$. Under urgency variation, the gap is the constant $d_{S,L} - d_{S,H}$, so selection moves both thresholds equally. Under fallback variation, the gap declines with b , so high-value selection narrows it, though it remains bounded below by $\int_{v_L}^{v_H} \phi(F(t)) dt$. Under late-participation variation, high-value selection can instead preserve or widen the gap: above the common fallback, the derivative of $R_H(b) - R_L(b)$ is $[1 - F(b)][\phi'_H(F(b)) - \phi'_L(F(b))]$, which is positive for sufficiently large $F(b)$.

The ban comparison in Section 8 applies to all three microfoundations. Disclosure likewise applies when the continuation state is verifiable; its numerical application isolates participation information, for which market-level certification is especially natural.

The quantitative baseline combines stronger late participation, a higher fallback, and lower seller urgency in the high-continuation state, retaining their distinct economic roles.

6.3 Seller Urgency Alone

With seller urgency alone, $Q_L = Q_H$ and $v_L = v_H$, so $u_L = u_H$ and $w_L = w_H$. Screening operates through $C_s = R - d_{S,s}$, giving the following explicit probing window.

Corollary 6.3 (Urgency-only probing window). *Let Q and v be common across states, let $d_{S,L} > d_{S,H} \geq 0$, and write $\Delta z = d_{S,L} - d_{S,H}$ and Γ for the common wedge. Then Assumption 1 holds, with*

$$\bar{D}_L = \frac{\kappa}{1 - \pi} - d_{S,L}, \quad \Delta_D = \Delta z - \frac{\pi \kappa}{1 - \pi}.$$

Suppose

$$\frac{\pi \kappa}{1 - \pi} < \Delta z, \quad d_{S,L} < \min \left\{ R(b), \Gamma(b) + \frac{\kappa}{1 - \pi} \right\}, \quad \bar{D}_L < \bar{d} \leq \frac{(1 - \pi)d_{S,L} - d_{S,H}}{\pi}.$$

Then Proposition 5.2 applies with support $\{W, P\}$: no early offer, a rejected attempt, and successful pre-emption occur with positive probability, the probing offer being accepted exactly by the urgent seller. If in addition $d_{S,L} \leq \kappa/(1 - \pi)$, the equilibrium has a D1-compatible completion.

Proof. With common (Q, v) , $w_H = w_L = w$ and $C_s = R - d_{S,s}$, so $C_H - C_L = \Delta z$ pointwise and Assumption 1 holds once $d_{S,L} < R(b)$. Since $\Gamma(\bar{b}) = 0$, $C_s(\bar{b}) = w(\bar{b}) - d_{S,s}$; substituting into (5.9) and (5.10) gives the displayed $\bar{D}_L$ and Δ_D , and $\bar{D}_L + \Delta_D/\pi = [(1 - \pi)d_{S,L} - d_{S,H}]/\pi$. The second primitive bound is $\chi_L > 0$; together with $\Delta z > \pi\kappa/(1 - \pi)$,

⁴With continued bargaining, seller patience can also affect negotiation before the round.

it also implies $d_{S,H} < \Gamma(\underline{b}) + \kappa$, or $\chi_H > 0$. The D1 statement is the sufficient condition (5.15) of Corollary 5.3. $\square$

The urgency gap must exceed the screening premium $\pi\kappa/(1-\pi)$ for a state-splitting price to exist, while the urgent seller's cost must stay below the level at which even the least attractive buyer could buy it out. The window's upper endpoint is positive exactly when $(1-\pi)d_{S,L} > d_{S,H}$, which the first condition already implies whenever $d_{S,H} \leq \kappa$. Beyond the displayed sufficient condition, Corollary 5.3 gives the exact D1 requirement $\Gamma(b_L^0) \geq d_{S,L} - \kappa/(1-\pi)$ at an interior marginal value b_L^0 . The urgency-only specification can therefore support both $\{W, P\}$ and $\{W, P, K\}$. The one-price support admits the explicit primitive window above; once knockout is active, the two prices and selected buyer pools must be determined jointly. The following calibration verifies the richer full-menu support.

Example 6.4 (Private seller urgency as the screening state). Let $F = U[0, 1]$, let the common terminal environment have $n = 2$ late bidders and no-sale continuation value $v = .4$, and let seller urgency be $d_{S,s} \in \{0, .02\}$ with equal probabilities, state L being the urgent seller. Let buyer urgency have the rate-10 truncated-exponential distribution on $[0, 1]$, and set $\kappa = .01$. An endpoint-inclusive numerical assessment has

$$(q_L, q_H) = (.586757, .643708)$$

and probabilities

$$(.550185, .092644, .357171)$$

of no early offer, a rejected attempt, and successful pre-emption. It satisfies all audited PBE and criterion-D1 conditions in the seller-plan representation.

The lower price is accepted only by the urgent seller, while the higher price is accepted by both types. Seller urgency can therefore reproduce the baseline screening pattern even without private information about late-participation prospects. The equilibrium equations are analytical; the example's fixed point and cross-type rankings are numerical. Appendix J gives the proof and audit.

7 Aligned-Composite Three-Action Example

We now illustrate Proposition 5.4 with action support $\{W, P, K\}$. Let all match values be uniform on $[0, 1]$, and take the aligned seller states

$$(n_L, v_L, d_{S,L}) = (2, 0.38, 0.02), \quad (n_H, v_H, d_{S,H}) = (3, 0.42, 0).$$

For $n \geq 0$ late buyers and no-sale continuation value $v > 0$, direct integration under the missing-order-statistic convention gives

$$w_{n,v}(b) = b - \frac{\max\{b^{n+1} - v^{n+1}, 0\}}{n+1}, \quad (7.1)$$

$$R_{n,v}(b) = \begin{cases} \frac{n-1}{n+1} + v^n - \frac{n-1}{n+1}v^{n+1}, & b \leq v, \\ \frac{n-1}{n+1} + b^n - \frac{n}{n+1}b^{n+1} + \frac{v^{n+1}}{n+1}, & b > v. \end{cases} \quad (7.2)$$

At $n = 0$, these expressions reduce to $w_{0,v}(b) = \min\{b, v\}$ and $R_{0,v}(b) = v$. State H has stronger late participation, a higher no-sale fallback, and a more patient seller, so all three microfoundations raise its net continuation value. The pre-selection no-sale probabilities are $0.38^3 = 0.054872$ and $0.42^4 = 0.031116960$.

Let urgency follow the truncated exponential distribution

$$G(d) = \frac{1 - \exp(-\lambda d)}{1 - \exp(-\lambda \bar{d})}, \quad 0 \leq d \leq \bar{d}, \quad (7.3)$$

and take

$$\pi = 0.5, \quad \bar{d} = 1, \quad \lambda = 10, \quad \kappa = 0.05. \quad (7.4)$$

The decreasing urgency density represents a market in which most buyers can wait, while a minority faces an acute acquisition or financing deadline.

An outward-rounded interval certificate verifies a fixed point at

$$q_L = 0.609022919, \quad q_H = 0.715349727. \quad (7.5)$$

It also verifies $0 < D_L(b) < D_H(b) < \bar{d}$ and $Z(b) > 0$ for every buyer value, so waiting, probing, and knockout all occur with positive probability.

The action masses are

$$(M_W, M_P, M_K) = (0.763873955, 0.100548544, 0.135577501).$$

They imply the following observed outcomes.

Table 7.1: Pre-emption incidence in the aligned-composite example

Observed outcome	Probability
No early offer	0.763873955
Attempt and rejection	0.050274272
Attempt and completed pre-emption	0.185851773

Consistent with the decreasing urgency density, most buyers wait. Among the minority who act, the higher knockout price accounts for most completed early sales, while probing

produces both rejected and accepted attempts. In this calibration, 23.6 percent of buyers attempt pre-emption, and 78.7 percent of attempts succeed. The selected price distribution produces the same sign as the conditional price difference documented by Flått et al. (2022): completed pre-emptions have a higher mean price than completed sales through the planned round.⁵

The same certificate verifies the strict cross-state response margins—seller H rejects q_L and seller L accepts q_H —the D1 inequalities, and the inward-pointing price-map conditions; the displayed root is locally unique. After accounting for early sales, the state-conditional no-sale probabilities are 0.054348677 and 0.030986927, with prior-weighted probability 0.042667802. Appendix B gives the finite corner reduction, and Appendix C reports the certified rectangles and numerical margins.

The example completes the baseline equilibrium analysis. We next use the one-offer equilibrium to derive policy implications.

8 Information Disclosure, Commitment, and Welfare

A ban removes the seller’s discretion to accept an early offer while leaving the terminal bidding rule unchanged. We first compare the baseline one-offer equilibrium with a ban and then study committed disclosure of the seller’s continuation state. The general results allow all three state components to vary. The ban illustration holds seller timing costs equal to isolate the role of the value-distribution tail, while the disclosure illustration holds fallback values and timing costs equal to isolate participation information.

8.1 A Ban on Early Offers

Consider a fully effective rule requiring the seller to reject every offer before the announced competitive sale. Compliance and substitution toward other contact channels are held fixed; implemented regulation may change both the incidence and welfare effects of early contracting through those margins. We evaluate the selected boundary-price full-menu equilibrium, so seller L ’s expected continuation value binds at the probing price and seller H ’s expected continuation value binds at the knockout price. We hold fixed the buyer population, information, resolution benefits, seller timing costs, and terminal-auction environment. Let B denote the ban and D early-contracting discretion, and let M_P and M_K be the equilibrium probabilities of probing and knockout in (5.21).

⁵In normalized units, weighting q_L and q_H by their state-contingent completion masses gives a mean price of 0.686588 among completed pre-emptions. Integrating $P_s(b)$ over waiting buyers and rejected probes, conditional on sale, gives 0.574124 among completed bidding-round sales. These levels are illustrative. The sign is parameter-dependent: more accepted probing at the lower price reduces the pre-emption mean, while more rejected probing in the high-continuation state raises the bidding-round mean; conditioning on completed sales also changes the composition by removing no-sale histories.

Proposition 8.1 (Payoff effects of a ban). *Under the fixed terminal environment:*

(i) *the seller's ex ante payoff satisfies*

$$U_S^D - U_S^B = (1 - \pi)M_K \int_{\underline{b}}^{\bar{b}} [C_H(b) - C_L(b)] h_K(b) db > 0; \quad (8.1)$$

(ii) *the early buyer's ex ante payoff satisfies*

$$U_E^D - U_E^B = \int_{\underline{b}}^{\bar{b}} f(b) \left\{ \int_{D_L(b)}^{D_H(b)} V_L(b, d) dG(d) \right. \quad (8.2)$$

$$\left. + \int_{D_H(b)}^{\bar{d}} V_H(b, d) dG(d) \right\} db \geq 0; \quad (8.3)$$

(iii) *the late buyers' aggregate expected payoff satisfies*

$$U_{\text{late}}^B - U_{\text{late}}^D = \mathbb{E}[\mathbf{1}\{A\} \Lambda_s(b)] \geq 0, \quad (8.4)$$

where A is the event that an early offer is accepted and

$$\Lambda_s(b) = \mathbb{E}_s[\max\{Y_{1N} - \max\{v_s, b, Y_{2N}\}, 0\}] \quad (8.5)$$

is the expected surplus of the winning late buyer in the terminal auction.

The first inequality uses Assumption 1 and positive knockout mass. The second is strict when a positive measure of buyer types makes an offer with strictly positive payoff. The third is strict when an offer is accepted with positive probability in a state in which a late buyer can earn positive surplus.

Proof. At the selected probing price, seller L 's expected gain relative to the auction is zero. At the selected knockout price, seller H 's corresponding expected gain is also zero. In state L , the same knockout offer yields the conditional gain

$$q_H - \int C_L(b) h_K(b) db = \int [C_H(b) - C_L(b)] h_K(b) db > 0.$$

Weighting the three accepted offer–state combinations proves (8.1). The result is Bayesian: for a particular realized b , the seller may earn less than in the counterfactual auction, but not on average conditional on its information.

Under the ban, the early buyer waits. Under discretion, its equilibrium incremental payoff is 0, $V_L(b, d)$, or $V_H(b, d)$ on the three regions in (5.27). Integrating over (b, d) gives (8.3). Finally, an accepted offer eliminates the terminal surplus $\Lambda_s(b)$ available to the winning late buyer; otherwise their outcome is unchanged. This proves (8.4). $\square$

Proposition 8.1 separates the early buyer from the late buyers. At the information set where it chooses an action, the early buyer weakly prefers discretion because waiting remains available. Conditional on a high-continuation realization, however, a probing buyer would ex post prefer to have avoided the rejected-offer cost. Late buyers favor the ban because it restores their opportunity to win. These payoff signs hold in the selected boundary-price full-menu equilibrium; the prices and selected buyer pools determine their magnitudes. Aggregate buyer surplus is higher under the ban if and only if the gain in (8.4) exceeds the early buyer's option value in (8.3).

The seller and early buyer compare their own payoffs when deciding whether to trade early. A late buyer may value the object more highly, but only part of that value reaches the seller through the auction price; the remainder is the late buyer's surplus. Early agreement eliminates this surplus as well as the contracting parties' competition wedge. The welfare comparison must therefore include the value of preserving access for late buyers.

For total surplus, count the buyer's resolution benefit d_B and the seller's timing cost $d_{S,s}$ at their social values, measured in monetary units. The buyer's intrinsic benefit enters welfare even when it involves no resource saving. Suppose v_s is also the social value of the no-sale continuation, rather than merely revenue from a later transfer. The buyer's effort cost κ reduces total surplus whether the offer is accepted or rejected. Conditional on state s , reaching the auction instead of allocating early to value b yields the expected allocative gain

$$L_s(b) = \mathbb{E}_s[\max\{\max\{v_s, Y_{1N}\} - b, 0\}]. \quad (8.6)$$

Writing $\Gamma_s(b)$ for the competition wedge, the pointwise identity

$$\begin{aligned} \max\{\max\{v_s, Y_{1N}\} - b, 0\} &= \max\{\max\{v_s, Y_{2N}\} - b, 0\} \\ &\quad + \max\{Y_{1N} - \max\{v_s, b, Y_{2N}\}, 0\} \end{aligned}$$

implies

$$L_s(b) = \Gamma_s(b) + \Lambda_s(b). \quad (8.7)$$

The allocative value of the auction is the sum of the wedge faced by the contracting buyer and seller and the surplus of the late buyer whom their agreement excludes.

Let O denote the event that the buyer makes an offer. Under risk neutrality:

Proposition 8.2 (Welfare effect of a ban). *Holding the terminal environment fixed,*

$$W^B - W^D = \mathbb{E}[\kappa \mathbf{1}\{O\} + \mathbf{1}\{A\}(L_s(b) - d_B - d_{S,s})]. \quad (8.8)$$

Hence a ban raises total surplus if and only if the allocative gain from the auctions it preserves, together with the avoided attempt costs, exceeds the buyer's resolution benefit

and seller costs saved by accepted early trades.

Proof. If an offer is rejected, both regimes reach the same auction, but discretion incurs κ . If an offer is accepted, the ban additionally replaces an early allocation worth b with a terminal allocation or retained asset worth $\max\{v_s, b, Y_{1N}\}$, while forgoing the buyer's resolution benefit d_B and incurring the seller's timing cost $d_{S,s}$. Taking expectations across accepted and rejected attempts gives (8.8). Transaction prices cancel as transfers. $\square$

$\Pr(O \setminus A) = \pi M_P$ in the pure full-menu equilibrium. Equation (8.8) can therefore be written as

$$W^B - W^D = \pi \kappa M_P + \mathbb{E}[\mathbf{1}\{A\}(\kappa + L_s(b) - d_B - d_{S,s})]. \quad (8.9)$$

The first term is the effort cost of rejected probing offers. Among accepted offers, discretion raises surplus precisely when the direct gains from early resolution exceed the allocative gain from competition plus the effort cost of the attempt. A ban is therefore favored by frequent rejected attempts, a large late-buyer allocation gain, or small resolution gains among accepted types. Discretion is favored when completed early trades deliver large resolution gains and rarely displace a higher-value late buyer. The identity is evaluated at the selected boundary-price equilibrium.

Greater upper-tail probability can reverse this welfare ranking even when early and late buyers draw from the same distribution. Set $(n_L, n_H) = (2, 6)$, $(v_L, v_H) = (.2, .4)$, $\pi = .5$, $d_{S,L} = d_{S,H} = 0$, $\kappa = .02$, and $\bar{d} = 1$. Let buyer urgency follow $G^P(d) = 10^{-4}d + (1 - 10^{-4})[1 - (1 - d)^{30}]$, and compare the common early- and late-buyer value distributions

$$F^S = .01U[0, 1] + .99 \text{Beta}(40, 40), \quad F^T = .01U[0, 1] + .99 \left[\frac{16}{17} \text{Beta}(38, 42) + \frac{1}{17} \text{Beta}(9, 1) \right].$$

Both distributions have mean .5, so the comparison changes the shape rather than the average of buyer values; F^T places substantially more probability in the upper tail.

In both cases, some buyers wait, some make probing offers, and others make knockout offers. Table 8.1 reports the resulting outcome incidence and welfare decomposition.

Table 8.1: Upper-Tail Values Reverse the Welfare Ranking

F	Rejection	Pre-emption	Attempt effort	Allocation gain	Early gains	$W^B - W^D$
F^S	0.05631	0.15435	0.00421	0.00388	0.01162	-0.00353
F^T	0.06674	0.06839	0.00270	0.00276	0.00516	0.00030

Notes: Entries are per object. “Attempt effort” is $\kappa \Pr(O)$, “Allocation gain” is $\mathbb{E}[\mathbf{1}\{A\}L_s(b)]$, and “Early gains” is $\mathbb{E}[\mathbf{1}\{A\}(d_B + d_{S,s})]$. This exercise sets $d_{S,L} = d_{S,H} = 0$ so that the rows differ only in the common early- and late-buyer value distribution. Outward-rounded interval arithmetic verifies both welfare signs.

Under F^S , the allocation gain and avoided attempt effort sum to .00810, less than the .01162 in resolution gains delivered by completed early trades, so discretion raises

Mean-Matched Value Distributions in the Ban Comparison

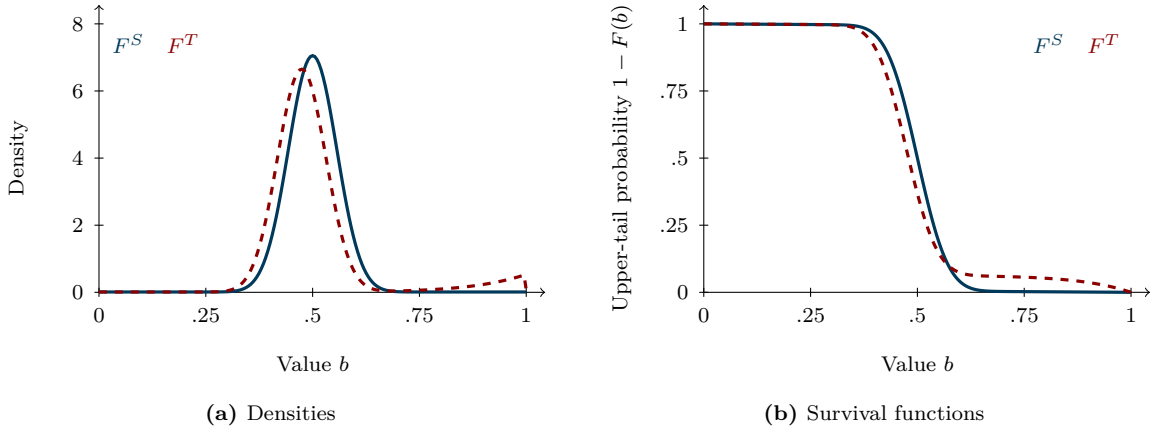


Figure 8.1: Notes: The solid blue line is F^S ; the dashed red line is F^T . The right panel plots the probability that one value draw exceeds b . In each case, the designated early buyer and every late buyer draw from the same distribution.

surplus. Under F^T , the corresponding benefits of a ban are .00546, above the .00516 resolution gain, so the ban raises surplus. The probability that one value draw exceeds .7 rises from .00310 to .05890, and excluded late-buyer surplus per completed early sale rises from .01909 to .03689. A single exceptional late value can substantially raise the value of preserving competition without comparably raising the seller's auction revenue: the best late match determines the allocative loss, whereas the auction price is governed by the runner-up, the early buyer, or the no-sale fallback. The upper tail can therefore increase the external cost of pre-emption more than the joint opportunity cost internalized by the early buyer and seller. Completed pre-emption falls from .15435 to .06839, limiting the total allocative loss. Even so, the allocative gain from the auctions a ban preserves, together with the avoided attempt effort, exceeds the gains from early resolution.

The regulatory assessment in [Finansdepartementet \(2021, ch. 6.6.3\)](#) emphasizes the unresolved tradeoff between preserving an open competitive process and avoiding transaction and search costs. Proposition 8.2 compares allocation with early resolution in a fixed-participation environment: the planned auction can improve allocation, early agreement delivers resolution benefits and saves seller continuation costs, and a ban avoids attempt costs. The timing of late buyers' preparation sharpens this comparison.

Corollary 8.3 (Late-buyer preparation). *Let $S_s(\tau) \geq 0$ denote the object-specific due-diligence, financing, travel, and related preparation resources that remain unspent when an early offer is accepted at time τ . Holding participation and the terminal auction fixed,*

$$W^B - W^D = \mathbb{E} \left[\kappa \mathbf{1}\{O\} + \mathbf{1}\{A\} (L_s(b) - d_B - d_{S,s} - S_s(\tau)) \right]. \quad (8.10)$$

Preparation incurred before τ , or useful independently of this object, cancels from the comparison. Only object-specific resources that the early sale actually avoids make

discretion more attractive.

Proof. Relative to Proposition 8.2, an accepted early sale saves exactly $S_s(\tau)$ in addition to the buyer's resolution benefit and the seller's saved timing cost. Costs already incurred are common to the two policy regimes and cancel. $\square$

Corollary 8.3 formalizes the timing margin emphasized in [Finansdepartementet \(2021, ch. 6.6.3\)](#) at fixed participation.

8.2 Committed Disclosure of the Seller State

Suppose a verifiable signal identifies the seller's continuation state s before the buyer acts and therefore reveals the state-contingent objects $(Q_s, v_s, d_{S,s})$. The result below applies whether the ordering originates in late participation, fallback, seller urgency, or their aligned combination. The numerical application later in the section sets $v_L = v_H$ and $d_{S,L} = d_{S,H}$ to isolate disclosure of participation prospects. Maintain $C_H(b) > C_L(b)$.

Whereas a ban suppresses all early contracting, disclosure preserves the option of early agreement while revealing the seller's continuation state. Consider first an automatic disclosure regime: s becomes verifiably public before the buyer acts, either because reporting is mandatory or because the seller chose the disclosure technology before observing s . There is no interim reporting choice in this regime. Voluntary certification after observing s is analyzed separately below.

Disclosure removes one reason for rejection: the buyer knows which seller state it faces before deciding whether to incur the offer cost. With a positive attempt cost, an offer known to be rejected is strictly worse than waiting, so every on-path offer is accepted. Early-sale incidence and welfare remain ambiguous because disclosure changes both prices and offer selection and removes the information rents an opaque seller earns from screening.

Conditional on state s , define

$$\widehat{D}_s(b; q) = q - w_s(b) + \kappa, \quad (8.11)$$

$$\widehat{m}_s(b; q) = 1 - G(\widehat{D}_s(b; q)), \quad (8.12)$$

and, whenever the denominator is positive,

$$\widehat{T}_s(q) = \frac{\int_{\underline{b}}^{\bar{b}} C_s(b) \widehat{m}_s(b; q) f(b) db}{\int_{\underline{b}}^{\bar{b}} \widehat{m}_s(b; q) f(b) db}. \quad (8.13)$$

The map is the seller's posterior continuation value among buyers who offer at price q . Let

$$\bar{D}_s^{\text{FD}} = C_s(\bar{b}) - w_s(\bar{b}) + \kappa. \quad (8.14)$$

Proposition 8.4 (Automatic disclosure of the seller state). *Fix a disclosed state s under Assumption 1.*

(i) *A no-early-offer PBE exists if and only if*

$$\bar{d} \leq \bar{D}_s^{\text{FD}}, \quad (8.15)$$

and it has a D1-compatible completion.

(ii) *If $\bar{d} > \bar{D}_s^{\text{FD}}$, then a no-early-offer PBE does not exist and $\hat{T}_s$ has a fixed point $\hat{q}_s \in (C_s(\underline{b}), C_s(\bar{b}))$. There is a PBE in which the buyer waits below $\widehat{D}_s(b; \hat{q}_s)$, offers $\hat{q}_s$ above it, and the seller accepts.*

(iii) *If $\kappa > 0$, every on-path submitted offer is accepted in any pure-strategy PBE under disclosure of the seller state.*

(iv) *Suppose $\kappa > 0$ and the set of marginal values is nonempty. Define*

$$b_s^0 = \sup \left\{ b : 0 \leq \widehat{D}_s(b; \hat{q}_s) \leq \bar{d} \right\}. \quad (8.16)$$

The pooling PBE in part (ii) has a D1-compatible completion if and only if

$$\hat{q}_s \leq C_s(b_s^0). \quad (8.17)$$

A sufficient condition is $\widehat{D}_s(\bar{b}; \hat{q}_s) \geq 0$.

Public disclosure reduces the problem to a state-specific pooling fixed point. With $\kappa > 0$, a knowingly rejected offer is strictly dominated by waiting; Appendix G establishes existence and the D1 boundary.

Combine the two disclosed-state problems by using no-early-offer behavior in a state satisfying (8.15) and the pooling equilibrium otherwise. Write FD for this canonical assessment with full disclosure of the seller state, set $\widehat{M}_s = 0$ in a no-early-offer state, and otherwise let

$$\widehat{M}_s = \int_{\underline{b}}^{\bar{b}} \widehat{m}_s(b; \hat{q}_s) f(b) db. \quad (8.18)$$

Then

$$\Pr_{\text{FD}}(O) = \Pr_{\text{FD}}(A) = (1 - \pi)\widehat{M}_L + \pi\widehat{M}_H, \quad \Pr_{\text{FD}}(O \setminus A) = 0. \quad (8.19)$$

Under opacity, early trade occurs above $D_L(b)$ in state L and above $D_H(b)$ in state H . Disclosure replaces these screening thresholds with the disclosed-state cutoff $\widehat{D}_s(b; \hat{q}_s) = \hat{q}_s - w_s(b) + \kappa$. A lower disclosed cutoff brings less urgent buyers—and, at a given urgency, buyers with lower match values—into early trade; a higher cutoff instead makes marginal early traders wait. Because the disclosed prices adjust endogenously, either ordering is possible.

The fixed-point condition leaves the seller with zero expected gain relative to the auction within each disclosed-state offer pool. The selected private full-menu equilibrium likewise gives zero rent to the seller state at the margin of each active price. Full disclosure therefore removes the extra rent seller L earns from a knockout price high enough to induce seller H to accept:

$$U_S^{\text{FD}} - U_S^D = -(1 - \pi)M_K \int_b^{\bar{b}} [C_H(b) - C_L(b)]h_K(b) db \leq 0. \quad (8.20)$$

Corollary 8.5 (Ex-ante disclosure commitment). *Suppose that, before observing s , the seller chooses between the selected opaque equilibrium and commitment to the canonical disclosed-state assessment. If disclosure costs $\delta \geq 0$, the seller weakly prefers opacity. The preference is strict if $M_K > 0$ or $\delta > 0$.*

Proof. The disclosure commitment yields $U_S^{\text{FD}} - \delta$. Equation (8.20) gives

$$U_S^D - (U_S^{\text{FD}} - \delta) = \delta + (1 - \pi)M_K \int [C_H(b) - C_L(b)]h_K(b) db \geq 0,$$

with the stated strictness conditions. □

For total surplus, couple the two regimes on the same realization and let A_j and O_j denote acceptance and offer under $j \in \{D, \text{FD}\}$. Since $O_{\text{FD}} = A_{\text{FD}}$,

$$W^{\text{FD}} - W^D = \kappa \Pr(O_D \setminus A_D) + \mathbb{E}[(A_{\text{FD}} - A_D)\{d_B + d_{S,s} - L_s(b) - \kappa\}]. \quad (8.21)$$

The first term is the effort saved by avoiding rejected probes: under opacity the buyer sometimes incurs κ and learns through rejection that the seller has the high continuation state, whereas disclosure reveals that state before the buyer decides whether to make an offer. The second term sums the welfare contributions of the added and removed early sales described by the cutoff changes above. An added early sale raises surplus exactly when $d_B + d_{S,s} > L_s(b) + \kappa$, because its timing gain then exceeds the foregone allocation through competition and the effort cost of the offer; removing an early sale has the opposite sign.

To isolate participation disclosure and the role of the prior, hold $F = U[0, 1]$, the common no-sale continuation value $v_0 = .4$, $(n_L, n_H) = (1, 3)$, $d_{S,L} = d_{S,H} = 0$, $\kappa = .02$, $\bar{d} = 1.2$, and $\lambda = 10$ fixed, and change only the probability π of the high-participation state. Table 8.2 shows that raising π from .65 to .775 reverses the welfare effect of disclosure.

The disclosed state-specific problems do not depend on the prior π , but the opaque screening menu does. At $\pi = .65$, avoiding rejected probes saves .003064, more than the .001446 loss from the changed early-sale set. Raising π to .775 reduces the opaque probing mass and hence the rejection saving to .001947. It also places more weight on the high-participation state, where early agreement displaces the stronger late competition;

Table 8.2: Opposite Welfare Effects of Participation-State Disclosure

Case	$\Pr_D(A)$	$\Pr_{FD}(A)$	$\Pr_D(O \setminus A)$	Saved rejections	Sale-set effect	$W^{FD} - W^D$
$\pi = .650$	.22533	.36103	.15322	.003064	-.001446	.001618
$\pi = .775$	.24259	.34158	.09736	.001947	-.002283	-.000336

Notes: “Saved rejections” is the first term in (8.21); “Sale-set effect” is the second. Entries are per object. Under disclosure every on-path offer is accepted, so disclosed offer and completion probabilities coincide.

the total sale-set loss rises to .002283, of which .001607 arises in that state. Disclosure then lowers welfare by .000336 even though it eliminates every rejected attempt.

Proposition 8.4 establishes a state-specific pooling equilibrium, with D1 survival under (8.17); accepted price menus may also carry further information about b . Corollary 8.5 concerns ex-ante commitment to automatic disclosure of the seller state.

8.2.1 Interim Voluntary Hard Evidence

Now suppose the seller observes s and then either remains silent or pays δ to present a state-specific certificate. A certificate is feasible only for the matching state. Truthfulness is therefore imposed by the evidence technology, not generated by the seller’s reporting incentives. Silence selects the canonical opaque continuation, while a certificate selects the corresponding disclosed-state continuation.

Proposition 8.6 (Costly voluntary hard evidence). *Conditional on these canonical continuations, for every $\delta > 0$ the unique pure reporting PBE has both seller states remain silent.*

Appendix G proves the result and gives the zero-cost classification. At $\delta = 0$, silence pooling and weak selective disclosure can coexist. Conditional on the named continuations, every $\delta > 0$ selects silence among pure reporting profiles. The result applies to truthful state-specific certificates and a common seller reporting objective.

Both policy comparisons hold the buyer population, late participation, resolution benefits, seller timing costs, and terminal auction fixed. Endogenous entry, preparation, information acquisition, or search would add further equilibrium responses. Continued bargaining also changes rejection values and replaces the single attempt cost with the effort cost of all proposals (Appendix H). The welfare accounting must then use the resulting allocation, early agreements, and total costs.

9 Risk Aversion and Bargaining

Early agreement can also insure parties against auction uncertainty. We first establish this alternative source of gains and then examine how continued bargaining changes the baseline outcomes.

9.1 Risk Aversion and Equilibrium Selection

Risk aversion changes the comparison between a certain early agreement and the payoff lottery generated by the announced round. Conditional on a known continuation state s , let the buyer's monetary risk premium ρ_B and the seller's monetary risk premium ρ_S equal expected terminal payoff minus its certainty equivalent, on the normalized payoff scale. A larger premium means that the party is willing to surrender more expected payoff to replace the lottery with certainty. A mutually acceptable early price exists exactly when

$$d_B + d_{S,s} + \rho_B + \rho_S \geq \Gamma_s(b) + \kappa. \quad (9.1)$$

The left-hand side collects the buyer's resolution benefit, the seller's saved timing cost, and the parties' insurance values; these gains must cover the competition wedge $\Gamma_s(b)$ and the offer cost κ . Equation (9.1) is a state-conditional bilateral-feasibility condition, not an equilibrium cutoff. In the signaling game the buyer does not observe s , and probing and knockout insure different compound lotteries. Their relevant risk premia are therefore action-specific and cannot generally be inserted into the baseline cutoffs as a common additive term.

A positive seller risk premium lowers the seller's minimum acceptable early price, but seller risk aversion also makes its acceptance threshold a nonlinear certainty equivalent of the posterior payoff lottery rather than the posterior mean in (4.8). We therefore establish the general feasibility effect for both parties and construct the signaling equilibrium using privately heterogeneous buyer risk aversion with a risk-neutral seller.

Seller-posted buyout models show that a certain early purchase can insure risk-averse buyers or sellers (Budish and Takeyama, 2001; Reynolds and Wooders, 2009; Mathews and Katzman, 2006). Equation (9.1) formalizes the insurance interpretation proposed by Flått et al. (2022); the baseline supplies heterogeneous benefits from early resolution as a second mechanism consistent with a higher conditional pre-emption price. In our buyer-proposed setting, the buyer's privately observed risk aversion determines who offers, while the offered price screens the seller's continuation state. This interaction produces the probing equilibrium below without buyer or seller urgency.

Proposition 9.1 (Probing with buyer risk aversion). *Consider the late-participation-only specification with*

$$(n_L, n_H, \pi, \kappa) = (2, 7, .5, .01), \quad b \sim U[0, 1], \\ v_L = v_H = 0, \quad d_B = d_{S,L} = d_{S,H} = 0.$$

The seller and late bidders are risk neutral. The early buyer has constant absolute risk aversion (CARA) utility $u_\alpha(x) = -e^{-\alpha x}$ over monetary payoff and privately observes an

independent coefficient of absolute risk aversion

$$\alpha \in \{\alpha_L, \alpha_H\}, \quad \Pr(\alpha = \alpha_L) = \Pr(\alpha = \alpha_H) = \frac{1}{2},$$

where $\alpha_L = .37$ and α_H is the unique root in the certified interval $[2.78802798023663, 2.78802798043663]$. There is a waiting-probing PBE at $q \simeq .655960$. Every α_L -buyer waits, while an α_H -buyer offers q if and only if $b \geq .811700$. Seller L accepts and seller H rejects. The probabilities of no early offer, a rejected attempt, and successful pre-emption are

$$(.905850, .047075, .047075).$$

The assessment survives criterion D1 over the complete rationalizable seller-plan set.

The auction exposes the early buyer to uncertainty about both whether it wins and how much it pays. A probing offer replaces that lottery with the certain payoff $b - q - \kappa$ in state L , while preserving the auction after rejection in state H . The construction uses the resulting demand for partial insurance: the less-risk-averse class waits, whereas sufficiently high-value buyers in the more-risk-averse class probe. Within either class, the gain from probing rises with match value.

Criterion D1 attributes an unused lower price to the top-value type in the less-risk-averse class and an unused higher price to the top-value type in the more-risk-averse class. The corresponding seller responses make neither deviation profitable. Outward-rounded interval arithmetic certifies the price, match-value cutoff, risk-aversion coefficient, and the continuum inequalities used in this refinement argument. The construction extends to a nearby calibrated family and to two sufficiently narrow continuous risk-aversion bands, with the upper band's center adjusting to preserve seller indifference. CARA is tractable here because a certain cash shift factors out of exponential expected utility. Appendix K gives the proof, the precise perturbation construction, and the bilateral-feasibility condition.

9.2 Continued Pre-emption-Stage Bargaining

The one-offer protocol rules out revisions after rejection. With continued early bargaining, rejection retains an option value, and proposals can reveal buyer types. Appendix H allows either party to make another binding proposal or end bargaining and activate the auction. Proposition H.1 extends the result that attempts require gains from early resolution and constructs a PBE with positive proposal costs and all three outcomes: no early offer, an attempt followed by the auction, and early agreement. Patient high-value buyers with $d_B = 0$ can pool with urgent initiators and earn information rents. Thus the outcome pattern can survive while buyer selection changes. The construction imposes no round cap and does not establish D1 survival. Welfare must also account for the effort costs of further proposals.

9.3 Post-Bidding Counteroffers

The seller may also retain discretion after bidding. Appendix I studies one final counteroffer, including a specification in which the seller identifies an early buyer who wins and conditions on whether it waited or probed. Proposition I.1 constructs a D1-compatible waiting–probing equilibrium with all three early outcomes. This calibrated result allows the earlier action to affect terminal terms; its refinement covers the stated one-counteroffer game.

10 Conclusion

An early private offer asks the seller to abandon an announced competitive sale. Competition creates an opportunity cost for the seller and early buyer jointly: late buyers may bid the price above the early buyer’s value, or the seller may have a more valuable no-sale use. Early-resolution benefits, saved seller costs, or insurance gains can compensate for this loss.

Private seller information determines which gains are realized. Buyers select into offers according to their match value and resolution benefit, so the offered price signals the same buyer’s likely auction bid. A moderate probe buys agreement in the weak seller state while preserving access to competition after rejection. More urgent buyers may instead pay for a knockout. Late participation, fallback, and seller urgency can each support screening, but differ in what rejection reveals and whether rejection-free equilibria survive D1.

Private gains do not establish efficiency because the contracting parties exclude late buyers’ surplus. Under risk neutrality, a fully effective ban raises welfare when improved allocation and avoided real attempt costs outweigh the direct resolution gains from completed early trades. This comparison holds the terminal environment fixed, uses social values for resolution benefits, seller costs and the fallback, and evaluates the selected equilibrium; its sign may differ across boundary-price fixed points. Disclosure eliminates on-path rejection in pure strategies but need not improve welfare: its saved attempt costs must be weighed against changes in which early trades occur. With positive certification cost and the canonical continuation equilibria, silence is the unique pure reporting equilibrium.

Privately heterogeneous buyer risk aversion supports D1-compatible probing even without direct resolution benefits or seller timing costs. Continued bargaining and a final counteroffer can also preserve the three early outcomes in the specified constructions, while changing buyer selection or terminal terms. The central tradeoff remains between securing early agreement and retaining the opportunities created by competition.

Computational materials and reproduction instructions accompany the numerical calibrations in Appendix A.

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A Numerical Calibrations

Table A.1 summarizes the numerical calibrations. In each calibration based on the baseline model, the designated early buyer and late buyers share the listed value distribution. Write $\text{TE}(\lambda; \bar{d})$ for the truncated-exponential urgency distribution in (7.3).

Table A.1: Numerical Calibrations

Exercise	Value distribution	Buyer heterogeneity	Other primitives and role
Waiting-probing regime	$F = U[0, 1]$	$G = \text{TE}(10; .045)$	$(n_L, v_L, d_{S,L}) = (2, .38, .02)$, $(n_H, v_H, d_{S,H}) = (3, .42, 0)$, $\kappa = .02$; no offer, rejection, and completed pre-emption all occur.
Aligned full menu	$F = U[0, 1]$	$G = \text{TE}(10; 1)$	The same three-channel aligned states, $\kappa = .05$; probing and knockout offers both occur.
Seller urgency and D1 survival	$F = U[0, 1]$	$G = \text{TE}(10; 1)$	Common $(n, v) = (2, .4)$, $(d_{S,L}, d_{S,H}) = (.02, 0)$, $\kappa = .01$; seller urgency screens offers, and a $\{W, K\}$ equilibrium survives D1.
Ban comparison	F^S and F^T in Figure 8.1	$G = 10^{-4}U[0, 1] + (1 - 10^{-4})\text{Beta}(1, 30)$	$(n_L, v_L) = (2, .2)$, $(n_H, v_H) = (6, .4)$, $d_{S,L} = d_{S,H} = 0$, $\kappa = .02$; a ban lowers surplus under F^S and raises it under F^T .
Participation disclosure	$F = U[0, 1]$	$G = \text{TE}(10; 1.2)$	$(n_L, n_H) = (1, 3)$, common $v = .4$, $d_{S,L} = d_{S,H} = 0$, $\kappa = .02$, with $\pi \in \{.65, .775\}$; disclosure raises surplus at $\pi = .65$ and lowers it at $\pi = .775$.
Post-bidding counteroffers	$F = U[0, 1]$	$G = \text{TE}(10; .15)$	$(n_L, n_H) = (2, 3)$, common $v = .4$, $d_{S,L} = d_{S,H} = .01$, $\kappa = .01$; probing survives action-linked counteroffers.
Continued bargaining	$F = U[0, 1]$	Atom-continuous G_η	Positive proposal costs; all three outcomes survive continued bargaining. The type distribution is described in Appendix H.
Buyer risk aversion	$F = U[0, 1]$	$d_B = 0$, binary CARA coefficient	$(n_L, n_H) = (2, 7)$, common $v = 0$, $\kappa = .01$; risk aversion supports probing, which survives D1.

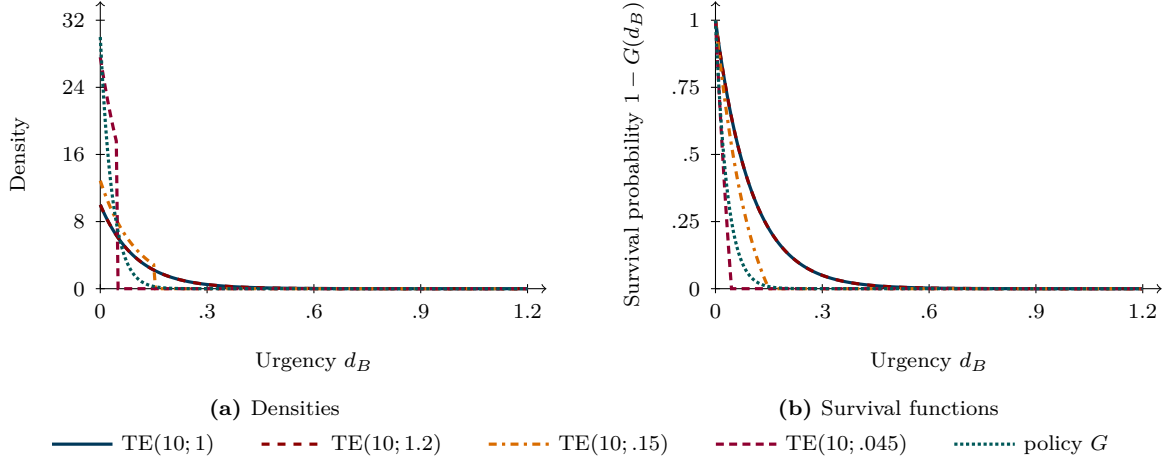


Figure A.1: Continuous Urgency Distributions Used in the Numerical Exercises

Notes: $\text{TE}(\lambda; \bar{d})$ denotes a rate- λ exponential truncated to $[0, \bar{d}]$. The policy distribution is $10^{-4}U[0, 1] + (1 - 10^{-4})\text{Beta}(1, 30)$. The continued-bargaining specification contains an atom, and the risk-aversion exercise uses two discrete risk-aversion types; neither has a continuous urgency density to plot.

Computational materials. The accompanying `ComputationalSupplement.zip` contains the claim-to-code map, source programs, precision settings, pinned dependencies, and execution records for the numerical certificates and audits. With Python 3.12, install `code/requirements-publication.txt` and run `python code/run_reproduction.py --mode publication` from the package root. The claim map distinguishes interval certificates, exact-arithmetic checks, floating-point audits, and diagnostics. Appendix C displays the main certificate; its tables and machine-readable interval records are regenerated by `code/generate_aligned_certificate_appendix.py`. The execution records identify software versions, source hashes, commands, and outputs.

B Proof of Proposition 5.4

The candidate ranges for q_L and q_H may overlap, so membership in $\mathcal{Q}$ alone does not order the two prices. Since $w_H(b) - w_L(b)$ is weakly increasing, requiring $Z(b; q) > 0$ for every buyer type is equivalent to placing the price pair above the boundary $Z(\bar{b}; q) = 0$. On that boundary, $q_H - q_L = \pi[w_H(\bar{b}) - w_L(\bar{b})] + \pi\kappa/(1 - \pi) > 0$; hence the admissible region lies strictly above the price-equality line $q_H = q_L$ and contains no pair with $q_L > q_H$. This orders both prices and the buyer's action cutoffs.

The residual vector $\Psi(q) = T(q) - q$ compares the continuation values generated by the selected buyer pools with the candidate prices. Figure B.1 locates a rectangle $\mathcal{Q}_0$ inside the price-ordering region, with each residual positive on its lower face and negative on its upper face. The Poincaré–Miranda theorem, the multidimensional intermediate-value theorem, then gives a fixed point.

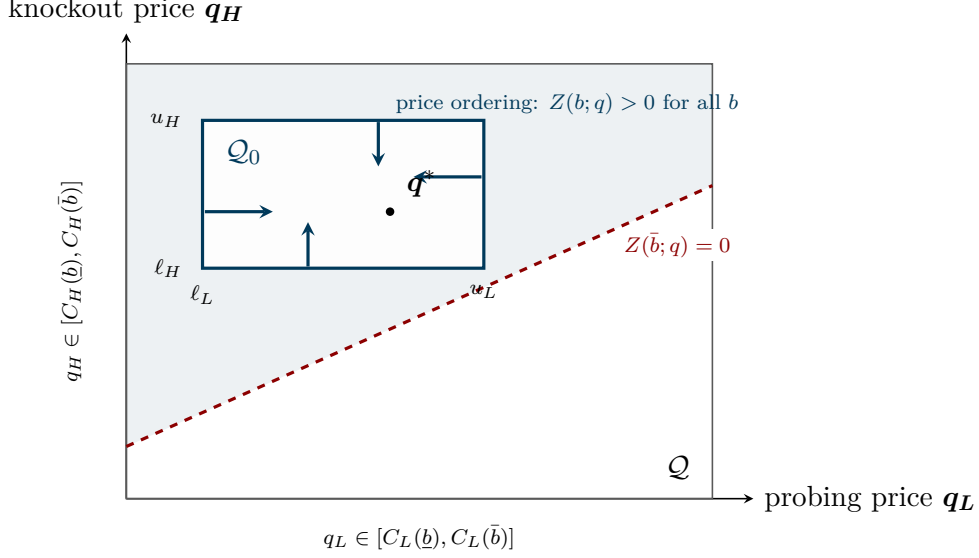


Figure B.1: Price-space geometry for the three-action fixed point. The outer rectangle $\mathcal{Q}$ contains all boundary-price candidates. Since $w_H(b) - w_L(b)$ is weakly increasing, $Z(b; q) > 0$ for every b is the region above $Z(\bar{b}; q) = 0$. The proof places $\mathcal{Q}_0$ inside this region. The horizontal arrows show the residual $T_P(q) - q_L$ on the two vertical faces, and the vertical arrows show $T_K(q) - q_H$ on the two horizontal faces. Their inward directions imply a fixed point q^* . Other interiority conditions are omitted.

By monotonicity, the pointwise interiority condition (5.25) can be checked at the support endpoints:

$$\begin{aligned}
\ell_L - w_L(\bar{b}) + \frac{\kappa}{1 - \pi} &> 0, \\
\frac{u_H - (1 - \pi)\ell_L}{\pi} - w_H(\underline{b}) &< \bar{d}, \\
\ell_H - u_L - \pi[w_H(\bar{b}) - w_L(\bar{b})] - \frac{\pi}{1 - \pi}\kappa &> 0.
\end{aligned} \tag{B.1}$$

These inequalities ensure all three action intervals at every match value, uniformly over $\mathcal{Q}_0$. Seller acceptance remains nested by the common-posterior ordering in Section 4.

For every $q \in \mathcal{Q}_0$, condition (5.25) and the positive density of G imply that all three actions have positive probability for every b . The conditional action probabilities in (5.20) and their integrals $M_a(q)$ are continuous in q . Since each $M_a(q) > 0$, the posterior densities in (5.22) and the price map T are continuous as well. Let $\Psi(q) = T(q) - q$. Condition (5.26) makes the first coordinate of Ψ positive and negative on the lower and upper q_L -faces, respectively, and does the same for the second coordinate on the corresponding q_H -faces. The Poincaré–Miranda theorem therefore gives a zero of Ψ in $\mathcal{Q}_0$; the strict face inequalities put it in the interior.

Fix such a point. At q_L , seller L 's posterior continuation value is

$$\int C_L(b) h_P(b; q) db = q_L,$$

so it accepts. Seller H rejects because

$$\int C_H(b)h_P(b; q) db > \int C_L(b)h_P(b; q) db = q_L.$$

At q_H , seller H accepts at equality, while seller L accepts strictly because

$$\int C_L(b)h_K(b; q) db < \int C_H(b)h_K(b; q) db = q_H.$$

Finally, $Z(b; q) > 0$ implies $q_H - q_L > \pi[w_H(b) - w_L(b)] + \pi\kappa/(1 - \pi) > 0$, so $q_L < q_H$.

For each b , equation (5.17) and $Z(b; q) > 0$ imply

$$D_L(b) < \widetilde{D}_H(b) < D_H(b).$$

Below $D_L(b)$, both submitted-offer payoffs are negative. Between $D_L(b)$ and $\widetilde{D}_H(b)$, only probing is profitable. Between $\widetilde{D}_H(b)$ and $D_H(b)$, both offers are profitable and probing yields the larger payoff. Above $D_H(b)$, knockout yields the larger payoff. These comparisons prove (5.27).

It remains to rule out off-path deviations. First consider sufficiently small, disjoint left neighborhoods of the two on-path prices. After an undercut of q_L , retain the on-path probing posterior. Seller L 's continuation value is then $q_L > p$, so both states reject. After an undercut of q_H , retain the on-path knockout posterior. Seller H 's continuation value is $q_H > p$, so H rejects; if seller L accepts, the deviation induces the same L -only response as the strictly cheaper on-path probe q_L . No local undercut is profitable, and these posteriors converge to their on-path counterparts as required by Assumption 2.

At every remaining off-path price, concentrate the posterior on $b = \bar{b}$. A price below $C_L(\bar{b})$ is rejected by both sellers, a price in $[C_L(\bar{b}), C_H(\bar{b}))$ is accepted only by seller L , and a price at least $C_H(\bar{b})$ is accepted by both. A rejected deviation yields $-\kappa$. Since

$$q_L = \int C_L(b)h_P(b; q) db \leq C_L(\bar{b}),$$

any off-path price accepted only in state L is weakly more expensive than probing. Likewise,

$$q_H = \int C_H(b)h_K(b; q) db \leq C_H(\bar{b}),$$

so any off-path price accepted in both states is weakly more expensive than knockout. No buyer can improve on its best on-path action. Seller actions are sequentially rational by (4.8), completing the proof.

Finite Corner Test for the Aligned Example

For the truncated-exponential urgency distribution used below, the four face conditions have a finite corner test. On the interior branch, let $\delta = q_H - q_L$ and $A = 1 - \pi$. Up to a factor common across b ,

$$m_P(b; q) \propto e^{\lambda w_L(b)} - \exp\left[-\lambda\left(\frac{\delta}{\pi} - \frac{\kappa}{A}\right)\right] e^{\lambda w_H(b)}.$$

The log-weight score with respect to δ is nondecreasing in b because $w_H(b) - w_L(b)$ is nondecreasing. Since $C_L(b)$ is also nondecreasing, with both functions strict above v_L , the covariance formula for a posterior mean implies that T_P increases in $q_H - q_L$. Similarly, m_K is proportional to $e^{-\lambda[r - w_H(b)]} - e^{-\lambda\bar{d}}$, where $r = [q_H - Aq_L]/\pi$, and its log-weight score implies that T_K increases in r . Hence (5.26) is implied by

$$T_P(\ell_L, \ell_H) > \ell_L, \quad T_P(u_L, u_H) < u_L, \quad T_K(u_L, \ell_H) > \ell_H, \quad T_K(\ell_L, u_H) < u_H.$$

Strict Ordering under Late Participation

Suppose the fallback and seller timing cost are common across states. First-order stochastic dominance of Q_H over Q_L gives a monotone coupling of bidder counts and hence $C_H(b) \geq C_L(b)$. Strictness below the fallback requires competition between late buyers. For $x \in (0, 1)$, let $a_n(x)$ be the probability that at least two of n independent draws exceed a value with quantile x . Then $a_0 = a_1 = 0$ and

$$a_n(x) - a_{n-1}(x) = (n-1)(1-x)^2 x^{n-2} > 0, \quad n \geq 2.$$

The tail-sum identity therefore gives

$$\Pr_H(Y_{2N} > t) - \Pr_L(Y_{2N} > t) = \sum_{k=2}^{\infty} (k-1)[1-F(t)]^2 F(t)^{k-2} \{\Pr_H(N \geq k) - \Pr_L(N \geq k)\}.$$

For $v_0 < t < \bar{b}$, this difference is strictly positive if at least one count-tail inequality is strict at $k \geq 2$. When $b \leq v_0$, integrating over $t \in (v_0, \bar{b})$ gives $C_H(b) > C_L(b)$. When $v_0 < b < \bar{b}$, the same argument applies above b ; below b , the seller's price exceeds t whenever at least one late value exceeds t , which gives a weakly positive contribution. At $b = \bar{b}$, nontrivial count dominance makes this latter contribution strictly positive on $(v_0, \bar{b})$. Conversely, a shift only from zero to one late buyer leaves $C_H(b) = C_L(b)$ for $b \leq v_0$. The strict tail condition in Section 4.2 is therefore necessary and sufficient for strict ordering at those buyer values under count dominance.

C Proof of Corollary 5.5

Write $b = \zeta(u)$, where u is uniform on $[0, 1]$, and hold (Q_L, Q_H, v_L, v_H) fixed. Let $(U_i)_{i \geq 1}$ be iid uniform draws, independent of $N \sim Q_s$, and let $U_{1N} \geq U_{2N}$ be the two highest among the first N draws, with $U_{1,0} = U_{2,0} = U_{2,1} = 0$. The auction objects can be written over this fixed probability space as

$$\begin{aligned} w_s^\zeta(u) &= \mathbb{E}[\min\{\zeta(u), \max\{v_s, \zeta(U_{1N})\}\}], \\ R_s^\zeta(u) &= \mathbb{E}[\max\{v_s, \zeta(U_{2N}), \min\{\zeta(u), \zeta(U_{1N})\}\}]. \end{aligned}$$

The min and max operators are Lipschitz. Hence w_s^ζ , R_s^ζ , and $C_s^\zeta = R_s^\zeta - d_{S,s}$ vary uniformly continuously with ζ under the sup norm, including when probability moves across either standing value.

In quantile coordinates, the action masses and both coordinates of T_ζ are integrals over a fixed uniform measure. Strict global interiority keeps all urgency cutoffs in a compact subset of $(0, \bar{d})$ in a neighborhood of (ζ^0, q^0) . Since G has a continuously differentiable positive density and both action masses are positive, $\Psi(\zeta, q) = T_\zeta(q) - q$ is continuous in (ζ, q) , continuously differentiable in q , and $D_q \Psi$ is jointly continuous. Differentiability with respect to the quantile is not required.

Apply the parameterized implicit-function theorem to

$$\Phi(\zeta, r, q) = T_\zeta(q) - q + r.$$

Nonsingularity of $D_q \Psi(\zeta^0, q^0)$ gives a unique nearby solution $q(\zeta, r)$ to $\Phi = 0$, continuous in (ζ, r) . At this solution,

$$q_L - T_{\zeta,P}(q) = r_L, \quad q_H - T_{\zeta,K}(q) = r_H,$$

so the relevant marginal seller accepts weakly, and strictly when the corresponding rent is positive. Every cutoff, cross-state seller response, positive-mass condition, and endpoint inequality used to deter off-path deviations is strict at $(\zeta^0, 0, q^0)$: positive action masses and strict monotonicity imply $q_L^0 = T_{\zeta^0,P}(q^0) < C_L^{\zeta^0}(\bar{b})$ and $q_H^0 = T_{\zeta^0,K}(q^0) < C_H^{\zeta^0}(\bar{b})$. These inequalities are continuous in (ζ, r, q) . The solution therefore remains globally interior and implements the assessment in Proposition 5.4. For every admissible common distribution, pointwise $C_L < C_H$ nests seller responses. When $\kappa > 0$, the strict endpoint margins in the nested-plan argument in Appendix D persist and supply the stated D1 completion in the contingent-plan representation.

At $r = 0$, the solution is the locally passive boundary-price branch. Positive rent vectors establish the surrounding PBE and D1 robustness result but violate Assumption 2 by the undercut argument following that assumption.

Certificate for the Aligned Uniform Calibration

For the calibration in Section 7, the certified price rectangle is

$$\mathcal{Q}_0 = [.605022919, .613022919] \times [.711349727, .719349727].$$

The truncated-exponential likelihood-ratio comparisons reduce all four continuum face inequalities to the corner evaluations in Appendix B. Table C.2 reports their strictly positive lower bounds and the global-interiority margins, including support endpoints and both standing-value kinks. Table C.1 reports the root box, Krawczyk image, and interval Jacobian. The residual system used for the local calculation is

$$\mathcal{E}(q) = (1 - e^{-10}) \begin{pmatrix} M_P(q)[T_P(q) - q_L] \\ M_K(q)[T_K(q) - q_H] \end{pmatrix}, \quad J = D_q \mathcal{E}.$$

Since both action masses are positive, its zeros coincide with the boundary-price fixed points. Strict Krawczyk inclusion establishes a locally unique zero, and the determinant enclosure excludes zero. At a root, nonsingularity also holds for the Jacobian of $T(q) - q$, since the two residual systems differ by a nonsingular diagonal factor. This certifies the required price box at the displayed primitives, while the corollary’s distributional neighborhood remains qualitative: no explicit perturbation radius is computed.

Table C.1: Root Enclosure and Local Uniqueness Certificate

Quantity	Outward enclosure
Root-box coordinate q_L	[0.609022919151066, 0.609022919153068]
Root-box coordinate q_H	[0.715349726667131, 0.715349726669133]
Krawczyk image, coordinate L	[0.609022919152066548242, 0.609022919152066548244]
Krawczyk image, coordinate H	[0.715349726668132175635, 0.715349726668132175636]
Jacobian entry J_{11}	[−0.154272567267, −0.154272567240]
Jacobian entry J_{12}	[0.053728588475, 0.053728588488]
Jacobian entry J_{21}	[−0.000041801644, −0.000041801637]
Jacobian entry J_{22}	[−0.135487742935, −0.135487742915]
$\det J$	[0.020904287872, 0.020904287879]

Notes: The calculation uses Arb through `python-flint 0.9.0`, at 60 decimal digits. These intervals are outward-rounded displays of the computed balls. Strict Krawczyk inclusion is checked before display rounding. The Jacobian is for the unnormalised zero-rent equations defined in the text, not for the price map T itself.

Table C.2: Uniform Price-Box and Equilibrium Margins

Quantity	Certified lower bound or range
$T_P(\ell_L, \ell_H) - \ell_L$	0.004000000
$u_L - T_P(u_L, u_H)$	0.003999999
$T_K(u_L, \ell_H) - \ell_H$	0.003996513
$u_H - T_K(\ell_L, u_H)$	0.003996068
$\min_{b \in [0,1], q \in \mathcal{Q}_0} D_L(b; q)$	0.020065585
$\min_{b \in [0,1], q \in \mathcal{Q}_0} [\bar{d} - D_H(b; q)]$	0.166323464
$\min_{b \in [0,1], q \in \mathcal{Q}_0} Z(b; q)$	0.011915854
$\ell_L - C_L(0)$	0.165580252
$C_L(1) - u_L$	0.051934414
$\ell_H - C_H(0)$	0.152820206
$C_H(1) - u_H$	0.038429512
$\min_b [C_H(b) - C_L(b)]$	0.068872090
$\mathbb{E}[C_H(b) \mid P] - q_L$	0.085924816
$q_H - \mathbb{E}[C_L(b) \mid K]$	0.086482870
$C_L(1) - q_L$	0.055934414
$C_H(1) - q_H$	0.042429513
Root-box range of D_L	[0.024065585, 0.709022920]
Root-box range of $\tilde{D}_H$	[0.043981439, 0.765349727]
Root-box range of D_H	[0.063897294, 0.821676535]

Notes: The first eleven bounds hold on the entire price rectangle $\mathcal{Q}_0$. The state-ordering bound holds for all buyer values. Seller-response, D1 and cutoff bounds hold on the certified root box. The finite corner argument in Appendix B extends the four face checks to whole faces; the cutoff identities reduce interiority to endpoint checks. All displayed lower bounds are rounded down.

D Action Supports and Equilibrium Refinement

D.1 On-Path Prices and Action Support

Proof. Because the buyer does not observe s and s is independent of buyer type, an on-path price induces the same posterior distribution of b in either seller state. Because $C_L(b) < C_H(b)$ pointwise, the seller's posterior continuation payoff is therefore higher in state H . Pure acceptance decisions are consequently nested: an offer is rejected in both states, accepted only in L , or accepted in both. An offer rejected in both gives the buyer $-\kappa < 0$ relative to making no early offer. Finally, suppose $p < p'$ are accepted by exactly the same seller states, and let $a_s \in \{0, 1\}$ indicate acceptance in state s . Every buyer type gains

$$\left[(1 - \pi)a_L + \pi a_H \right] (p' - p) > 0$$

by replacing p' with p . Hence two buyer types cannot choose distinct prices that are accepted in the same seller states. This argument also covers continuous variation in the pure price rule across buyer types. If a lower and a higher price interval were each chosen by a positive mass of types under the same acceptance pattern, every type choosing from the higher interval could profitably switch to a lower on-path price without changing which seller states accept. The price is therefore constant almost surely across types conditional on its acceptance pattern. If a probing price q_L is used, seller L 's sequential rationality gives $q_L \geq \mathbb{E}[C_L(b) \mid P] \geq C_L(\underline{b})$. If a knockout price q_H is used, seller H 's sequential rationality gives $q_H \geq \mathbb{E}[C_H(b) \mid K] \geq C_H(\underline{b})$. Substituting these price bounds and using the definitions in (5.1) gives

$$(1 - \pi)[w_L(\underline{b}) - q_L] - \kappa \leq (1 - \pi)[w_L(\underline{b}) - C_L(\underline{b})] - \kappa = -(1 - \pi)\chi_L < 0,$$

and

$$\bar{w}(\underline{b}) - q_H - \kappa \leq \bar{w}(\underline{b}) - C_H(\underline{b}) - \kappa = -\chi_H < 0.$$

At the corner type $(\underline{b}, 0)$, waiting gives the normalized payoff zero, while the two displayed inequalities make probing and knockout strictly worse. Buyer payoffs are continuous in (b, d) , so the same strict ranking holds throughout some neighborhood of $(\underline{b}, 0)$. Because the maintained value and urgency distributions have positive densities, that neighborhood contains a positive mass of buyer types; hence W is used with positive probability. The price bound above permits at most one probing price and at most one knockout price. With W present, the only possible action supports are therefore $\{W\}$, $\{W, P\}$, $\{W, K\}$, and $\{W, P, K\}$. $\square$

D.2 Waiting–Probing Equilibria and D1

Lemma D.1 (Contingent-plan representation for criterion D1). *Suppose the buyer does not observe s , takes no action after the seller responds, and the distribution of s is independent of buyer type. After any price p , the one-offer game is outcome- and payoff-equivalent to a signaling game in which the seller chooses a contingent acceptance plan $a = (a_L, a_H) \in [0, 1]^2$. Criterion D1 in this representation is the strict-gain set-inclusion test used below.*

Proof. Let μ be the posterior over buyer type after p , and let $\lambda_s > 0$ be the prior probability of state s . The seller's expected payoff from plan a is

$$\sum_{s \in \{L, H\}} \lambda_s \left\{ a_s p + (1 - a_s) \int C_s(b) d\mu(b, d) \right\}.$$

This objective separates across states. A plan is therefore a best response if and only if each a_s is optimal at the corresponding seller information set (p, s) in the original game. The

buyer's expected payoff from the same acceptance probabilities is unchanged. Moreover, independence and the buyer's inability to condition its offer on s imply $\mu(\cdot \mid p, s) = \mu(\cdot \mid p)$: one buyer tremble generates the same posterior in both seller states.

The receiver responses in the mixed-best-response formulation of D1 are therefore precisely these rationalizable contingent plans. Formally, let $\mathcal{A}(p)$ be the union, over posteriors μ , of the seller's best-response plans after p . For buyer type $\theta = (b, d)$, let $U^*(\theta)$ denote its equilibrium payoff and let $u_\theta(p, a)$ denote its payoff from deviating to price p when the seller uses plan a . Define

$$\begin{aligned}\mathcal{D}(\theta, p) &= \{a \in \mathcal{A}(p) : u_\theta(p, a) > U^*(\theta)\}, \\ \mathcal{D}^0(\theta, p) &= \{a \in \mathcal{A}(p) : u_\theta(p, a) = U^*(\theta)\}.\end{aligned}$$

The strict-gain D1 rule deletes θ if another type θ' satisfies

$$\mathcal{D}(\theta, p) \cup \mathcal{D}^0(\theta, p) \subsetneq \mathcal{D}(\theta', p).$$

Thus every rationalizable plan that makes the deleted type weakly willing to deviate makes the dominating type strictly willing. This is the criterion D1 set-inclusion formulation used below for the stated contingent-plan representation; a different extensive-form representation may induce a different refinement test. $\square$

Proof of Proposition 5.2. For sufficiency of the no-early-offer condition, assign every off-path price to $b = \bar{b}$. The cheapest prices inducing L -only and both-state acceptance are then $C_L(\bar{b})$ and $C_H(\bar{b})$. Their payoffs are increasing in (b, d) , so the two inequalities in (5.13) deter every buyer. The same highest-type posterior survives D1 in the contingent-plan representation: under every rationalizable plan with positive acceptance probability, the gain from the unused price is weakly increasing in urgency and in the state-contingent auction willingness $w_s(b)$, while rejection in both states loses κ for every type. Hence no other type can D1-dominate $(\bar{b}, \bar{d})$.

For necessity, first suppose $\bar{d} > \bar{D}_H$. A price just above $C_H(\bar{b})$ is accepted in both states under every posterior and is profitable for types near $(\bar{b}, \bar{d})$. If instead $\bar{d} > \bar{D}_L$, a price just above $C_L(\bar{b})$ is accepted by seller L under every posterior and its L -only payoff is positive for the top type. If seller H also accepts under the induced posterior, this only raises that type's payoff: at the limiting price,

$$w_H(\bar{b}) + \bar{d} - C_L(\bar{b}) = w_H(\bar{b}) - w_L(\bar{b}) + \bar{d} + d_{S,L} > 0,$$

where $R_L(\bar{b}) = w_L(\bar{b})$. Thus violation of either inequality creates a profitable deviation even when continuation-value supports overlap.

Now impose (5.14). Its strict lower bound rules out no-early-offer PBE. On $[C_L(b), C_L(\bar{b})]$, the probing map is continuous wherever its denominator is positive. At

the lower endpoint its posterior mean is strictly above $C_L(b)$: indeed, $\bar{d} > \bar{D}_L$ and $C_L(\underline{b}) < C_L(\bar{b})$ imply

$$D_L(\bar{b}; C_L(\underline{b})) < \bar{D}_L < \bar{d},$$

so a positive interval of high values on which $C_L(b) > C_L(\underline{b})$ probes. At the upper endpoint, $D_L(\bar{b}; C_L(\bar{b})) = \bar{D}_L < \bar{d}$, so a positive interval of high- b types probes and the posterior mean is strictly below $C_L(\bar{b})$. The intermediate value theorem gives an interior fixed point q_L . At that point the cutoff $D_L(b; q_L)$ implements the stated buyer strategy, Bayes' rule makes seller L indifferent at q_L , and seller H rejects because, under the same probe posterior μ_B ,

$$\int C_H(b) d\mu_B(b) > \int C_L(b) d\mu_B(b) = q_L.$$

After prices in a sufficiently small left neighborhood of q_L , retain the on-path probing posterior. Seller L 's continuation value is then q_L , so an undercut is rejected by both seller states. Assign every remaining unused price to $b = \bar{b}$. A lower price is rejected, while a higher price that only seller L accepts is dominated by q_L . The cheapest price accepted in both states is $C_H(\bar{b})$. Relative to probing, its payoff satisfies

$$\begin{aligned} \bar{w}(b) + d - C_H(\bar{b}) - \kappa - \{(1 - \pi)[w_L(b) + d - q_L] - \kappa\} \\ = \pi[w_H(b) + d] + (1 - \pi)q_L - C_H(\bar{b}) \\ \leq \pi[w_H(\bar{b}) + \bar{d}] + (1 - \pi)C_L(\bar{b}) - C_H(\bar{b}) \leq 0, \end{aligned}$$

where the final inequality is equivalent to the upper bound on $\bar{d}$ in (5.14). Thus no knockout deviation gains, including for a type that waits, since a positive knockout payoff would imply a positive probing payoff.

The final condition in (5.14) gives $D_L(\underline{b}; q_L) > 0$, while $\bar{d} > \bar{D}_L > D_L(\bar{b}; q_L)$. Both waiting and probing therefore have positive probability. Since both continuation states have positive probability, probing generates both rejection and acceptance with positive probability.

It remains to verify criterion D1 in the contingent-plan representation for $\kappa > 0$. Pointwise state ordering and a common posterior imply that the union of rationalizable seller plans after any price is contained in

$$\mathcal{A}(p) \subseteq \{(x, 0) : x \in [0, 1]\} \cup \{(1, y) : y \in [0, 1]\}, \quad (\text{D.1})$$

where the coordinates are the (L, H) acceptance probabilities. For an undercut $p < q_L$, the wait-probe marginal type $(b, D_L(b; q_L))$ requires, on the first branch, the acceptance probability

$$x^*(p) = \frac{\kappa}{\kappa + (1 - \pi)(q_L - p)}, \quad (\text{D.2})$$

independently of b ; every nonmarginal type requires strictly more. Along the marginal

curve, the second branch weakly favors higher b because $w_H(b) - w_L(b)$ is weakly increasing. Hence the highest feasible marginal value

$$b_L^0 = \sup\{b \in [\underline{b}, \bar{b}] : D_L(b; q_L) \geq 0\}$$

is D1-permitted and has the largest continuation value among permitted marginal types. Belief on it deters every undercut exactly when $q_L \leq C_L(b_L^0)$; conversely, if $q_L > C_L(b_L^0)$, some $p \in (C_L(b_L^0), q_L)$ is accepted by seller L under every surviving posterior and benefits every marginal type, as in the argument for Proposition 8.4(iv).

To verify this inequality under (5.15), define

$$\Xi_L(b) = C_L(b) - w_L(b) + \frac{\kappa}{1 - \pi} = \Gamma_L(b) - d_{S,L} + \frac{\kappa}{1 - \pi}.$$

The competition wedge makes Ξ_L weakly decreasing. If $b_L^0 < \bar{b}$, then $D_L(b_L^0; q_L) = 0$, so

$$C_L(b_L^0) - q_L = \Xi_L(b_L^0) \geq \Xi_L(\bar{b}) = \bar{D}_L \geq 0.$$

If $b_L^0 = \bar{b}$, interiority of the fixed point directly gives $q_L < C_L(\bar{b})$. For an overcut $p > q_L$, no type gains on the first branch of (D.1), since it gives no more L -acceptance than the cheaper equilibrium probe. On the second branch, the required H -acceptance probability is minimized by the highest-value, highest-urgency probing type. Belief on $\bar{b}$ makes H reject below $C_H(\bar{b})$; at or above that price, the top-type inequality in (5.14) rules out a gain even with acceptance in both states. These beliefs cover every off-path price and give the claimed D1 completion. $\square$

Boundary Pricing and Probing-Price Multiplicity

Boundary pricing selects a convenient zero-rent member of a larger PBE and D1-compatible set. An exact example uses the late-participation-only uniform specialization

$$(n_L, n_H) = (2, 7), \quad \pi = \frac{1}{2}, \quad d_S = .01, \quad \kappa = .05, \quad \bar{d} = .15,$$

and the rate-10 truncated-exponential urgency distribution. Besides the boundary fixed point, consider

$$\tilde{q}_L = C_L(1) = \frac{197}{300}.$$

The top-value cutoff is $D_L(1; \tilde{q}_L) = 9/100 < \bar{d}$, while sufficiently low values wait, so both actions have positive mass. The probing pool is nondegenerate below $b = 1$; strict monotonicity of C_L therefore gives

$$\mathbb{E}[C_L(b) \mid \tilde{q}_L] < C_L(1) = \tilde{q}_L,$$

and seller L accepts strictly. Seller H rejects because $\tilde{q}_L < C_H(0) = 37/50$.

Assign every off-path price to $b = 1$. An undercut is rejected because it is below $C_L(1)$, and an accepted L -only overcut is more expensive than the equilibrium probe. The cheapest both-state offer accepted under every seller posterior is $C_H(1)$. For the top type, its payoff relative to probing is

$$V_H(1, \bar{d}; C_H(1)) - V_L(1, \bar{d}; \tilde{q}_L) = -\frac{29}{1200} < 0,$$

and the difference is weakly smaller for every other buyer type. The complete contingent-plan comparison permits the same endpoint belief: here $b_L^0 = 1$ and the exact D1 condition holds at equality,

$$\tilde{q}_L = C_L(b_L^0).$$

All remaining seller-response, participation, and deviation inequalities are strict. Continuity therefore yields a nondegenerate interval of waiting-probing, D1-compatible prices immediately below $\tilde{q}_L$, with seller L earning a positive rent throughout. This establishes local price multiplicity under D1; these prices are outside the locally passive selection in Assumption 2.

Full-Menu Refinement

When $\kappa > 0$, the full-menu PBE in Proposition 5.4 likewise admits a D1-compatible completion in the contingent-plan representation. Boundary pricing equates each on-path price to the corresponding seller's posterior continuation value. Global interiority gives positive probing and knockout mass at values arbitrarily close to $\bar{b}$; because C_L and C_H are strictly increasing on a top interval, their conditional means are strictly below their endpoint values. Thus $q_L < C_L(\bar{b})$ and $q_H < C_H(\bar{b})$. For $p < q_L$, the argument leading to (D.2) makes the highest-value wait-probe marginal type $(\bar{b}, D_L(\bar{b}))$ D1-permitted. Belief on that type makes seller L reject because $p < q_L < C_L(\bar{b})$, and pointwise ordering makes seller H reject as well.

For an intermediate price $q_L < p < q_H$, no type gains on the first branch of (D.1). On the second branch, the probe-knockout marginal type $(b, D_H(b))$ requires

$$y^*(p) = \frac{(1 - \pi)(p - q_L)}{\pi(r_0 - p)}, \quad r_0 = w_H(b) + D_H(b) = \frac{q_H - (1 - \pi)q_L}{\pi}, \quad (\text{D.3})$$

which is independent of b ; types on either side of the boundary require weakly more H -acceptance. The highest-value marginal type is therefore D1-permitted. Belief on $\bar{b}$ makes H reject because $p < q_H < C_H(\bar{b})$, while any L -acceptance cannot generate a gain at a price above q_L . At $p \geq q_H$, even acceptance in both states is weakly worse than the equilibrium knockout. These cases exhaust the nested response set. D1 therefore supports

the equilibrium without uniquely selecting it or making implementation belief-independent.

D.3 Waiting–Knockout Construction

The remaining one-price support is $\{W, K\}$: some buyers wait, while the others make a knockout offer accepted in both seller states. It produces attempts and completed pre-emption but no rejected attempts, so the seller state does not generate different on-path responses, although it still affects the equilibrium price and buyer cutoff.

The waiting–knockout cutoff and conditional action probability are given in (5.7).

Buyers with $d < \widetilde{D}_H(b; q)$ wait, while those with $d \geq \widetilde{D}_H(b; q)$ make the knockout offer; hence $m_K(b; q)$ is the probability of knockout conditional on b . Whenever the aggregate knockout probability is positive, seller H 's expected continuation value after observing the offer is

$$T_K(q) = \frac{\int_{\underline{b}}^{\bar{b}} C_H(b) m_K(b; q) f(b) db}{\int_{\underline{b}}^{\bar{b}} m_K(b; q) f(b) db}. \quad (\text{D.4})$$

The equality $q = T_K(q)$ is the waiting–knockout boundary-price condition: seller H is indifferent between accepting the knockout price and continuing to the bidding round. It is the price selected by Assumption 2 when knockout is the active accepted-offer category. Two further checks ensure that both actions occur and that no buyer prefers an unused probing offer. First,

$$0 < \widetilde{D}_H(\bar{b}; q) \quad \text{and} \quad \widetilde{D}_H(\bar{b}; q) < \bar{d}. \quad (\text{D.5})$$

Because the cutoff decreases with b , these inequalities leave fully patient, low-value buyers who wait and sufficiently urgent, high-value buyers who choose knockout. Second, for every buyer type (b, d) , require

$$(1 - \pi)[w_L(b) + d - C_L(\bar{b})] - \kappa \leq \max\{0, \bar{w}(b) + d - q - \kappa\}. \quad (\text{D.6})$$

Under the off-path belief $b = \bar{b}$, $C_L(\bar{b})$ is the cheapest price seller L would accept. The left side of (D.6) is the buyer's payoff from that cheapest probe; the right side is its best payoff from waiting or the on-path knockout. The inequality therefore keeps probing out of the buyer's optimal action set.

Local passivity adds a separate nearby-deviation check. An infinitesimal undercut inherits the knockout pool's posterior, so seller H rejects it while seller L accepts it; the undercut is therefore a probe rather than a cheaper knockout. A strict sufficient condition for deterring these local probes is

$$(1 - \pi)[w_L(b) + d - q] - \kappa < \max\{0, \bar{w}(b) + d - q - \kappa\} \quad \text{for every } (b, d). \quad (\text{D.7})$$

At a fixed point satisfying (D.5)–(D.7), Bayes' rule makes seller H indifferent at q , and

state ordering makes seller L accept. After sufficiently small undercuts, assign the on-path knockout posterior; the strict inequality (D.7) deters the resulting local probes. At all remaining unused prices, assign $b = \bar{b}$. Any unused offer accepted in both states must then cost at least $C_H(\bar{b}) \geq q$, while (D.6) deters an offer accepted only in L . Buyers wait below $\widetilde{D}_H(b; q)$ and offer q above it. These strategies and beliefs form a locally passive PBE with support $\{W, K\}$.

The fixed point exists under simple endpoint conditions. For every $q \in [C_H(b), C_H(\bar{b})]$, $T_K(q)$ is a weighted average of $C_H(b)$ and therefore lies in the same interval. Moreover, $\chi_H > 0$ gives $\widetilde{D}_H(b; C_H(b)) > 0$, while $C_H(\bar{b}) - \bar{w}(\bar{b}) + \kappa < \bar{d}$ gives $\widetilde{D}_H(\bar{b}; C_H(\bar{b})) < \bar{d}$. These endpoint inequalities imply positive waiting and knockout probabilities throughout the interval, so T_K is continuous and has a fixed point. A root implements the selected $\{W, K\}$ support when it also satisfies (D.6) and (D.7). Unlike the $\{W, P\}$ result, these additional checks do not reduce to a single interval for $\bar{d}$: they must deter both distant and infinitesimal probing offers for every buyer type. For example, holding $\bar{d}$ and the waiting–knockout fixed point fixed, a lower continuation value for seller L can make an unused probing offer attractive to intermediate buyer types and invalidate the $\{W, K\}$ support.

Pre-emption can therefore be driven by gains from certain early resolution without any on-path offer being used to screen the seller state. Whether this rejection-free outcome survives criterion D1 depends on what the state changes. When the state changes late-participation prospects, it also changes the buyer’s continuation payoff, and D1 rules out the $\{W, K\}$ equilibrium. When the state changes only the seller’s fallback or urgency, buyer willingness may be locally insensitive to that information, allowing the equilibrium to survive under the conditions in Section 6.

E Proof of Proposition 6.1

Let q_H be the knockout price and define the wait–knockout cutoff

$$D_K(b) = q_H - \bar{w}(b) + \kappa, \quad \mathcal{B}_K = \{b : 0 \leq D_K(b) \leq \bar{d}\}, \quad \beta_K = \inf \mathcal{B}_K.$$

Positive waiting and knockout mass, continuity, and full interval urgency support make this marginal set nonempty, and positive knockout mass gives $\beta_K < \bar{b}$. Write $A = 1 - \pi$, $\Delta(b) = w_H(b) - w_L(b)$, and consider an undercut $p = q_H - \delta$, where $\delta > 0$. By the common-posterior ordering, every rationalizable seller plan $a = (a_L, a_H)$ is nested and belongs to

$$\mathcal{A}(p) \subseteq \{(x, 0) : x \in [0, 1]\} \cup \{(1, y) : y \in [0, 1]\}.$$

Let $\rho(a) = Aa_L + \pi a_H$. At the hypothetical wait-knockout marginal type $(b, D_K(b))$, the deviation gain is

$$m_b(a) = \rho(a)(\delta + \kappa) - A\pi(a_L - a_H)\Delta(b) - \kappa. \quad (\text{E.1})$$

For a type with urgency $d = D_K(b) + z$, the equilibrium payoff is $\max\{z, 0\}$, so its deviation gain is exactly

$$h_{b,z}(a) = m_b(a) + \rho(a)z - \max\{z, 0\}. \quad (\text{E.2})$$

Under the aligned composite specification, equation (4.6) gives

$$\Delta(b) = 0 \quad \text{for } b \leq v_L, \quad \Delta'(b) = \begin{cases} \phi_L(F(b)) > 0, & v_L < b < v_H, \\ \phi_L(F(b)) - \phi_H(F(b)) > 0, & b > v_H. \end{cases}$$

In the late-participation-only specialization $v_L = v_H = v_0$, the same expressions hold with the middle branch empty, and the argument below applies verbatim. Define

$$\mathcal{B}_K^* = \begin{cases} \{\beta_K\}, & \beta_K \geq v_L, \\ \mathcal{B}_K \cap [\beta_K, v_L], & \beta_K < v_L. \end{cases}$$

For any $b_* \in \mathcal{B}_K^*$, every $b \geq \beta_K$, and every nested plan,

$$m_{b_*}(a) - m_b(a) = A\pi(a_L - a_H)[\Delta(b) - \Delta(b_*)] \geq 0. \quad (\text{E.3})$$

The inequality is strict at a partial plan whenever $b > v_L$ and $b_* \leq v_L$, or whenever $b > b_* \geq v_L$. Marginal types in $\mathcal{B}_K^*$ have identical gain sets when $\beta_K < v_L$. At full acceptance all marginal types gain the same $\delta > 0$. Moreover, $\rho z - \max\{z, 0\} \leq 0$, with strict inequality whenever a nonmarginal type could otherwise tie at a payoff-relevant partial plan. Thus the marginal type at a given b is weakly most willing.

If $b < \beta_K$, then $D_K(b) > \bar{d}$. When $\beta_K > \underline{b}$, continuity gives $D_K(\beta_K) = \bar{d}$, and the deviation payoff is strictly increasing in both b and d whenever $\rho > 0$; rejection alone yields $-\kappa < 0$. Hence the same strict comparison holds below β_K . If $\beta_K = \underline{b}$, that region is empty. Equations (E.2)–(E.3) therefore leave the single marginal type $(\beta_K, D_K(\beta_K))$ D1-undominated when $\beta_K \geq v_L$. When $\beta_K < v_L$, the D1-undominated set is

$$\Theta_K^* = \{(b, D_K(b)) : b \in \mathcal{B}_K^*\};$$

these plateau types tie rather than delete one another. On the branch $(1, y)$, their common marginal gain is zero at

$$y_K^* = 1 - \frac{\delta}{\pi\{\delta + \kappa + A\Delta(\beta_K)\}} \in (0, 1)$$

for δ sufficiently small, while no type outside the surviving set gains strictly there. Hence no type in the set is itself deleted.

Positive knockout mass and strict increase of $\bar{w}$ give the on-path pool positive probability above $\max\{\beta_K, v_H\}$. Since C_H is constant below v_H and strictly increasing above it, seller H 's on-path optimality gives

$$C_H(\beta_K) < \mathbb{E}[C_H(b) \mid q_H] \leq q_H.$$

Moreover, $q_H \leq C_H(\bar{b})$: otherwise a slightly lower price above $C_H(\bar{b})$ would be accepted in both states under every posterior and would strictly dominate q_H for its users. Because $C_H(\beta_K) > C_L(b) > 0$, choose a positive $p \in (C_H(\beta_K), q_H)$ sufficiently close to q_H . Continuity and strict increase above v_H give a unique $b_p > v_H$ satisfying $C_H(b_p) = p$. A posterior concentrated on b_p makes seller H indifferent and seller L strictly willing to accept, so the whole branch $(1, y)$ is rationalizable. Strict-gain D1 restricts the buyer posterior to the singleton marginal type if $\beta_K \geq v_L$, and to Θ_K^* if $\beta_K < v_L$. In the latter case C_H has the same value $C_H(\beta_K)$ at every permitted type. Every permitted posterior therefore makes both sellers accept p , and every surviving marginal buyer saves $q_H - p > 0$. The $\{W, K\}$ assessment fails D1 in the contingent-plan representation.

Fallback-only boundary. The exclusion is false without late-participation variation. Let $F = U[0, 1]$, let both states have exactly two late buyers, $(v_L, v_H) = (0.2, 0.4)$, $\pi = 1/2$, $d_S = 0.01$, $\kappa = 0.05$, and $d \sim U[0, 0.15]$. The boundary knockout map has a numerically bracketed fixed point $q_H = 0.654466963$, with lowest marginal value $\beta_K = 0.623111832 > v_H$ and knockout mass 0.204305347. For every $b \geq v_H$, $w_H(b) - w_L(b) = 7/375$, so all feasible marginal types have the same D1 gain set. Belief on the top marginal type makes both sellers reject every undercut because $C_L(1) = 989/1500 > q_H$; no overcut is profitable. Hence this fallback-only knockout equilibrium has positive waiting and knockout mass and admits a D1-compatible completion in the contingent-plan representation.

Parametric diagnostic. Specialize to the late-participation-only case $v_L = v_H = 0$. In the uniform benchmark $b \sim U[0, 1]$, $(n_L, n_H) = (2, 7)$, $\pi = 1/2$, and $d_S = 0.01$, the restrictions

$$\bar{d} > \frac{319}{600}, \quad \frac{7}{16} < \kappa < \bar{d} - \frac{113}{1200}$$

support a nondegenerate boundary-price PBE with support $\{W, K\}$ before refinement. Indeed, the knockout posterior map continuously maps $[C_H(0), C_H(1)]$ into itself and the upper inequality gives positive knockout mass. The lower inequality also gives

$$\max_b Z(b; C_L(1), q_H) \leq C_H(1) - C_L(1) - \kappa = \frac{5}{24} - \kappa < 0,$$

so the top-value off-path belief deters probing. The proposition then eliminates this raw

equilibrium. This appendix calculation is only a diagnostic; the general result does not cover mixed acceptance or an all-knockout assessment.

F Proof of Proposition 6.2

Maintain the urgency-only specification: common (Q, v) , hence a common w , R , and wedge $\Gamma = R - w$, with $C_s = R - d_{S,s}$ and $d_{S,L} > d_{S,H} \geq 0$. Fix a pure PBE with support $\{W, K\}$ and price q_H , so a buyer attempts exactly when $w(b) + d \geq q_H + \kappa$ and both seller states accept q_H . Note first that the first survival condition is equivalent to

$$C_L(b_K^0) \geq q_H - \frac{\pi\kappa}{1-\pi}, \quad (\text{F.1})$$

since $C_L = C_H - (d_{S,L} - d_{S,H})$ pointwise; when $b_K^0 < \bar{b}$, the marginal identity $w(b_K^0) = q_H + \kappa$ gives $C_s(b_K^0) = q_H + \kappa + \Gamma(b_K^0) - d_{S,s}$, so (F.1) reads $d_{S,L} \leq \Gamma(b_K^0) + \kappa/(1-\pi)$, and $\Delta_D \geq 0$ then delivers the second condition because $d_{S,H} \leq d_{S,L} - \pi\kappa/(1-\pi) \leq \Gamma(b_K^0) + \kappa$.

Scalar reduction and the tied curve. Because $w_H \equiv w_L$, a contingent plan $a = (a_L, a_H)$ matters to a deviating buyer only through its acceptance weight $\rho = (1-\pi)a_L + \pi a_H \in [0, 1]$: a type with scalar $z = w(b) + d$ obtains $\rho(z - p) - \kappa$ from deviating to p , against the equilibrium value $U^*(z) = \max\{0, z - q_H - \kappa\}$. For $p < q_H$ the weak-gain set is $\{\rho \geq \tau(z; p)\}$ with $\tau(z; p) = [\kappa + U^*(z)]/(z - p)$, strictly decreasing in z below $q_H + \kappa$ and strictly increasing above it. The deviation-eager types after every undercut therefore form the tied curve $\Theta_K = \{(b, d) : w(b) + d = q_H + \kappa\}$, whose value range is $[\beta_K, b_K^0]$ with $b_K^0 = \min\{\bar{b}, w^{-1}(q_H + \kappa)\}$. All curve types have identical gain sets, and whenever some rationalizable plan gives the curve types a strict gain, the strict-gain comparison deletes every off-curve type. On the $(x, 0)$ branch this bite occurs exactly when $p < q_H - \pi\kappa/(1-\pi)$, so that $\tau(q_H + \kappa; p) < 1 - \pi$; on the $(1, y)$ branch it extends to every undercut $p \in [C_H(b), q_H)$, where a posterior making the patient state indifferent rationalizes an interior y . D1-permitted posteriors after an undercut are therefore the distributions on Θ_K , and $\mathbb{E}[C_s(b)]$ over permitted posteriors is maximized at the singleton on the top curve type $(b_K^0, D_K(b_K^0))$.

Survival. Suppose both survival conditions hold. Place the off-path belief on $(b_K^0, D_K(b_K^0))$ at every $p \neq q_H$ and let each seller respond sequentially rationally, accepting at equality. For $p < C_L(b_K^0)$, both states reject and every type's gain is at most $-\kappa < 0$. For $p \in [C_L(b_K^0), q_H)$, the urgent state accepts and the patient state rejects, since $p < q_H \leq C_H(b_K^0)$ by the second condition; thus $\rho = 1 - \pi$, and the supremum over all types of $(1 - \pi)(z - p) - \kappa - U^*(z)$ is attained on the curve, where it equals

$$(1 - \pi)(q_H + \kappa - p) - \kappa \leq (1 - \pi)(q_H + \kappa - C_L(b_K^0)) - \kappa \leq 0$$

by (F.1). For $p > q_H$, every type's gain is negative under every plan, so gain sets are empty and D1 places no restriction. The belief is D1-permitted at every off-path price and no deviation strictly gains: the equilibrium survives. At equality in (F.1) the supremum is zero, weak rather than strict, so the strict-gain criterion still passes; the survival region is closed.

Failure. Suppose (F.1) fails. Pick $p \in (C_L(b_K^0), q_H - \pi\kappa/(1 - \pi))$, a nonempty interval. Deletion has bite and leaves exactly Θ_K , so under every permitted posterior $\mathbb{E}[C_L(b)] \leq C_L(b_K^0) < p$: the urgent state must accept, giving $\rho \geq 1 - \pi$, and every curve type gains at least $(1 - \pi)(q_H + \kappa - p) - \kappa > 0$, strictly. If instead the second condition fails, $q_H > C_H(b_K^0)$, any $p \in (C_H(b_K^0), q_H)$ forces acceptance by both states under every permitted posterior, and every knockout type gains $q_H - p > 0$. In either case every D1-consistent completion admits a profitable deviation, so the assessment fails strict-gain D1. $\square$

Numerical instance and boundary. At the primitives of Example 6.4, the boundary fixed point for support $\{W, K\}$, $q_H = \mathbb{E}[C_H(b) \mid \text{attempt}]$, equals .637358061, with waiting mass .615537 and knockout mass .384463; $b_K^0 = .790989$ and $\Gamma(b_K^0) = .003044$, so both survival conditions hold with margins .003044 and .013044, and this refined equilibrium coexists with the Example's full-menu equilibrium. Because the fixed point does not involve $d_{S,L}$, the survival boundary there is the vertical cut $d_{S,L}^* = \Gamma(b_K^0) + \kappa/(1 - \pi) = .023044$: certified failures occur at $d_{S,L} \in \{.025, .03, .05\}$, while raw $\{W, K\}$ PBEs exist up to $d_{S,L} = .060642$, so on $(.023044, .060642]$ the exclusion is driven by criterion D1 exactly as in the aligned case. Raising the offer cost to $\kappa = .05$ yields a variant with $D_K(\bar{b}) \geq 0$ and $C_L(\bar{b}) > q_H = .643707$ (numerically close to, but distinct from, the Example's full-menu price .643708), in which the belief on the top-value marginal type makes both states reject every undercut—the exact analogue of the fallback-only construction above. Because $R' \leq w'$ above v , the first-condition residual $q_H - \pi\kappa/(1 - \pi) - C_L(b_K^0(q_H))$ is nondecreasing in q_H , so survival of any waiting-knockout price is decided at the lowest, boundary-priced fixed point, and the D1-surviving price set is an interval. An outward-rounded ball-arithmetic calculation applies the Krawczyk theorem to both boundary fixed points—for $\kappa = .01$ in the marginal coordinate b_K^0 , and for $\kappa = .05$ directly in the price—and proves existence and local uniqueness with root enclosures of radius below 2×10^{-12} . The same calculation certifies strict signs, with enclosure radii below 10^{-11} , for both survival-condition margins, the survival boundary, the waiting and knockout masses, the partial-acceptance deterrence bound, the killer-interval widths and raw-persistence margins at $d_{S,L} \in \{.025, .03, .05\}$, and the corner and full-rejection margins of the $\kappa = .05$ variant. Every strategic margin used is certified strict except the intended boundary equalities.

G Disclosure Proofs

Proof of Proposition 8.4. Under no-early-offer behavior, $C_s(\bar{b})$ is the lowest price accepted under every posterior. Because its payoff increases in (b, d) , it is unprofitable for all buyers exactly when it is unprofitable at $(\bar{b}, \bar{d})$, yielding (8.15). If the inequality holds, assign lower off-path prices to $b = \bar{b}$; this D1-permitted belief induces rejection. If it fails, types near $(\bar{b}, \bar{d})$ gain from $C_s(\bar{b})$ under every posterior.

If the inequality fails, offer mass is positive at the upper endpoint of $[C_s(\underline{b}), C_s(\bar{b})]$. At $q = C_s(\underline{b})$, buyers in an upper-value set of positive measure submit an offer, and $C_s(b) > C_s(\underline{b})$ on a positive-measure subset because C_s is strictly increasing on a top interval. Hence $\hat{T}_s(C_s(\underline{b})) > C_s(\underline{b})$. At $q = C_s(\bar{b})$, the posterior mean lies below the upper endpoint, so the intermediate value theorem gives an interior fixed point. Its cutoff is optimal for the buyer, and Bayes' rule makes the seller indifferent at $\hat{q}_s$. Assign every other price to $b = \bar{b}$: undercuts are rejected, while accepted deviations cost at least $C_s(\bar{b}) > \hat{q}_s$, so none is profitable.

With public s , a surely rejected offer is dominated by waiting, proving (iii). For (iv), marginal type $(b, \widehat{D}_s(b; \hat{q}_s))$ needs acceptance probability $\kappa/(\kappa + \hat{q}_s - p)$ for an undercut $p < \hat{q}_s$. Types with cutoffs outside $[0, \bar{d}]$ need more and fail the strict-gain test, so the largest surviving continuation value is $C_s(b_s^0)$. Under (8.17), belief on this type rejects every undercut. Otherwise, some $p \in (C_s(b_s^0), \hat{q}_s)$ is accepted under every surviving posterior and benefits all marginal types. Under the sufficient condition $\widehat{D}_s(\bar{b}; \hat{q}_s) \geq 0$, positive offer mass also implies $\widehat{D}_s(\bar{b}; \hat{q}_s) < \bar{d}$. Hence $\bar{b}$ is a marginal value, $b_s^0 = \bar{b}$, and the interior fixed point satisfies $\hat{q}_s < C_s(\bar{b})$. $\square$

G.1 Voluntary Hard Evidence

Proof of Proposition 8.6. Normalize each seller's terminal-auction payoff to zero. Under the selected opaque continuation after pooled silence, seller L earns $K_L^D \geq 0$ and seller H earns zero. Under a correctly matched state-specific continuation, either seller earns zero before paying the certification cost. An H -seller mistaken for L can reject the induced offer and obtain zero; let $J_{LH} \geq 0$ denote an L -seller's payoff from being mistaken for H .

Pooling on silence is a PBE for every $\delta \geq 0$: relative to disclosure, the two seller states gain $K_L^D + \delta$ and δ , respectively. For $\delta > 0$, any profile in which H certifies gives that type $-\delta$, while silence followed by rejection guarantees zero. If only L certifies, it obtains $-\delta$ rather than the nonnegative imitation payoff J_{LH} . Full disclosure of the seller state likewise gives each state $-\delta$, while off-path silence and rejection guarantee zero. Hence no pure reporting profile involving disclosure is a PBE when $\delta > 0$, proving uniqueness within the named continuation class. $\square$

At $\delta = 0$, the inequalities become weak. Besides pooling on silence, H -only disclosure

is a PBE. L -only disclosure is supportable if and only if $J_{LH} = 0$, and disclosure by both seller states is supportable when off-path silence is assigned to L and invokes the corresponding state-specific continuation. Reporting-stage D1 and the intuitive criterion do not select between silence pooling and H -only disclosure: each certificate has a singleton feasible-type set, and H 's deviation to silence is an on-path indifference. This appendix classification is conditional on the named offer continuations and does not establish refinement of the reporting profile with disclosure by both seller states.

H Pre-emption-Stage Bargaining

The baseline makes the buyer's early offer final: rejection leads directly to the planned auction. Suppose instead that either party may make another binding proposal, at nonnegative proposal costs c_B and c_S , while either party may always end bargaining and activate the unchanged auction. If continued bargaining is valuable to urgent buyers, patient high-value buyers may imitate them and share the resulting information rent. Renegotiation can therefore change both whether early agreement occurs and which buyer types initiate. Formally, if $\Omega_s(p) \geq 0$ is seller type s 's incremental value of retaining this option after rejecting p , its effective rejection value is

$$\mathcal{R}_s(p) = \mathcal{C}_s(p) + \Omega_s(p), \quad (\text{H.1})$$

$$\mathcal{R}_L(p) \leq p < \mathcal{R}_H(p). \quad (\text{H.2})$$

The second line is necessary for the initial offer to separate the seller states. Full equilibrium additionally requires jointly sequentially rational proposals, exit decisions, posteriors, and continuation play, as in the bargaining literature with two-sided incomplete information (Cramton, 1984; Rubinstein, 1985; Grossman and Perry, 1986; Chatterjee and Samuelson, 1987; Fudenberg et al., 1987; Chatterjee and Samuelson, 1988; Cramton, 1992; Watson, 1998); Ausubel et al. (2002) survey this literature.

Proposition H.1 (Urgency and pooling under continued pre-emption-stage bargaining). *Consider the unrestricted alternating-offer extension above, with an initial attempt cost $\kappa > 0$.*

- (i) *Under a common prior and risk neutrality, if $d_B = d_{S,s} = 0$ almost surely, the auction remains available after every proposal, and perpetual bargaining cannot raise joint material payoff above the auction, every PBE has zero early-attempt probability.*
- (ii) *In the late-participation-only specialization with common $U[0, 1]$ private values, there are strictly positive proposal costs and an atom-continuous buyer-urgency distribution for which an unrestricted-game PBE has no early offer, an attempt followed by the*

auction, and early agreement, each with positive probability. A positive mass of buyers with $d_B = 0$ initiates. No round cap is imposed, although the selected path ends after at most three proposals.

The proof below specifies strategies and beliefs after every finite history. Urgent buyers create gains from early resolution, while patient high-value buyers pool with them and earn information rents through the revision path. Commitment to proceed directly to the bidding round after rejection removes this rent. Pooling can reduce allocative efficiency, but urgent types obtain an intrinsic benefit from early resolution, leaving the overall welfare effect ambiguous.

H.1 Proof of Proposition H.1

For part (ii), specialize to $v_L = v_H = 0$, deterministic late counts $(n_L, n_H) = (2, 5)$, $d_S = 0$, and $\pi = \Pr(s = H) = 3/10$. Let b and the i.i.d. sequence $(Y_i)_{i \geq 1}$ be mutually independent and uniform on $[0, 1]$; the first n_s late buyers participate in state s . Independently draw buyer urgency from

$$G_\eta = \eta\delta_0 + (1 - \eta)U[0, 1]. \quad (\text{H.3})$$

Set

$$\kappa = \frac{1}{2000}, \quad c_B = \frac{1}{1000}, \quad c_S = \frac{1}{500}, \quad \beta = \frac{9}{10},$$

where β is the patient buyer's initiation cutoff, and use initial offer $q = 1/2$, seller counteroffer $r = 3797/5000$, and buyer revision $p = 573/875$. The construction applies for every

$$0 \leq \eta < \eta^\dagger, \quad \eta^\dagger := \frac{117204341}{120480341}. \quad (\text{H.4})$$

Proof. For part (i), let I be the event that the buyer makes a binding initial offer. Fix the realized private types and bargaining history while integrating over terminal-auction randomness. Let $K_B + K_S \geq \kappa$ be the total proposal costs on an attempted history, and define gains relative to the planned auction by $g_B = U_B - u_s(b)$ and $g_S = U_S - C_s(b)$. If bargaining fails, then

$$g_B + g_S = -(K_B + K_S).$$

If agreement occurs, the price cancels between the parties and

$$g_B + g_S = b - u_s(b) - C_s(b) - (K_B + K_S) = -\Gamma_s(b) - (K_B + K_S).$$

The assumed completion gives the same upper bound on an infinite history. Hence every attempted history satisfies $g_B + g_S \leq -\kappa$.

At the buyer's initial information set, waiting gives its conditional expected auction payoff. Initial optimality therefore implies $\mathbb{E}[g_B \mathbf{1}_I] \geq 0$. At the seller's first response,

condition on its full information, including its type, the observed proposal, and the preceding contact history. The seller can invoke the unchanged auction before incurring any voluntary continuation cost. Under regular conditional beliefs, defined almost everywhere and updated by Bayes' rule on reached histories, seller optimality and iterated expectations imply $\mathbb{E}[g_S \mathbf{1}_I] \geq 0$. Because both constraints use the common prior,

$$0 \leq \mathbb{E}[(g_B + g_S) \mathbf{1}_I] \leq -\kappa \Pr(I),$$

so $\Pr(I) = 0$.

For part (ii), use the mutually independent construction above. Here $1 - \pi = 7/10$, and Equation (7.2) gives $w_s = w_{n_s,0}$ and $C_s = R_{n_s,0}$. Since $w_L(\beta) = 657/1000$, the stated prices satisfy

$$p = w_L(\beta) - \frac{c_B + \kappa}{1 - \pi} = \frac{573}{875}, \quad r = (1 - \pi)w_L(\beta) - \kappa + \pi = \frac{3797}{5000}, \quad q = \frac{1}{2} < p < r.$$

After the seller counteroffer r , termination, revision to p , and acceptance of r yield, respectively,

$$0, \quad P(b, d) = (1 - \pi)[w_L(b) + d - p] - c_B, \quad E(b, d) = (1 - \pi)w_L(b) + \pi w_H(b) + d - r.$$

Define

$$a(b) = p + \frac{c_B}{1 - \pi} - w_L(b), \quad x(b) = a(b) + \frac{\kappa}{1 - \pi} = w_L(\beta) - w_L(b),$$

$$y(b) = \frac{r - (1 - \pi)p - c_B - \pi w_H(b)}{\pi} = 1 - w_H(b).$$

The buyer waits initially when $d < \max\{x(b), 0\}$ and offers q otherwise, with initiation prescribed at the null tie $(\beta, 0)$. Every type that reaches r , including types arriving only through a tremble, follows the complete strategy

$$\begin{cases} \text{terminate,} & d < a(b), \\ \text{revise to } p, & a(b) \leq d < y(b), \\ \text{accept } r, & d \geq y(b). \end{cases}$$

Indeed, a equates revision and termination, while y equates acceptance and revision. Moreover,

$$\frac{d}{db}[y(b) - x(b)] = b^5 - b^2 < 0, \quad y(\beta) > x(\beta) = 0.$$

Thus $x(b) < y(b)$ below β , while $a(b) < x(b) < 0 < y(b)$ above it. Every on-path initiator has $d \geq \max\{x(b), 0\} > a(b)$, so termination is off path. A patient type strictly waits below β and strictly initiates and revises above it.

For the continuous urgency component, waiting, revision, acceptance of r , and initiation have conditional weights

$$\begin{aligned} e_0(b) &= \max\{x(b), 0\}, & m_0(b) &= y(b) - \max\{x(b), 0\}, \\ h_0(b) &= 1 - y(b), & z_0(b) &= 1 - \max\{x(b), 0\}. \end{aligned}$$

The patient atom waits below β , revises above it, and never accepts r . Hence

$$\begin{aligned} e_\eta(b) &= (1 - \eta)e_0(b) + \eta \mathbf{1}\{b < \beta\}, & m_\eta(b) &= (1 - \eta)m_0(b) + \eta \mathbf{1}\{b \geq \beta\}, \\ h_\eta(b) &= (1 - \eta)h_0(b), & z_\eta(b) &= (1 - \eta)z_0(b) + \eta \mathbf{1}\{b \geq \beta\}. \end{aligned}$$

Integrating over $b \sim U[0, 1]$ gives

$$\begin{aligned} Z_e &= \frac{9639 + 26361\eta}{40000}, & Z_p &= \frac{237581 - 153581\eta}{840000}, \\ Z_h &= (1 - \eta)\frac{10}{21}, & Z_q &= \frac{30361 - 26361\eta}{40000}, \end{aligned}$$

where $Z_e + Z_p + Z_h = 1$ and $Z_p + Z_h = Z_q$. Bayes' rule makes the post- q density z_η/Z_q and the post- p density m_η/Z_p . For $s \in \{L, H\}$, define the corresponding unnormalized auction values

$$N_{sq} = \int_0^1 z_\eta(b) C_s(b) db, \quad N_{sp} = \int_0^1 m_\eta(b) C_s(b) db.$$

After q , both seller states counter at r ; after p , seller L accepts and seller H invokes the auction. Before subtracting c_S , the counteroffer payoffs are $V_L = pZ_p + rZ_h$ and $V_H = N_{Hp} + rZ_h$. The complete on-path seller inequalities are

$$V_s > (q + c_S)Z_q, \quad V_s > c_S Z_q + N_{sq} \quad (s = L, H),$$

$$pZ_p > N_{Lp}, \quad N_{Hp} > pZ_p.$$

Exact integration gives the six unnormalized margins

$L : r \text{ versus } q$	$(97493167 - 88505167\eta)/588000000$
$L : r \text{ versus auction}$	$(209471659 - 211036159\eta)/1470000000$
$H : r \text{ versus } q$	$(734314608737 - 601422352487\eta)/409500000000$
$H : r \text{ versus auction}$	$(117204341 - 120480341\eta)/16380000000$
$L : p \text{ versus auction}$	$(266643493 - 271217293\eta)/5292000000$
$H : \text{auction versus } p$	$(387410369909 + 104670423841\eta)/2866500000000$

The fourth margin is the first to reach zero, at $\eta = \eta^\dagger$ in (H.4); every other margin remains strictly positive there. The stated seller actions are therefore strict throughout the interval

in the proposition. At equality the same path is a weak PBE if seller H counters at r when indifferent; above it, H strictly prefers the auction after q .

It remains to complete later histories. Write $\bar{C}_{sq} = N_{sq}/Z_q$ and $\bar{C}_{sp} = N_{sp}/Z_p$. For a preceding on-path buyer offer $j \in \{q, p\}$, let

$$\varepsilon = \frac{c_S - c_B}{2} = \frac{1}{2000}, \quad \tau_j = \bar{C}_{Lj} + \varepsilon, \quad T_j = \tau_j + c_B.$$

After an unexpected seller price ζ , the buyer believes state L . Writing $v = w_L(b) + d$, a type with $v > T_j$ accepts when $\zeta \leq T_j$ and offers τ_j otherwise; a type with $v \leq T_j$ accepts when $\zeta \leq v$ and terminates otherwise. These rules compare $v - \zeta$, $v - T_j$, and zero and therefore define a best response for every actual buyer type.

Every type in either on-path posterior has $v \geq w_L(\beta) > T_j$. Following a high seller price, all such types pool on τ_j , and the selected assessment retains the preceding posterior. Seller L strictly accepts by ε , while seller H strictly invokes the auction. The unnormalized versions of

$$w_L(\beta) - T_j, \quad \bar{C}_{Hj} - \tau_j, \quad C_L(1) - \tau_j$$

are affine in η and positive at both zero and $\eta^\dagger$; at the latter endpoint their smallest values across $j \in \{q, p\}$ are, respectively, 0.005684921, 0.167132113, and 0.016351588. Moreover, a seller price at or below T_j yields at most

$$T_j - c_S = \bar{C}_{Lj} + \varepsilon + c_B - c_S < \bar{C}_{Lj}.$$

Above T_j , seller L receives $\tau_j - c_S < \bar{C}_{Lj}$, while seller H invokes the auction and receives $\bar{C}_{Hj} - c_S < \bar{C}_{Hj}$. No further seller counteroffer therefore beats the posterior auction value.

At every nonspecial buyer price, use the off-path belief $\delta_{(1,1)}$. The seller's acceptance thresholds are then $C_L(1) = 2/3$ and $C_H(1) = 5/6$. Apply the same response rule to every actual buyer type with

$$\tau_* = C_L(1), \quad T_* = C_L(1) + c_B.$$

The believed type can use this tail with payoff margin 999/1000. Seller L accepts τ_* while indifferent to the auction, seller H strictly auctions, and, because $c_S > c_B$, every further seller counteroffer is strictly worse. The four binding global buyer-price comparisons are

$$(1 - \pi)[C_L(1) - p] - c_B = \frac{109}{15000}, \quad C_H(1) - r = \frac{1109}{15000},$$

$$(1 - \pi)[C_L(1) - p] = \frac{31}{3750}, \quad C_H(1) + c_B - r = \frac{281}{3750},$$

and are strictly positive. They cover initial and revised offers accepted only in state L or in both seller states; rejected prices are dominated by not making the proposal. Together with the recursive tail, these inequalities exhaust buyer deviations at every continuation

history.

For a Bayes-limit check, preserve atoms on the prescribed q - and p -history tails and add vanishing full-support price densities. At a nonspecial buyer offer, choose type-dependent trembles whose likelihood concentrates on $\{(b, d) : b > 1 - \delta_n, d > 1 - \delta_n\}$, $\delta_n \downarrow 0$, while every tremble probability vanishes. The seller's posterior then converges to $\delta_{(1,1)}$. At every finite history h followed by an unexpected seller price ζ , choose positive full-support subdensities $\lambda_{L,h}^n(\zeta)$ and $\lambda_{H,h}^n(\zeta)$ with vanishing total mass and relative rates such that the full L/H likelihood ratio, including the likelihood of h , diverges pointwise in ζ . They depend on the state and public history but not on the buyer's hidden type, and therefore leave the seller's conditional posterior over buyer types unchanged. Apply the construction recursively after every longer finite history. The buyer's posterior then converges to state L after every unexpected seller price. This supplies one atom-preserving, full-support regularization delivering the stated posteriors; it does not establish sequential equilibrium or dynamic D1 refinement.

Finally, for $\eta > 0$, patient buyers initiate with mass $\eta/10$. Their failure and agreement masses are $3\eta/100$ and $7\eta/100$. The total outcome probabilities are

$$\Pr(\text{no early offer}) = \frac{9639 + 26361\eta}{40000},$$

$$\Pr(\text{attempt then auction}) = \frac{237581 - 153581\eta}{2800000},$$

$$\Pr(\text{early agreement}) = \frac{1887689 - 1691689\eta}{2800000}.$$

All three are positive throughout (H.4). The buyer strategies, seller inequalities, recursive completion, and Bayes' rule thus define the claimed PBE, completing part (ii). $\square$

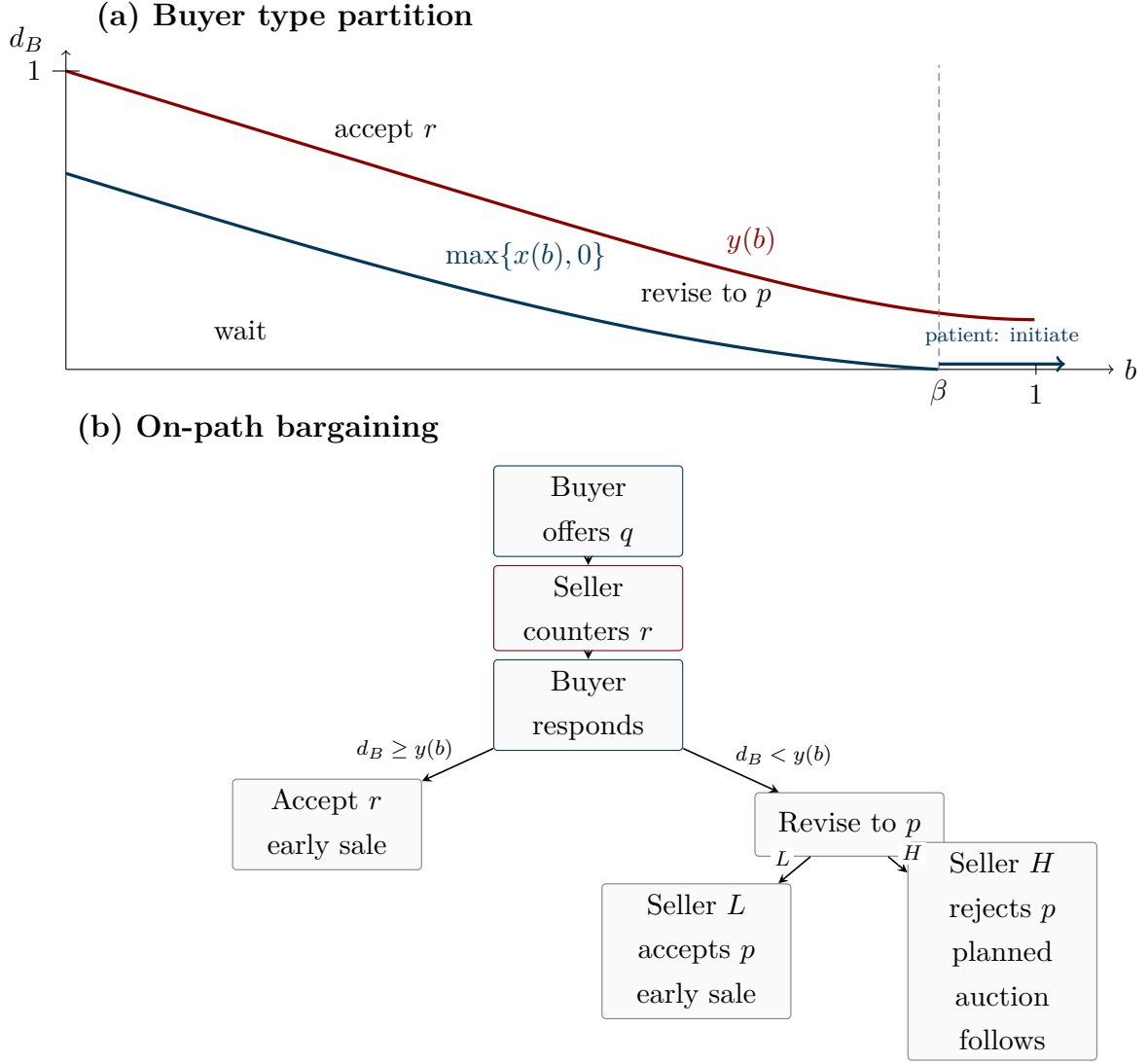


Figure H.1: Buyer type partition and on-path proposals in the repeated-bargaining PBE of Proposition H.1(ii).

I Post-Bidding Counteroffers

A separate commitment concern arises after the bidding round. We study a continuous public English clock followed by one possible seller counteroffer. As a benchmark, the terminal decision cannot condition on the earlier pre-emption action. The seller observes the clock history and the standing price x , set by the last losing dropout, but not the survivor's unsubmitted exit value. It may accept x , or release the survivor from that bid and make one final counteroffer $r > x$. Rejection of r ends without sale and gives the seller the public fallback v ; the survivor cannot revise and losing bidders cannot return (Koska and Stähler, 2021). We assume that the terminal seller has no action-linked selection information and therefore treats the survivor as an otherwise unselected iid draw from F .

Conditional on x , the survivor's value then has distribution $F(\cdot \mid \cdot \geq x)$. If

$$r = \rho(v) \in \arg \max_{p \geq v} (p - v)[1 - F(p)]$$

is the unique interior maximizer and $(p - v)[1 - F(p)]$ is weakly decreasing for $p \geq r$, the seller counters at r when $x < r$ and accepts x otherwise, and truthful clock exit remains a weak best response. Thus terminal discretion creates an endogenous, unannounced threshold and can inefficiently exclude willing buyers. The no-action-linkage calculation below derives the survivor's conditional distribution and the resulting continuation objects.

Suppose instead that the seller can identify whether the survivor is the early buyer and, if so, whether that buyer waited or probed. The seller may then condition its counteroffer on that action.

Proposition I.1 (Post-bidding counteroffer with action linkage). *Let $F = U[0, 1]$, $(n_L, n_H) = (2, 3)$, $\pi = 1/2$, $v = .4$, $d_{S,L} = d_{S,H} = .01$, $\kappa = .01$, $\lambda = 10$, and $\bar{d} = .15$. There exists a waiting-probing PBE with*

$$(q, r_P, r_W) \simeq (.738406, .780957, .661140),$$

where r_P and r_W are the action-specific counteroffer thresholds after probing and waiting. Its outcome probabilities are

$$(\Pr(\text{no early offer}), \Pr(\text{rejection}), \Pr(\text{success})) \simeq (.865398, .067301, .067301).$$

Every on-path initiator has positive urgency. The assessment survives criterion D1 in the stated one-counteroffer contingent-plan representation, over the complete set of posterior-rationalizable early responses and sequentially optimal one-counteroffer rules, including irregular posterior-revenue rules.

The action-linkage analysis below gives the fixed-point system and refinement proof. An outward-rounded interval certificate establishes the root, optimality of both on-path counteroffers, and the inequalities used in the D1 argument. Hence screening and all three early outcomes can survive even when the seller identifies the winner's early action and tailors its terminal counteroffer accordingly.

I.1 No Action Linkage

When the terminal decision cannot condition on the earlier action, let $X_{(m)} \geq X_{(m-1)}$ denote the two highest values among the $m \geq 2$ active bidders. Conditional on the final

English-clock price $X_{(m-1)} = x$, the survivor's value $X_{(m)}$ has density

$$f_{X_{(m)}|X_{(m-1)}=x,m}(y) = \frac{f(y)}{1 - F(x)} \mathbf{1}\{y \geq x\}.$$

This density is independent of m . For $x < \bar{b}$, a terminal price $p \geq \max\{x, v\}$ gives expected seller payoff $v + (p - v)[1 - F(p)]/[1 - F(x)]$. Thus, for a general distribution, the seller chooses

$$\sigma(x) \in \arg \max_{p \geq \max\{x, v\}} (p - v)[1 - F(p)].$$

Under the post-peak monotonicity condition stated above, r maximizes this expression when $x < r$, and x is a maximizer when $x \geq r$. Choosing acceptance in the latter case gives $\sigma(x) = \max\{x, r\}$. Uniqueness of the unconstrained maximizer alone does not imply this rule: revenue could rise again beyond a standing price above r . The continuation formulas below apply to the threshold rule; the constrained maximization remains valid without the shape restriction.

Let $Y_{1,n} \geq Y_{2,n}$ be the two highest values among n late buyers, and write $Y_{1s} = Y_{1,n_s}$. Define

$$R_F(n, b; r) := \mathbb{E}[\max\{r, Y_{2,n}, \min\{b, Y_{1,n}\}\}],$$

the seller's gross expected payoff when both the committed threshold and the no-sale payoff equal r . With counteroffer threshold r , the early buyer's willingness and the seller's continuation objects are

$$w_s^{CO}(b) = b - \mathbb{E}[\max\{b - \max\{r, Y_{1s}\}, 0\}],$$

$$R_s^{CO}(b) = R_F(n_s, b; r) - (r - v)F(r)^{n_s} \mathbf{1}\{b < r\}, \quad C_s^{CO}(b) = R_s^{CO}(b) - d_S. \quad (\text{I.1})$$

I.2 Action Linkage

Suppose instead that the terminal seller can identify an early-buyer survivor and remember whether that buyer waited or probed. If $\alpha_a(b)$ is the probability of action $a \in \{W, P\}$ at value b , define the unnormalized selected tail

$$T_a(z) = \int_z^1 \alpha_a(t) dF(t).$$

After the last losing bidder drops out at x , the action-specific terminal price satisfies

$$\sigma_a(x) \in \arg \max_{z \geq \max\{x, v\}} (z - v)T_a(z).$$

The normalizing probability $T_a(x)$ cancels from this maximization. Thus the early action affects both the terminal counteroffer and the buyer's payoff after a rejected early offer.

At the uniform primitives of Proposition I.1, let $Y_{1,n} \geq Y_{2,n}$ be the two highest values among n late buyers and let $\rho = (1 + v)/2$. For a sequential terminal-price rule σ with $\sigma(x) \geq \max\{x, v\}$, define

$$S_n^\sigma(b) = \mathbb{E}[\max\{b - \sigma(Y_{1,n}), 0\}].$$

The seller's corresponding gross terminal payoff conditional on b is $R_n^\sigma(b) = \mathbb{E}[P_n^\sigma(b)]$, where

$$\begin{aligned} P_n^\sigma(b) = & \mathbf{1}\{Y_{1,n} \leq b\} [\mathbf{1}\{b \geq \sigma(Y_{1,n})\} \sigma(Y_{1,n}) + \mathbf{1}\{b < \sigma(Y_{1,n})\} v] \\ & + \mathbf{1}\{Y_{1,n} > b\} [\mathbf{1}\{Y_{1,n} \geq \rho\} \max\{\rho, b, Y_{2,n}\} + \mathbf{1}\{Y_{1,n} < \rho\} v]. \end{aligned}$$

For the scalar rule $\sigma_r(x) = \max\{x, r\}$, write $S_n(b; r) = S_n^{\sigma_r}(b)$, $R_n(b; r) = R_n^{\sigma_r}(b)$, and

$$S_n(b; r) = \frac{\max\{b^{n+1} - r^{n+1}, 0\}}{n+1}, \quad w_s(b; r) = b - S_{n_s}(b; r), \quad C_s(b; r) = R_{n_s}(b; r) - d_S.$$

Thus r changes the counteroffer faced by an early-buyer winner; a late winner still faces the common threshold ρ .

For the waiting-probing assessment, let r_P and r_W denote the selected terminal thresholds after probing and waiting. Let $\tau = (b, d)$ denote a buyer type and set $A = 1 - \pi$. Its on-path waiting and probing payoffs are

$$\begin{aligned} U_W(b, d) &= AS_{n_L}(b; r_W) + \pi S_{n_H}(b; r_W) - d, & U^*(b, d) &= \max\{U_W(b, d), U_P(b, d)\}. \\ U_P(b, d) &= A(b - q) + \pi\{S_{n_H}(b; r_P) - d\} - \kappa, \end{aligned}$$

We also write $U^*(\tau) = U^*(b, d)$. The wait-probe urgency cutoff and probing probability are

$$\begin{aligned} D(b) &= q - w_L(b; r_W) + \frac{\kappa - \pi\{w_H(b; r_W) - w_H(b; r_P)\}}{A}, \\ \alpha_P(b) &= 1 - G(D(b)). \end{aligned}$$

The full-linkage fixed point solves

$$\begin{aligned} q &= \frac{\int \alpha_P(b) C_L(b; r_P) dF(b)}{\int \alpha_P(b) dF(b)}, \\ r_P &\in \arg \max_{r \geq v} (r - v) \int_r^1 \alpha_P(b) dF(b), \\ r_W &\in \arg \max_{r \geq v} (r - v) \int_r^1 [1 - \alpha_P(b)] dF(b). \end{aligned}$$

At the primitives in Proposition I.1, an outward-rounded ball-arithmetic calculation applies the Krawczyk theorem to the system above and proves a root in the box centered

at

$$(q, r_P, r_W) = (.738406414005472, .780957223173591, .661140319939824)$$

with radius 10^{-10} in each coordinate. The Krawczyk image has coordinate radii below 5.8×10^{-17} , so the root is unique within this box. Interval differentiation also makes each selected posterior-revenue derivative strictly decreasing on its support. The two root coordinates are therefore the global maxima of their respective posterior-revenue curves. Hence $\sigma_a(x) = \max\{x, r_a\}$ is sequentially optimal after every on-path clock dropout.

After an unused price p , let ν be the marginal posterior over b , let $T_\nu(z) = \nu([z, 1])$, and let a receiver plan be $\alpha = (\sigma, a_L, a_H)$, where

$$\sigma(x) \in \arg \max_{z \geq \max\{x, v\}} (z - v)T_\nu(z)$$

and a_s is the probability that seller state s accepts the early price. Define

$$K_n(\nu, \sigma) = \int R_n^\sigma(b) d\nu(b) - d_S.$$

Sequential rationality requires $a_s = 1$ if $p > K_{n_s}(\nu, \sigma)$, $a_s = 0$ if $p < K_{n_s}(\nu, \sigma)$, and permits mixing at equality. Type $\tau = (b, d)$'s deviation payoff is

$$\begin{aligned} u_\tau(p; \alpha) &= A\{a_L(b - p) + (1 - a_L)[S_{n_L}^\sigma(b) - d]\} \\ &\quad + \pi\{a_H(b - p) + (1 - a_H)[S_{n_H}^\sigma(b) - d]\} - \kappa. \end{aligned}$$

A scalar-reserve witness uses $\sigma = \sigma_r$.

It remains to verify off-path incentives when the same posterior determines early acceptance and the subsequent terminal rule. Let

$$\tau_M = (1, D(1)), \quad D(1) \simeq .040637366, \quad \tau_U = (1, \bar{d}),$$

and let

$$p^\dagger = 1 - \kappa - U^*(1, \bar{d}) \simeq .865699572.$$

Equal-revenue posterior tails and mixing at seller indifference generate scalar-reserve witness plans. In the following table, (a_L, a_H) are early-acceptance probabilities and x is the mixing probability of the indifferent seller state:

unused-price interval	terminal threshold	(a_L, a_H)	undominated type
$p \leq .644733060$	.632096261	$(0, 0)$	τ_M
$p \in [.644733060, .727669887]$	r_W	$(x, 0)$	τ_M
$p \in [.727669887, q)$	.678	$(x, 0)$	τ_M
$p \in (q, .824480449]$	r_P	$(1, x)$	τ_U
$p \in [.80, p^\dagger]$	.85	$(1, x)$	τ_U .

In each row $x \in [0, 1]$ makes the displayed type indifferent, while no other buyer type gains strictly. The intervals overlap and cover every unused price at or below $p^\dagger$. Consequently, enlarging the receiver's plan set cannot D1-dominate the displayed type: its witness plan would have to belong to a proposed dominator's strict-gain set, but it belongs to no type's strict-gain set.

For completeness, consider an arbitrary posterior ν as above, possibly with several posterior-revenue peaks. Define

$$\Phi_\nu(z) = (z - v)T_\nu(z), \quad r_\nu = \min \arg \max_z \Phi_\nu(z), \quad M_\nu(x) = \max_{z \geq \max\{x, v\}} \Phi_\nu(z).$$

Every sequentially optimal counteroffer is at least $\max\{x, r_\nu\}$, and therefore

$$S_n^\sigma(b) \leq S_n(b; r_\nu).$$

Let Λ_n be gross seller revenue when the early buyer's value is zero:

$$\Lambda_n = v\rho^n + n(1 - \rho)\rho^n + (n - 1)(1 - \rho^n) - \frac{n(n - 1)}{n + 1}(1 - \rho^{n+1}).$$

Conditioning on the highest late value and integrating the late-winner term by parts gives

$$\begin{aligned} K_n(\nu, \sigma) + d_S &= \Lambda_n + \int_0^1 nx^{n-1}M_\nu(x) dx \\ &\quad + \int_\rho^1 nx^{n-1}(1 + v - 2x)T_\nu(x) dx. \end{aligned}$$

The final integrand is nonpositive, while $M_\nu(x) \leq (r_\nu - v)T_\nu(r_\nu) \leq r_\nu - v$. Hence

$$K_n(\nu, \sigma) \leq \Lambda_n + r_\nu - v - d_S$$

for every posterior-optimal one-counteroffer rule, without a regularity or scalar-reserve restriction.

Here $(\Lambda_{n_L}, \Lambda_{n_H}) = (.562, .62005)$. At $p = p^\dagger$, rejection by L requires $r_\nu \geq p^\dagger - .152 \simeq .713699572$, while rejection by H requires $r_\nu \geq p^\dagger - .21005 \simeq .655649572$. Combining these floors with the buyer-surplus bound gives the following rigorous upper bounds:

(a_L, a_H)	$(1, 1)$	$(1, 0)$	$(0, 1)$	$(0, 0)$
$\sup_\tau \{u_\tau(p^\dagger; \alpha) - U^*(\tau)\}$	0	$< -.036828520$	$< -.032651821$	$< -.008706110$

Higher unused prices raise the relevant threshold floor and make accepted trade more expensive, so no type weakly gains above $p^\dagger$.

Finally, assign belief δ_{τ_M} below q and δ_{τ_U} above q through $p^\dagger$. Both beliefs put probability one on $b = 1$, induce terminal threshold 1 and seller continuation $1 - d_S = .99$,

and deter every deviation. Prices above $p^\dagger$ have empty weak-gain sets, so D1 imposes no further restriction. The same outward-rounded calculation encloses the endpoint comparisons, overlaps between witness branches, and the four high-price bounds above with strict slack. This completes the criterion-D1 verification in the one-counteroffer contingent-plan representation.

J Seller Urgency as the Screening State

Verification of Example 6.4. Write $A = 1 - \pi = \pi = 1/2$. Because the terminal environment is common across seller types, write its buyer willingness and seller revenue as $w(b)$ and $R(b)$. The urgent and patient seller continuations are $R(b) - .02$ and $R(b)$. For candidate prices (q_L, q_H) , buyer optimality gives cutoffs

$$D_L(b) = q_L - w(b) + \frac{\kappa}{A}, \quad D_H(b) = \frac{q_H - Aq_L}{\pi} - w(b).$$

The buyer waits below D_L , probes between the cutoffs, and makes the knockout offer above D_H . Setting q_L equal to the urgent seller's posterior continuation in the probe pool and q_H equal to the patient seller's continuation in the knockout pool gives the displayed fixed point. A 131,072-panel endpoint-inclusive audit gives residuals below 1.7×10^{-14} , action masses

$$(.550185181, .185288486, .264526333),$$

and the positive constant cutoff gap .093902289. At q_L , the urgent seller is indifferent and the patient seller rejects with margin .02. At q_H , the patient seller is indifferent and the urgent seller accepts with the same margin.

For a raw off-path completion, belief on $b = 1$ gives continuation thresholds (.668, .688). Prices below .668 are rejected, prices in [.668, .688) buy only the urgent seller at a higher price than q_L , and prices weakly above .688 buy both types at a higher price than q_H . No deviation is profitable.

For D1, buyer payoffs depend on (b, d) only through $z = w(b) + d$. A contingent seller plan matters only through its aggregate acceptance probability r , and deviation payoff is $r(z - p) - \kappa$. For $.452 \leq p < q_L$, strict-gain sets are ordered by their crossing threshold

$$\frac{\kappa + U^*(z)}{z - p}, \quad U^*(z) = \max\{0, A(z - q_L) - \kappa, z - q_H - \kappa\},$$

which is minimized at $z_L = q_L + \kappa/A$. D1 leaves all types with this same z_L tied. Belief on their largest feasible value $b_L^0 = .699522891$ makes both sellers reject because

$$R(b_L^0) - .02 - q_L = .009043008 > 0.$$

For $q_L < p < q_H$, the crossing threshold is minimized at $z_H = (q_H - Aq_L)/\pi$. Its largest feasible value is $b_H^0 = 1$, and

$$R(1) - .02 - q_H = .024292340 > 0.$$

Both sellers again reject. Below .452, rejection is the only rationalizable seller plan, so every buyer's gain set is empty and the first belief remains D1-permitted. Above q_H , the on-path knockout is strictly cheaper than any deviation under the same or narrower acceptance. These regions exhaust the off-path price set and complete the D1 proof. $\square$

K Risk Aversion

K.1 Bilateral Feasibility with Risk Premia

Condition on a known continuation state s . The early buyer's terminal-auction surplus is

$$X_s(b) := \max\{b - \max\{v_s, Y_{1N}\}, 0\},$$

and the seller's terminal monetary payoff is $P_s(b)$ from (3.1). Their expected payoffs are $\mathbb{E}_s[X_s(b)] = b - w_s(b)$ and $\mathbb{E}_s[P_s(b)] = R_s(b)$. Let CE_B and CE_S denote monetary certainty equivalents at the relevant initial wealth. Monetary payoffs use the normalization in Section 3; subtracting d_B from the buyer's payoff also shifts its initial wealth by d_B , preserving its utility over outcomes. Suppressing dependence on (s, b) , define each monetary risk premium as expected net payoff minus its certainty equivalent:

$$\rho_B = \mathbb{E}_s[X_s(b)] - d_B - \text{CE}_B(X_s(b) - d_B), \quad (\text{K.1})$$

$$\rho_S = R_s(b) - d_{S,s} - \text{CE}_S(P_s(b) - d_{S,s}). \quad (\text{K.2})$$

Concavity implies $\rho_B, \rho_S \geq 0$. The buyer weakly prefers a certain early trade at price p when

$$b - p - \kappa \geq \text{CE}_B(X_s(b) - d_B) = b - w_s(b) - d_B - \rho_B,$$

so its maximum early price is $p_B^{\max} = w_s(b) + d_B + \rho_B - \kappa$. The seller weakly prefers early trade when $p \geq p_S^{\min}$, where

$$p_S^{\min} = \text{CE}_S(P_s(b) - d_{S,s}) = R_s(b) - d_{S,s} - \rho_S.$$

The interval $[p_S^{\min}, p_B^{\max}]$ is nonempty exactly when (9.1) holds. The risk premia, buyer resolution benefit, and saved seller timing cost increase what the parties can jointly devote to early agreement; the competition wedge and offer cost are what must be financed.

This calculation concerns bilateral feasibility conditional on s . In the equilibrium

game, the buyer offers before observing the seller's state. The certainty equivalent of the resulting compound lottery is generally not the probability-weighted average of state-specific certainty equivalents. Probing removes auction risk only when seller L accepts, whereas knockout removes it in both states, so the relevant buyer risk premium is action-specific. Recovering the baseline cutoffs by adding risk premia to the direct gains from early resolution requires the same monetary risk premium to apply across states and acceptance profiles.

Risk aversion changes the value of the auction lottery but not the allocation rule: truthful bidding remains weakly dominant in the committed second-price benchmark for any increasing utility function. Thus the extension changes the buyer's ex-ante choice between waiting and offering, not its optimal terminal bid.

K.2 Buyer Risk Aversion and Equilibrium Sorting

For Proposition 9.1, specialize to deterministic late-bidder counts $n \in \{2, 7\}$, zero no-sale value, and iid $U[0, 1]$ late values. Write $Y_{1n} \geq Y_{2n}$ for the two highest values among the n late buyers. For a buyer with CARA utility $u_\alpha(x) = -e^{-\alpha x}$, the coefficient $\alpha > 0$ measures absolute risk aversion. Its terminal-auction surplus against n late buyers is $X_n(b) = \max\{b - Y_{1n}, 0\}$. Define the exponential moment of that surplus and its value-normalized form by

$$M_n(b, \alpha) = \mathbb{E}[e^{-\alpha X_n(b)}], \quad Z_n(b, \alpha) = e^{\alpha b} M_n(b, \alpha) = \mathbb{E}[e^{\alpha \min\{b, Y_{1n}\}}].$$

Expected terminal utility is $-M_n(b, \alpha)$, so maximizing expected utility is equivalent to minimizing M_n . The normalization Z_n removes the common cash component $e^{-\alpha b}$ and simplifies the comparison of actions. For uniform late values,

$$M_n(b, \alpha) = 1 - b^n + n \int_0^b y^{n-1} e^{-\alpha(b-y)} dy.$$

The risk-neutral seller's expected terminal revenue conditional on the early buyer's value is

$$R_n(b) := \mathbb{E}[\max\{Y_{2n}, \min\{b, Y_{1n}\}\}] = \frac{n-1}{n+1} + b^n - \frac{n}{n+1} b^{n+1}.$$

Thus $R_L = R_2$ and $R_H = R_7$ in this late-participation-only specialization. With equal prior probability on the two states, waiting gives the exponential moment

$$W(b, \alpha) = \frac{1}{2} M_2(b, \alpha) + \frac{1}{2} M_7(b, \alpha).$$

A probe at price q is accepted in state L , rejected in state H , and incurs κ in either state. Its exponential moment is therefore

$$P(b, \alpha; q) = e^{\alpha\kappa} \left[\frac{1}{2} e^{-\alpha(b-q)} + \frac{1}{2} M_7(b, \alpha) \right].$$

The buyer prefers probing exactly when $P \leq W$. Because expected utility equals the negative of these moments, the following positive rescaling is the buyer's scaled expected-utility gain from probing:

$$\mathcal{F}(b, \alpha; q) = 2e^{\alpha b} \{W(b, \alpha) - P(b, \alpha; q)\} = Z_2(b, \alpha) - e^{\alpha(q+\kappa)} - (e^{\alpha\kappa} - 1)Z_7(b, \alpha).$$

Thus $\mathcal{F} \geq 0$ means probing. Its derivative with respect to match value is

$$\mathcal{F}_b = \alpha e^{\alpha b} \left[(1 - b^2) - (e^{\alpha\kappa} - 1)(1 - b^7) \right].$$

For both calibrated risk-aversion coefficients, $e^{\alpha\kappa} - 1 < 2/7$. Because $(1 - b^2)/(1 - b^7) \geq 2/7$, we have $\mathcal{F}_b > 0$ for $0 < b < 1$: within either risk-aversion class, a higher match value makes probing more attractive and there is at most one match-value cutoff.

Set $\alpha_L = .37$, and let β denote the match-value cutoff for the more-risk-averse class: an α_H -buyer probes exactly when $b \geq \beta$. We construct a parameter point and equilibrium by solving for (q, β, α_H) . The first equation makes the highest-match-value buyer in the α_L class indifferent, the second makes seller L indifferent at the selected probing price given the pool $b \geq \beta$, and the third makes the buyer with $(b, \alpha) = (\beta, \alpha_H)$ indifferent:

$$\mathcal{F}(1, \alpha_L; q) = 0, \tag{K.3}$$

$$q = \frac{1}{1 - \beta} \int_{\beta}^1 R_2(t) dt = \frac{3 + \beta + \beta^2 - \beta^3}{6}, \tag{K.4}$$

$$\mathcal{F}(\beta, \alpha_H; q) = 0. \tag{K.5}$$

Verified interval arithmetic, which expands every numerical bound to include possible rounding error, locates a solution at

$$\begin{aligned} q &= .655960451221777\dots, \\ \beta &= .811699930342779\dots, \\ \alpha_H &= 2.7880279803\dots \end{aligned}$$

Equation (K.3) and monotonicity imply that every α_L -type below $b = 1$ waits. Equations (K.4)–(K.5) imply that α_H -types probe exactly when $b \geq \beta$. Seller L is indifferent and accepts, whereas seller H 's rejection margin is .1954402.

For the D1 refinement, the two seller states have disjoint ranges of possible posterior

continuation values:

$$R_2(b) \in [1/3, 2/3], \quad R_7(b) \in [3/4, 7/8].$$

The separation orders all posterior-rationalizable seller responses to an unused price: low prices are rejected in both states, intermediate prices may be accepted only in state L , and sufficiently high prices may be accepted in both states, with mixing possible only at seller indifference. For every potentially profitable price below q , D1 leaves the top-value type in the less-risk-averse class, $\theta_L^* = (1, \alpha_L)$, undominated. Under that belief both sellers reject because $R_2(1) = 2/3 > q$. For potentially profitable prices above q , the corresponding D1-undominated type is the top-value type in the more-risk-averse class, $\theta_H^* = (1, \alpha_H)$, and seller H rejects. The remaining price intervals offer no buyer type a strict gain. Thus the beliefs

$$\mu(\cdot | p) = \begin{cases} \delta_{\theta_L^*}, & p < q, \\ \delta_{\theta_H^*}, & p > q, \end{cases}$$

together with sequentially optimal seller responses, cover the complete price line. The certificate verifies these D1 gain comparisons for every buyer type and every unused price, including the points at which the seller's optimal response changes, as well as every strict seller-response margin. This proves Proposition 9.1.

K.3 Calibrated Continuation and Continuous Risk-Aversion Bands

Let λ denote the probability of the more-risk-averse class. The equilibrium equations, with state and class probabilities restored, have a nonsingular Jacobian with respect to (q, β, α_H) . The implicit function theorem therefore gives a nearby calibrated family $(q, \beta, \alpha_H) = g(\pi, \kappa, \alpha_L, \lambda)$, preserving the sorting and strict seller and D1 margins. Here α_H adjusts jointly with the equilibrium price and cutoff. This result does not establish robustness to independent changes in all primitive coefficients, including α_H .

An explicit construction also replaces the two atoms by continuous bands. Hold the displayed $(\pi, \kappa, \alpha_L, q)$ fixed. For $\varepsilon > 0$, put half the probability uniformly on $[\alpha_L - \varepsilon, \alpha_L]$ and half uniformly on $[c(\varepsilon) - \varepsilon, c(\varepsilon) + \varepsilon]$. These distributions converge to the binary distribution as $\varepsilon \rightarrow 0$ and $c(\varepsilon) \rightarrow \alpha_H$. The lower band is one-sided because the top lower-class buyer is initially indifferent. The certified inequality $\mathcal{F}_\alpha(1, \alpha_L; q) > .01166$, together with $\mathcal{F}(1, \alpha_L; q) = 0$, makes all lower-band buyers wait for sufficiently small ε , apart from the indifferent endpoint. The strict condition $e^{\alpha\kappa} - 1 < 2/7$ preserves monotonicity in $b < 1$ throughout both bands.

For α near α_H , define the smooth cutoff $\beta(\alpha)$ by $\mathcal{F}(\beta(\alpha), \alpha; q) = 0$. Let

$$H(\alpha) = \int_{\beta(\alpha)}^1 [R_2(b) - q] db, \quad \mathcal{G}(c, \varepsilon) = \frac{1}{2} \int_{-1}^1 H(c + \varepsilon t) dt.$$

Seller L 's zero-rent condition is $\mathcal{G}(c, \varepsilon) = 0$. At $(c, \varepsilon) = (\alpha_H, 0)$, it holds by (K.4), and

$$\mathcal{G}_c(\alpha_H, 0) = -[R_2(\beta) - q]\beta'(\alpha_H), \quad \beta'(\alpha_H) = -\frac{\mathcal{F}_\alpha}{\mathcal{F}_b} < 0.$$

Interval arithmetic encloses $\mathcal{G}_c$ in $-.0005138999 \pm 4.42 \times 10^{-11}$, strictly below zero. The implicit function theorem therefore supplies $c(\varepsilon)$ for all sufficiently small positive ε , with q unchanged. The offer posterior weights (b, α) by $\mathbf{1}\{b \geq \beta(\alpha)\}$ on the upper band. Its normalization is positive; $\mathcal{G} = 0$ gives seller L 's indifference, while seller H 's strict rejection margin persists. The positive waiting, rejection, and agreement probabilities converge to those of the binary assessment.

To check D1 uniformly over unused prices, consider first $p \in [1/3, q)$, where only seller L can accept. Write $S_n = 1 - b^n$. A waiting buyer's threshold acceptance probability has derivative in b with the sign of

$$N_\alpha(b, p) = (S_2 + S_7)(Z_2 - e^{\alpha p}) - S_2(Z_2 + Z_7).$$

Throughout the lower band, $N_\alpha/(S_2 + S_7) \leq e^{\alpha/3}[(5/9)e^{2\alpha/3} - 1] < 0$ for $b < 1$, so every within-coefficient minimizer has $b = 1$. Near the upper band, the certified bound $N_\alpha(b, 1/3) < -.173$ for waiting values persists; the prober's threshold increases in b , so the upper-band minimizers are the cutoff types. To compare these with $(1, \alpha_L)$, put $\delta = q - p$, $\varphi(z) = (1 - e^{-z})/z$, with $\varphi(0) = 1$, and

$$A(b, \alpha, \delta) = \frac{\alpha e^{\alpha q}}{Z_2(b, \alpha) - e^{\alpha q}} \varphi(\alpha \delta).$$

For indifferent types the threshold is $1/[1 + \delta A]$. The binary certificate gives $A(1, \alpha_L, \delta) - A(\beta, \alpha_H, \delta) > 7.82$ uniformly on $0 \leq \delta \leq q - 1/3$. Continuity on this compact interval preserves the strict comparison with every nearby upper-band cutoff, including as $p \uparrow q$. D1 therefore removes all upper-band types and all lower-band types with $b < 1$ whenever a strict gain is possible. Every surviving belief has $b = 1$, under which both sellers reject since $p < q < R_2(1)$.

For $p \in [3/4, 7/8]$, seller L accepts and only seller H 's acceptance probability can vary. Among hypothetical probers the threshold is minimized at $b = 1$, where it is

$$x_T(\alpha; p) = \frac{e^{\alpha p} - e^{\alpha q}}{Z_7(1, \alpha) - e^{\alpha p}}.$$

A waiter has a weakly higher threshold. On the positive-gain region, x_T strictly decreases in α : after dividing by $\alpha e^{\alpha p}$, its numerator decreases and its denominator increases in α . Thus the top-value buyer with coefficient $c(\varepsilon) + \varepsilon$ survives D1, and seller H rejects. The highest potentially profitable price remains strictly below $7/8$, since the binary bound is $p^\dagger < .792$. The remaining price intervals either admit no acceptance or reproduce the probing response at a higher price. These beliefs deter every deviation. Hence the two-band assessment is a D1-compatible PBE for all sufficiently small $\varepsilon > 0$. The argument gives no explicit positive band width and retains a gap between the bands; it does not establish the result for an arbitrary connected distribution of risk aversion.